

Neur 101 Intro to Neuroscience

Numbered Transcript

NEUR 101 • Dr. Baland Jalal

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Lecture 1: The Brain

At a glance — Jalal introduces the brain.

- 1.1 All right. Well, very excited to be here to talk about the brain. Thank you so much for coming. I appreciate you guys coming from far distances in some cases. This will be fun.
- 1.2 So, I'm originally Kurdish from Iraq. I grew up in Copenhagen, Denmark. I grew up in a tough neighborhood over there, which you call the ghetto. It was really tough, tough beginnings. When I was a teenager in Copenhagen, I had an experience. It was weird. I was sleeping in my bedroom this one morning, and I felt like something was on my chest, strangling me. It was weird. I felt awake. I could see my surroundings. I could sense the room, the ceiling. My Tupac poster was there and everything, but I felt like there was a ghostly evil presence in the room. And then suddenly, I saw my legs going up and down, and I wanted to move, but I couldn't. I was literally paralyzed.
- 1.3 At this point, I thought, "I might die from this. What's going on? Am I dying? Is this what death looks like? This transition from this world to the next. Is that happening now? " It was crazy. And with each moment, I just felt things becoming more and more intense to the point I was being strangled, and I was like, "I'm going to die now. It's getting worse. I can't breathe. " I just wanted to scream: "Dad, mom." But I couldn't speak. I couldn't say anything. And then, finally, I was able to jolt myself out of this state. So, the next day, I was terrified. "What do I do? Do I go tell my parents? " At the time, I was, you can say, the black sheep in the family. I wasn't the best kid. So, what do I go tell my parents? I had a ghost visit me in my little room. Would that be nice? Probably not. That'd be pretty insane. And so, do I Google, "I saw a ghost?" That'd be weird what would happen there.
- 1.4 So, I later discovered that there's something called sleep paralysis, this condition, this state where you're paralyzed from head to toe, and as you are paralyzed, you literally might see a ghost in your room, and this ghost might strangle you. You may feel like you're dying, and it can come in all kinds of shapes. Some people have this creature; they see this being wearing a hat and this black outfit and looking like I'm looking now. So, they might see this guy like Freddy Kruger or in some cultures, it has other names, like space alien abduction might be one version of this.
- 1.5 So, I had this experience, and this took me on a journey to understand the human brain because I realized, well, this is actually something the human brain constructs. This is something that—the brain can create these sights, these vivid, intense, crisp visions of ghosts, and if the brain can create this, this must be fascinating. At the time, I was a really, really bad student. I was very bad in school. I had the worst grades, and science was the worst of all. Dahlia, physiology wasn't for me. I didn't like physiology, chemistry, all that stuff. Not my thing. But having this experience, I started to look into the brain, what's going on, and this took me on this journey to six different countries,

seven different countries, to study this phenomenon called sleep paralysis. I studied the brain. I went to California, studied with one of the legends in neuroscience under his wings, learning about the brain and all that stuff. So, from this bedroom, having this weird experience, I went all the way to all these places to study the brain. Is that clear? That's what happened. That was my background.

- 1.6** The brain is strange. It's weird. It's this organ. It's made up of 100 billion neurons. These cells that we have in the brain communicate; they chit-chat with each other; they use synapses. Each one of these neurons, they have about a 1,000 to 10,000 connections with other neurons. That makes the brain the most complex form of matter in the known universe. Everything we know in the whole universe, the brain is the most complex. It's just the most complex object in the known universe.
- 1.7** So, in this course, you'll be learning about the human brain, and we'll go from the nitty-gritty of neurons, synapses, and things like that. We just got to learn that stuff to have basics, like directions of the brain, what are the names of the different structures of the brain. So, we'll go from that all the way to things like Shakespeare and genius and how does spirituality come about in the brain. Is there something like a genius? So, that's the journey, going from the small stuff all the way to the big stuff. That's the plan.
- 1.8** The human brain, if you look at it, is unique compared to other animals. If you look something like the chimp, for example. If you look at a chimp brain, the chimp brain is not that different from ours in many ways. We share 98% of our genes with the chimp. The only difference is the following: when cells started to divide—so, you had a single cell that became two cells, and then divided, and more and more cells—that continued all the way to 100 billion neurons. That was the end product for the human brain.
- 1.9** But chimps, their brain, also had the same division going on, but because of a single gene—so, there's a single gene deciding when this division, this process of division and dividing of cells should stop. So, this single cell, this single gene, is different in human beings. It stops a few rounds earlier than in chimps. For that reason, their brain is one-third of our size. For that reason, you have all this massive complex difference in the sense that the human brain being that vastly unique. We're having language and things like that. It's from a single gene. That's quite interesting. I call this the police officer gene; that single gene making all that difference.
- 1.10** Now, neurons, they talk to each other. First, you have the neuron itself. This part of the talk, by the way, is going to be more technical. I just want to lay a foundation for you before we go further up. So, you have the neurons, and they talk to each other through these thread-like fibers. You have the axon, so the soma, the cell body, and then you have the axons, these fibers protruding, coming out of the neuron. And then, on other neurons, you have dendrites, other fibers. The axons and the dendrites allow the communication flow between neurons. Then there's a synapse in between. The synapse is the gap from where all the magic happens. That's where the communication

occurs through these synapses. Cells in the brain, neurons, cell bodies, axons, dendrites, synapses. That's where the communication takes place.

- 1.11** Now, the cell body itself is boring. Nothing much happens. You have the mitochondria, you have the DNA, you have the basic stuff. The unique part is really the axons and the dendrites in the brain cells compared to the cells of the rest of the body.
- 1.12** Now, in terms of the communication in the brain, it's chemical in the sense that it goes from—it's electrical. There's an electrical impulse called the action potential. This is what allows the communication to occur. You have an action potential occur, an electrical charge, and the neuron travels down the axon. And then, a chemical reaction occurs whereby synapsis, they have vesicles that are being released. So the molecules are released; these are the neuro transmitters, the chemicals, and they travel in this fluid-like substance and then go to the other side of the dendrite. This is the presynaptic neuron and then the postsynaptic neuron.
- 1.13** To be honest with you, when I had biopsychology early, very my first classes, I found this stuff so boring, to be honest, because it was like, well, here's the neuron, here is the synapse, here's this, here's that. It was boring because there's no context, really. That is boring. Then we got to the brain centers, and that's the next step. So, you go from a neuron, and then you go to the centers of the brain, the basal ganglia, here's the globus pallidus, here's the ventral dorsolateral prefrontal cortex, here's your brain stem, all that. Kind of boring because there's no context. It's just a lot of fancy Latin names for different structures.
- 1.14** But once you actually learn what these things do, you have this communication between these cells, and here's the magic of reality, all our dreams, poetry, pity, infinity, Shakespeare. It all comes about through this chit-chat. That's crazy. How is it that all our sensory experiences, when we fall in love, everything comes about through these neurons? It makes no sense. Have you ever held a brain in your bloody hands? It's just this [inaudible], little substance. There's nothing unique about it when you look at it. It's a jelly-like substance. It's like a cauliflower, really. The cortex, the outer layer where 30 billion neurons are placed, is like a folded book. It looks like a cauliflower because our brain has to have it all inside the skull. Otherwise, it just couldn't fit in there. It's just squeezing it all in, and nothing fancy about it. But yet it's so fancy. It's crazy. You agree, Clay? Okay. Very good.
- 1.15** So, the different types of neurons—and again, I'm going to the technicals because you have to learn this. You got to know this. There are different types of neurons. There are multipolar neurons, unipolar, bipolar, all this stuff. It just means that some neurons have the cell body and then one axon and one dendrite. Others have one of these processes extending from the cell body. Most of the neurons are called multipolar. That means, Jose, that, basically, there are multiple processes extending from the cell body. Does that make sense? Multi, got it? Five, two. Uni, one. Interneuron, none. Okay? Got it. Okay. Got to learn this, man.

- 1.16** And then, there is myelin, a fatty substance on the axon. The axon does what? Communicates, it sends signals. Dendrite? Receives. Dendritic spine? That 's the place where the thing, the neurotransmitter, goes to. Over here, you have the axon terminal. Technical names, got to learn them.
- 1.17** So, you have this fatty-like substance on the axon. This is actually pretty cool because until you have brain damage and you realize—let 's say you're depressed. So, for depression, you have all the stress and the glucocorticoids flooding the whole body, and then you have all t his myelin being removed, and so your neurons can't communicate that efficiently and fast. So, you realize this is very important. Or MS, multiple sclerosis, you don't have all this fatty substance on the axon, allowing the action potential to occur, the electrical charge.
- 1.18** The electrical charge itself, think about it. Here you have— chemicals are charged. They're called ions. Dahlia would know this. They're called ions. You have a bit of this—you have the sodium, which is positive. It's on the outside of the cell membrane. And then, you have potassium inside the cell, which is negative. So, the cell is more negative at rest, and the membrane, meaning just the little wall that separates the bloody neuron from the rest, is permeable, meaning that things can—semi-permeable, so things can go in at times, but other times they can't. Certain ions can cross, others can't—some chemicals, meaning.
- 1.19** Now, at this point, when there's a stimulus, this electrical stimulus, you have the sodium ions rushing. They rush to the cell; they go inside. As they rush to the cell, there are some gates, like channels. They open up, ions go inside the cell. The cell now becomes a little bit more positive. The potassium ions slowly go out. There's a flip from negative to positive, and in that flip, it goes back, and that's the whole electrical thing. Is that clear? Do I make sense? Not really. Do I? It kind of makes sense, right?
- 1.20** Outside: sodium, positive. Inside: potassium, negative. Resting state. Electrical stimulus. The gates open: go inside—just chemistry. Go inside, changing the electrical charge in the cell. And then, it goes from that electrical to the chemical neurotransmitter. Dancing, dancing, dancing, dancing, going over there to the dendrite, to the dendritic spine. Is that good? All right, so we have all that going on.
- 1.21** All right, these synapses can be on and off at any given point. They could be inhibitory excitatory. And these neurochemicals, these neurotransmitters that are being sent to the other side, crossing the big ocean, can be either inhibitory or excitatory in nature, meaning they can be less likely to start an action potential in the next neuron or not. Create a signal in the next neuron or not, pretty much.
- 1.22** You have different—actually, the brain has a lot of these neurotransmitters. These neurotransmitters, when I learned about them, like serotonin, dopamine, glutamate, aspartate, GABA, all this stuff, I was like, "What? Does that make sense? That's boring learning these names." But then I realized that these actually have functions. Every day,

we have a little bit more of this, and we go through depression, and then we might have glutamate go up, and then you have GABA, not enough GABA, to inhibit the brain, and serotonin, you feel a sense of well-being. You're excited about this talk, so you have a lot of dopamine, maybe or maybe not. So, these chemicals actually have functions.

1.23 There's something called neuropeptides, like oxytocin. You fall in love. A lot of oxytocin there, like the bonding. You have a child. Michael just had a child. You had a lot of oxytocin, helping your brain to remove and delete certain circuits and allowing you to bond with the child and be less selfish and less ego-centered and all that stuff. Less stressed. Absolutely, my man. You have all that.

1.24 These neurochemicals—about 14 we usually talk about—neurotransmitters, but they're maybe closer to 100. There's a lot of these. So, you have those, and as we said, their goal is to either inhibit or excite the next neuron. So, it's inhibitory or excitatory. There's this inhibitory and excitatory dance going on in the brain, really. That's what it is. The brain is just going inhibitory and excitatory. Certain neurochemicals, like glutamate, for example, which is the main excitatory neurotransmitter in the brain, just make the brain very excited. There are a lot of chemicals going on, a lot of action potentials, a lot of stuff going on. For neuroplasticity, you need a lot of glutamate; it helps you form your connections. Neuroplasticity is a term for the brain being able to change itself. We'll get to that.

1.25 Some more terms for you guys.

1.26 Amino acids. Amino acids are the names for a class of neurotransmitters, like glutamate, aspartate, GABA, glycine. Monamines is another name: dopamine, epinephrine, norepinephrine, serotonin. They have different functions. They have a different basis in terms of their molecular origin. And then, you have the acetylcholine and the neuropeptides. These are the classes. But when you learn about them, they're very cool.

1.27 Like acetylcholine is cool, acetylcholine is part of the attention system. So, when I learn—when I want to change my brain to be better at some skill or something—acetylcholine is that neurotransmitter that's part of the nucleus basalis system. In order to change my brain and to release growth factors from protein that allows me to create new circuits in the brain, I need a lot of this acetylcholine.

1.28 So, what I do is, for example, I have to pay a lot of attention to, let's say, learning a new skill. I don't know what I would learn that would be completely new, like ballet or something; that would be very new for me. So, I'd have to pay a lot of attention to this. It would be very new. So, I activate my nucleus basalis system, and in the nucleus basalis, you have this acetylcholine neurotransmitter being released. And when I release that, I have a lot of focus, I have a lot of attention, and when I have that, I

release nerve growth factors, which I will get to later on, and it allows my brain to be more plastic; it gets into the plastic state. Did you know that? That's interesting.

1.29 So, that's what I'm saying. We go from knowing the Latin names and all these names, and then we actually learn what it means to my life, to Jose's life, to Baland's life. How do I actually change my brain, for example? Well, you learn about this. So dopamine, when you have this reward neurotransmitter when you are excited, you have this excitement, you have a certain form of happiness. There are two forms of pleasure, and we'll get to that, too. So that's more like, "I feel satisfied;" t here's a consumatory pleasure. And then, there's this excitement pleasure when I'm having a gift or something. "I'm excited to unwrap the gift;" that's another type of dopaminergic pleasure. But the dopamine system is also part of the brain plastic system, so to speak. Those are some of the functions that these neurotransmitters have.

1.30 So, we can say that there are these neurotransmitters, and then there are neuropeptides, which are different. They're released in a different manner. They are a bit different in that way. And so, oxytocin would be one example. Endorphins—they're so interesting. I find endorphins really interesting. You know that Ronald Reagan, when he was shot, he was like, boom, blasted. But yet none of his security people knew that he was shot. Nobody knew because he didn't have pain. He felt no pain, and they took this chap to his car and nobody knew. And then later when they interviewed him, he said, "I've only been shot in the movies, and in the movies, I'm supposed to feel pain and agony. But when I was shot in real life, I didn't feel anything". That's because the endorphins have this system of quelling and inhibiting pain. And so, a lot of pain, as we will see—pain is one of the systems in the brain that you can manipulate.

1.31 Psychologically, there are a lot of ways to trick yourself not to feel pain. Pain is, in many ways, an illusion. The brain has certain gates and has the potential to actually kill pain. So, these are the endorphins. Acupuncture, for example. You know acupuncture, when you have, for example, in certain countries, in China, for example, it's been used for a very, very long time, for hundreds of years before surgery. So, they have somebody lying there, they don't do a painkiller, so what they do, they might do acupuncture on this chap, and then they do the surgery, and he won't feel pain.

1.32 Over here in the West, we're like, "What are these guys doing? That's some crazy stuff." But then, what they did, actually, they saw that when you do this on animals, it works, too. What's going on? You can do this on animals and you can do surgery on animals. Right, doctor? You do some surgery on animals, and it even works.

1.33 Then, they had some drugs, antagonist drugs, to basically antagonize the receptors of the endorphins, and lo and behold, the acupuncture doesn't work anymore. It doesn't have the pain- quelling, dampening abilities anymore. So, they thought, "Oh my God, it's working; the acupuncture may be working through the endorphin system." Neuropeptide—can you see? You go from the boring name all the way to the function. When you go to the function, that's when the interesting part comes in. Got it?

- 1.34** Some more systems and divisions. The brain goes from neurotransmitters, neurons, and then neurotransmitters and synapses and all that, and then we have— they will make circuits, so to speak. They make circuits, meaning you have a cluster, a set of— you have a cluster of neurons over here in the brain, set over here, clusters over here, and then they make circuits; they communicate. So, through the axons, these fibers allow these different parts of the brain to communicate. These are called circuits. And together, if there are several circuits, we call it a system.
- 1.35** Roughly, we talk about the central nervous system, the CNS, and the peripheral nervous system. It's a fancy nerd-speak for the brain and the spinal cord. The brain and the spinal cord, we call the central nervous system. If you have damage, a car accident, can you regenerate your spinal cord? No. Impossible, right? That's not possible. But if I cut the wires here, the nerves here, can I regenerate? Yeah, we can. We can, and that is because we have something called glial cells, Schwann cells, part of the glial family, that allow that to happen.
- 1.36** I don't think I mentioned glia, so you should know there are all these various kinds of neurons, but there are also glial supporting cells. It's not that exciting, but they help with some of the circuitry, help with some of the myelin. They outnumber the neurons by one to 10. So, there are a bunch of these guys, the glial cells. They have fancy names like Schwann cells and all that. You don't need to know that. The Schwann cells in the peripheral nervous system allow for the tracks to regenerate. Sometimes, actually, what occurs, you may have a track, like some nerves being cut, like, say, there's a medial, radial, and ulnar nerve. Is that right, doctor? The ulnar, medial, and radial nerve. Okay, so you got the nerves, right? You cut them, so the nerves are cut, but for some accident or something, you have the nerves crossing in the wrong way, and then you want to move this finger, and then you move this one instead. That can happen, too.
- 1.37** But in the peripheral nervous system, which is a system that brings information from the sensory receptors, you have thermal pain receptors, temperature receptors, touch receptors, all kinds of receptors here in the skin that send information, go to the spinal cord, and then go into the brain. The peripheral nervous system sends commands to your organs, the glands. Central nervous system, peripheral nervous system.
- 1.38** The somatic nervous system is the one that sends commands to the muscles to move and brings information from the sensory nerves. Somatic nervous system. The part of the peripheral nervous system that regulates the internal organs is the autonomic nervous system, the glands, the hormones, and all that stuff. That part can also be further divided into a parasympathetic and a sympathetic nervous system. Sympathetic meaning not that I'm being sympathetic with somebody, but I'm in a stressed, hypervigilant state, a fight-and-flight reaction thing. When you're agitated and all that stuff, you're in a sympathetic state. When you're in a relaxed, calm, meditative state,

it's called the parasympathetic state. You just got to learn this. This is the basics. Once we get out of this whole mess, it'll be more fun.

- 1.39** Also, you have to learn about directions of the brain. The directions meaning—if you're looking at this most complex form of matter in the known universe, you want to know if you're going this way, what does that mean. What is left? Even in my little ghetto in Copenhagen, we had to know what's the name of that street and that street. What about the human brain, the most complex form of matter? You got to know where it's going, which directions. But when you talk about the brain, you go lateral (meaning out), medial (inside), that's what you call the structures, dorsal (up), ventral (down), medial, anterior, posterior. Superior, sometimes also is used, inferior. Is that clear? All right, superior, inferior, dorsal, also ventral. You go lateral, medial.
- 1.40** And then, you cut the brain. You cut the brain here because you want to have a brain scan, and it's called a coronal view. Not Corona, but coronal view. And then, if you cut it from the middle right here, cut the middle here, it's called the sagittal view, horizontal section, and it looks like a wrinkled man, really. It's wrinkly, right? It's all wrinkly. These wrinkles, you call gyrus, gyri, the wrinkles, and the furrows inside them, we call sulci, a sulcus. And then, in between, you have the fissures, the fissure, like the longitudinal fissure, okay, here. You have that, and then you have the corpus callosum, which is a wall that separates the two hemispheres. Each hemisphere is like this mirror-like walnut, half on each side, and there's this wall allowing them to communicate.
- 1.41** Now, if you cut this wall, Baland does a karate chop, cut this wall, and then these two won't be able to communicate, these two hemispheres. And that's interesting because it's actually done sometimes when people have, or it used to be done a lot when people had epileptic seizures, the grand mal seizures, where they were like, a lot of movement and all that. Volleys of electrical activity in the brain, and so they have all this movement. So, to prevent that from spreading from one hemisphere to the other, they would cut the corpus callosum.
- 1.42** That was interesting because then, when you had this chap stand there with no corpus callosum, it has been surgically removed, and you ask his hemisphere over here, the left hemisphere, by communicating over here on a screen, you say, "Jalal, do you prefer vanilla or chocolate?" I might go, "I prefer vanilla ice cream." Then, you communicate with the other hemisphere over here on the screen. You ask the same question. He says, "Chocolate." Here he says vanilla, here he says chocolate.
- 1.43** Mind you, each hemisphere controls the other side of the body, meaning everything I see from this side of the world comes to my brain on the right side and vice versa. So, there's this—when I move my right arm here, it's the left hemisphere. When I move the left, right hemisphere. I get a stroke over here; it doesn't work. Right hemisphere. So, the corpus callosum, when it's cut, you have two individuals, two consciousnesses in the same body, in a sense. That's interesting. So, the corpus callosum allows the hemispheres to communicate.

- 1.44** I remember learning this and I thought, "No way!" I mean, this is boring. Telencephalon, diencephalon—it's the forebrain, upper brain, the cortex, that region), thalamus, just fancy names for structures and parts of the brain that are up here. Okay, got it? Then, you have the midbrain, the middle part of the brain, mesencephalon. And then you go to the hindbrain, the lower part of the brain, the brain stem, which is the lower part, and that's called metencephalon and myelencephalon. So, we have these names.
- 1.45** Let's start with the cortex. The cortex is the upper part of the brain. It's the outer layer. As we said, there are approximately 30 billion neurons. So, if each of these makes contact to about 1,000 to 10,000 other neurons through these synapses, that is calculated to something about 100 million billion synapses. That's pretty crazy. There are different lobes, there are different areas in this cortex, in this outer layer of the brain. Are you all familiar with the term 'cortex?' I just want to gauge you. Do you all know the term cortex? Cortex. You're on top of it. I didn't know cortex when I was starting out, but obviously, you're not starting out, and that 's great.
- 1.46** The cortex, the outer layer of the brain, is a fancy area because a lot of our higher faculties come about from the cortex. It's divided into four lobes, four areas. You have the frontal lobes, this part. In the frontal lobes, you have things like movement, the motor cortex. When I move an arm, it occurs in the frontal lobes. You have something like higher order thinking, decision making, being able to plan for the future, to think logically, rationally, all that stuff is in the prefrontal cortex, meaning the most frontal part.
- 1.47** You have the parietal lobes. Parietal is the place in the brain where you have spatial navigation, meaning you know yourself in space, where am I in space, my body image. I close my eyes. So, we all have a body image, a sense of self, a sense of I, me. I feel anchored in this body. I'm anchored in Baland's body. I move my body from A to B in space, and I know where it's positioned. These are the parietal lobes. When I move around in a room navigating space.
- 1.48** Then, you have the occipital lobes at the back of the brain involved in vision, seeing, seeing the world. So you have that.
- 1.49** And then, you have the temporal lobes. So, that's the region of the brain involved in, well, hearing because it's right next to the ears, but it has all kinds of functions. It's very close to the emotional part of the brain. It's on the outside surface. And a lot of vision occurs over there, too, our ability to see faces, recognize faces. A lot of that is there, too. That's the outer layer of the brain.
- 1.50** You go a little bit deeper; you look at other structures. Beneath it, you have places like the thalamus—it 's a relay station, meaning when you have any sensory information coming through the senses, which first is registered in the cortex: seeing, hearing, touch—touch is in the parietal lobes—all these things, it always goes through the

thalamus. There's a region that's a stop station in the brain called the thalamus. Actually, it's unknown exactly why we would need a stop station, but there is this stop station in the brain for all sensory information processing. All the sensory information coming in always goes through this thalamus region. For what? I don't know. Only God knows. But for some reason, we have the thalamus. The thalamus is an important structure lower down.

- 1.51** Hypothalamus. Hypo-thalamus means beneath. It's the hormone factory of the brain. We'll get to that because the hypothalamus is so fascinating. The hypothalamus is insane. This is the hormone factory, meaning it directs all the hormones. It signals another part of the brain called the pituitary further down in the brain. Then, it sends commands to the glands in the body, where hormones are eventually released. But this is the master gland. In this part of the brain, when you get sleepy, you need sunlight in order not to feel sleepy all the time. You need the sunlight to set your biological clock, for example. When you get that, my eyes don't like these lights, but when you get that, you communicate with the hypothalamus, setting your biological clock, the suprachiasmatic nucleus of the hypothalamus.
- 1.52** Now, the same region of the brain controls aggression. The same region of the brain controls sexual behavior, sexual motivation. Now, then you might ask, "Why is rape or sexual aggression a factor for male sexual motivation sometimes?" The same part of the brain, it's like the size of a marble, controls aggression and sexual behavior right next to each other. When places in the brain are close to each other, you often see cross-wiring and cross-activation. So, might it be a coincidence over here that you have sexual motivation and aggression coming from the same part of the brain? It's interesting.
- 1.53** On that note, in the parietal lobes, there's a homunculus, meaning there are neurons creating a map of an entire human body over there. It's called topographical organization. It has this human head, arm, but it looks like the way I'm looking now. It has that same—so, the feet area is next to my leg area. The leg area is next to the rest of the body, the stomach, the arm. So, it has that topographical organization, meaning it resembles the human form. Now, you might say, "Why does that come about, Baland? What is that about?" And we'll get to that because that's really fascinating. It turns out there's a very straightforward reason for this. I'm going to keep that for later, but it turns out that the feet area is right next to the sexual organs in the brain. So, next time you get a foot massage, you may have other sensations, too, because there's a lot of cross-wiring going on. When two structures are next to each other in the brain, they tend to overlap a lot. There's a lot of cross-wiring and cross-activity. So, some very interesting stuff. We'll get to it.
- 1.54** Now, you have the hypothalamus. You have the limbic structures of the brain, the emotional part of the brain. It's called the limbic structures. It's like a limbic ring. It's tucked behind your temporal lobes. Now, we're getting deeper inside the brain, a little

bit deeper inside it. In the limbic ring, you have things like the amygdala: processing fear, processing forms of aggression, fear, anxiety. It's called the amygdala, an almond-like structure behind your ears, the amygdala. I'll get to this point over and over, to the structure.

- 1.55** You have things like the fornix. The anterior cingulate, it's active when I'm vigilant and I'm vigilantly monitoring the world. In obsessive-compulsive disorder, the anterior cingulate goes off the roof. It's always active, "This is dirty. I need to wash my hands." The anterior cingulate goes on the top here, on the limbic ring.
- 1.56** You have a region called the septum where you have pleasure, excitement, dopamine. So you feel excited. It's in that limbic ring. Limbic structure. The limbic system is important. It's the emotional system of the brain. Limbic system. Remember this. It's going to come over and over in your life. You'll hear about the limbic, the emotional part of the brain.
- 1.57** Then, you have another structure, which I'll go into depth with a little bit later, but it's called the basal ganglia. Also, deeper in the brain. So, we went from the cortex deeper. Basal ganglia: movement, but not all kinds of movement because we said, up here in the frontal lobes, you have voluntary movement. That's in the cortex: command. In the cortex: move, Baland, move. But then, you have another part of the brain, the basal ganglia, involved in voluntary—sorry, not voluntary—involuntary, automatic movements. When I jump out of bed, it's involuntary. When I walk, I don't really think about it; I just walk. Or when you tell me a good joke, I might laugh. That's involuntary.
- 1.58** Or that I'm smiling, in fact. A good photographer, when he tells me a good joke before he takes a portrait, I might smile. But can I smile voluntarily, Michael? Can I smile voluntarily? Not a real smile, right? There are two kinds of smiles. There's the voluntary smile and the automatic, real, whole-hearted smile. Is that right? So, it turns out the voluntary smile is uncharming. It's not really charming, is it? It doesn't look that good. It turns out that it's because the muscles are different, using different muscles. You actually have certain wrinkles here around the eyes over here when you smile, the real smile. It's called the Duchenne smile, from a French neurologist, Duchenne. So, you see these muscles. So, if you want to know if your partner is actually happy about something, look at these muscles over here. If she's really smiling or if she's making it up, faking it. You can look at that, and scientists, sometimes when they do research, they'll actually film, they'll record these muscles, see if they're moving when you're smiling. It's interesting.
- 1.59** In Parkinson's disease or syndrome, you have a death of these dopaminergic neurons that allow movement to occur in the basal ganglia. The basal ganglia—there are a lot of fancy names for various structures inside it called the globus pallidus, the striatum, the nucleus accumbens, and things like that, the caudate, the putamen. Just fancy names for structures inside this basal ganglia. Just want to throw it out so you know the names. But here, if you have Parkinson's, there's a part of the brain in a slightly

different structure called the substantia nigra, and here you have pitch-black cells that actually die. They die off. The cells that produce dopamine die here. Okay, so because of that circuit going to the basal ganglia, you don't have movement. You become stiff.

- 1.60** Then, you have Sydenham chorea, another neurological disorder. You have too much dopamine, too much movement. You become like this. In Sydenham chorea, there are jerky movements everywhere because now the system is too excited, too much excitation in the system.
- 1.61** Let's go down in the brain, a little further down to the brain stem. The brain stem is interesting. You have the cerebellum. This structure is about 20% of the volume of the brain, yet about 80% of neurons are in this structure called the cerebellum. It's involved in things like memory, but procedural memories, meaning memories that are like a procedure or automatic, like riding a bicycle or, I don't know, walking up the stairs or something like that. It's automatic. That's all in the cerebellum. It's a very interesting structure. It's involved in balance, a sense of balance.
- 1.62** You have things like the medulla oblongata. I love that name, medulla oblongata. It's another structure over there, like a structure, and that structure is also with the pons, things like breathing, respiratory, blood pressure, all that stuff. It's in that region of the brain.
- 1.63** So, there's tremendous modularity in the brain. This is another point. There's been a conflict. There's been a lot of thought going back and forth in terms of: is the brain modular, or is it a whole thing that always works in a larger context where everything is just working with everything and not in isolation. Now, it turns out, you actually have a language module in the brain. There's a module over here, on the left side of the brain, involved in language. You have something called the Broca's area. It's around the frontal cortex because the frontal is involved in motor executions. So, this is called the Broca's area. You speak, speech, syntax, grammar, all that stuff.
- 1.64** Then, there's another region over here in the parietal called the Wernicke's area involved in understanding language. Grammar, not grammar, sorry. Semantics, the actual understanding of language in the Wernicke's. These are connected by a band of fibers called the articulate fasciculus.
- 1.65** But mind you, if you have a stroke, let's say in the Broca's area, you have a stroke, you have damage, you have trauma to that part of the brain, you will not be able to produce language. Your grammar will be all over the place. Not very good. You want to speak—the guy, Paul Broca, who discovered this, his first patient kept saying, "Tan, tan, tan" over and over. He had damage to his Broca's area. So, you can't produce language.
- 1.66** Now, it's not like you can't speak. You can still speak because the motor area here in the brain controls the muscles and the execution of being able to actually produce speech or have movements of your mouth. But it's just your language. You don't have

—the grammar is not there, your ability to produce grammar, which is very unique to humans.

- 1.67** In Wernicke's area, on the other hand, if that part of the brain is damaged, you speak very eloquently like Shakespeare, but everything is gibberish, nonsense; it makes no sense. The syntax is perfect, better than in some of these schools, English students in some of these Shakespeare schools. Very good grammar. But the ability to understand language is gone. You don't understand it; it's gibberish. You speak to them, they will act like they understand, but they're just speaking gibberish. It's very scary. I actually saw a patient with this recently. It was actually a dear friend who had a stroke over there. It's like meeting a schizophrenic all of a sudden, even though they are not schizophrenic, but somehow, because of that part of the brain not working, they seem so strange. It seems so strange. Even the logic.
- 1.68** I had this conversation with my mentor out there, my former mentor, my friend. He was a physician, neurologist, scientist. We were looking at this with this patient, and we were discussing, "To what extent is language influencing a thought?" because there's a lot of debate going on among researchers and in the world of neuroscience, going back to Chomsky and Noam Chomsky saying, "Well, language is so strange, it's so unique because it seems like there's something going on here." I mean, no animal, no chimp with 98% of our genes can speak. Can they produce Shakespeare? No. Can they give this lecture? Probably not. Can they have a conversation? No. Language, being able to understand and produce language, is so unique. And then, when you see that ability go away, you go like, "My God, what's going on?"
- 1.69** So, there's a region there, the Wernicke's area, and it works together in sync with the Broca's area. Paul Broca. Carl Wernicke discovered the other module. So, there are modules in the brain.
- 1.70** There's a vision module, as we talked about, being able to see. There's a hearing module and so forth. So, the brain is extremely modular. Yet there's also a lot of cross-activation and dynamic cross-talk between regions of the brain, and that's what makes the brain interesting. A lot of cross-talk, and that's why we went from the era of the Paul Broca's and the Carl Wernicke's and all that, where we said, "No, everything, every structure equals function." You speak; you have a speech module in the brain. You have the language, you have the vision module of the brain, you have the math module of the brain. There's actually a part of the brain that is specific for math, doing math.
- 1.71** Angular gyrus—there's a math module in the brain, in the left hemisphere, mainly. We have them on both sides, but on the left side, in the temporal lobes, there's a region called the angular gyrus. It's actually unique to human beings. It's involved in calculation. Some of these math savants, these people that have extraordinary math abilities, might have a hypertrophied, enlarged angular gyrus. The angular gyrus on the right side is involved in the more abstract understanding of math, not merely computation and calculation. The right side of the brain, that computational style, its

way of being is more holistic, big picture, spatial. That's the right hemisphere. So, the angular gyrus is more the bigger concepts in math in the right hemisphere. It turns out Einstein had huge angular gyri. When they looked at his brain, he had huge angular gyri, which is quite interesting, isn't it? Who knew? He was an interesting character, this Einstein.

- 1.72** So, there's one example we'll get to—there's something called the 'rubber hand illusion.' Do you know about the rubber hand illusion? It's an interesting one. You have your hand; place it here, your left hand. Right, yea, do it, please do it, like in a Tony Robbins's course. Go ahead. Do it. Put your left hand here, and then your right hand, you put it underneath the table right here. Right hand underneath the table. And then Baland comes and strokes and taps your hand. You go stroke, stroke, tap, tap, tap, stroke, stroke, tap, tap, tap. But he doesn't do it on this left hand; instead, he does it on the table in front of you. So, he actually strokes and taps, taps, taps, strokes the table in front of you. Your visual access, Amari, your visual access is now on your left hand and the table. Your right hand, where is it? It's underneath the table. As I'm stroking, tap, tap, stroke, stroke, tap, tapping the table in front of you, I will also stroke and tap, tap, stroke, stroke, stroke your right hand underneath the table. But mind you, I will do it in a precise, synchronous manner, so the stroking and tapping on the table and the stroking and tapping on your right hand underneath the table is completely in sync; it's matched, completely matched. Stroke, stroke, tap, tap, tap, occurring on your hand underneath the table and on the table itself is precise in the way it's being done. Is that clear?
- 1.73** Guess what happens in about a few minutes of me doing this. You will feel touch sensations arising from the table as if it was your hand right there. You literally feel touch sensations arising from the hand as if it was yours. Now, your hand is the table. Your right hand is no longer underneath the table. You feel it's here; it's the table. If I, Baland, would take a hammer and smash the table, you would have your pain sensation, pain receptors, pain cells, neurons in your brain, in the somatosensory cortex, become active from me afflicting pain to a table. That's the power of neuroscience. You can go in a few minutes— I can transform Jose into a table just in a few minutes like that. That's pretty amazing.
- 1.74** So, I was like, "Wow, this is a very cool trick, the rubber hand illusion." Because, normally, what you would do is, actually, you would have the left hand here, you have a little barrier, a little wall, it would actually be here, the wall, and then you'd have a fake rubber hand right here, and then you'd put your right hand over here, not underneath the table, but right over here. And so, your vision would be on the rubber hand and you'd feel the sensations in the rubber hand. So, you would incorporate a rubber hand into your sense of self, your body image. Okay? Very good.
- 1.75** Then later, science has discovered that you can do it with a table or you can even do it with a—if you have a mannequin doll or something standing over here and you stroke

and tap the doll in front of you and somebody strokes and taps your head behind you, you will feel that it's your head over there. This might cure my migraines and stuff because your head is temporarily transported into the doll's head. It's interesting. But normally, you would just use the rubber hand.

- 1.76** I thought, "Wow, this is a cool trick." I was living in California back in the day when life was good. I was living in California, and then, one day, I was driving on the freeway, and I went, "My God, this rubber hands trick is so cool." What if you had somebody with OCD, obsessive-compulsive disorder, and you were to contaminate this rubber hand while doing the trick? Will they feel contaminated in the rubber hand? That was my big question. So, I had to try this experiment. We did this with a lot of volunteers. We brought them to the laboratory, and we did the stroke, stroke, tap, tap, tap, tap, stroke, stroke on the rubber hand. And then, Baland came in and put some disgusting thing on the rubber hand. Guess what happened. They felt contamination and disgust in the rubber hand. They felt disgusted in the rubber hand as if their hand was disgusted. It was interesting.
- 1.77** It shows you that modules that have to do with touch, vision, a sense of feedback from limbs, and disgust centers—there's a region called the insula—they all operate together. So, even though you have a module, there's a bigger picture. They can communicate at all times and create dynamic holes. And then, Baland can contaminate a rubber hand and make them feel like it's their hand that's being contaminated. We replicated this study in Boston, at the Harvard Medical School, on OCD patients. My God, you should have seen them. I was looking at some of these patients, and they were shaking and they were just all over the place because of us contaminating a rubber hand in front of them. Does that even make sense? Yet it happened.
- 1.78** Of course, I don't do this to—I don't want to torture these patients. I do this for a clinical reason. It turns out the way you treat OCD these days is that you have a patient, and you have him do something he finds completely disgusting and aversive, like touching. Some people have contamination aversion, some OCD patients. The way you treat it is that you have the guy touch something disgusting over and over, like the bottom of your shoes or even a toilet seat. They touch it for 30 minutes, and what happens is that the disgust and anxiety go up, but then, eventually, they habituate, and the disgust goes down.
- 1.79** Now, a major issue with this treatment is that about 25% of patients won't start the therapy because they fear starting. It's too aversive. If I can't shake hands with you, I'm not going to go touch this toilet seat. I'm not going to put my hands in the toilet bowl. I'm not doing that. A bunch of them drop out. Huge percentages drop out; they won't complete the therapy, about 20%. So, it's a big problem.
- 1.80** So, I thought, well, if I can trick the brain by creating an illusion like this, then I can say, "Look, Joe," who has OCD, "Look, Joe, I won't contaminate your real hand, but I'll contaminate this rubber hand," knowing how the brain works. In that way, I can create

a sense of disgust and anxiety, but in an indirect way where I won't contaminate his skin. That's how you can take information and knowledge about neuroscience and apply it in very direct clinical ways and potentially help millions of people in the world. This is what we did. Very cool. So, this is the rubber hand illusion.

- 1.81** I remember one time I was over in Boston with one of my friends, a professor over there, and we did it in the air. We said, "Let's just do it in the air," meaning we did the stroking, tap, tap, stroke, stroke, stroke, tap, tap on a hand underneath the table, and then we stroked and tapped the air. The guy felt like there was a ghost hand. Kind of spooky; there's a ghost hand.
- 1.82** So, your body image—this is called the body image; not the psychological body image, in the magazines and health magazines and all that, but the neurologically scaffolded, anchored body image. The sense of being anchored here occurs in the brain in a module. I keep using the word 'module' because it's a word scientists use. A module called the superior parietal lobule, for example, is a fancy name for some neurons over here that have your entire body image mapped onto the abstract feeling of Baland being in this body.
- 1.83** We'll get to body image in the next lecture. How body image comes about in this superior parietal lobule, and how this can go wrong in some patients. Thank you.
- 1.84** 10
- 1.85** Seek backward
- 1.86** Pause
- 1.87** 10
- 1.88** Seek forward
- 1.89** Pause
- 1.90** 0:30 / 55:40
- 1.91** Mute
- 1.92** Quality
- 1.93** Playback rate
- 1.94** Captions
- 1.95** Enter fullscreen mode
- 1.96** Intro to Neuroscience

Lecture 2: Brain and Self

At a glance — The brain and the self.

- 2.1 In the previous talk, we discussed body image, a sense of self. I feel anchored in this body. I close my eyes; I have this vivid sense of body image. I go from A to B in space. I don't feel I'm a part of Jose's body. I don't feel I occupy Brad Pitt's body, unfortunately. I feel anchored here in my body. So, this feeling is created in the brain. This is actually a brain construct; the brain facilitates this.
- 2.2 There's a region called the superior parietal lobule, which is up here, in the parietal lobe, the spatial navigation center of the brain, that's involved in that function. And, in fact, 'body image,' that phrase, was coined by two gentlemen: Henry Head and Lord Russell Brain. Those are real names. They coined the phrase body image to refer to this phenomenon of body image. So, the superior parietal lobule, a huge epic name.
- 2.3 Now, you have a stroke in that part of the brain. So, stroke. Damage. A blood vessel bursts or something like that. Stroke happens. Now you have, let's say, paralysis in the left hand. It doesn't work; it doesn't move anymore because, we said, the right hand, the right hemisphere, controls the left side; the left controls the right side. Now, at this point, some patients with this condition, with this stroke in the superior parietal lobule, when they go to the doctor, the neurologist will speak to them, talk to them, and will say, "You had a stroke." Explains you had a stroke. "Your hand—you have to go to rehab. You have to do certain exercises, you have to use it a lot to make the neurons become active again in that part or have other regions overtake that part of the brain to make you able to use the arm."
- 2.4 A subset of them will say, "Doctor, you don't understand. This arm doesn't belong to me. It belongs to you," pointing at the doctor. Or they might say, "Doctor, doctor, this arm doesn't belong to me. It belongs to my sister in Italy." Okay? So, for some reason, this person has been completely sane their entire life, and they're still sane. You sit with them, you play chess with them, okay? They go on Facebook and Instagram, they play Pokemon Go, or whatever. They're normal, but they have this one quirk. Whenever they refer to the left hand, they might say, "It now belongs to my sister in Italy."
- 2.5 This is strange when you think about it, but it tells you that the brain, in general, hates incongruencies. It hates them. It hates them. It dreads them. And so, for that reason, it actually will invent things to sort of fill in these holes. It will literally come up with all these kinds of explanations, these illogical, absurd, idiotic explanations, to sort of account for the fact that the left hand is paralyzed. That's interesting, that superior parietal lobule.
- 2.6 Now, the right hemisphere is very much the sort of the body image hemisphere. It's much more involved in the body image, body type stuff, meaning, again, there is that kind of preferential processing style where the right will not only do spatial navigation

and overall big picture, but also body image stuff, sense of self stuff. Okay? Is that clear? Very good.

2.7 Then, there's another group of patients. These patients are very interesting. Ever since they could remember, they had this desire, this very, very strong feeling that their arm—that they just want to amputate their arm, which is healthy. They will go to the doctor and say, "Doctor, I want it amputated, and I want it to be amputated right here." So, they draw an exact line. Now, weeks later, you bring them in and you say, "Show me the line, Joe." And Joe will draw it in the exact same place. Okay?

2.8 For long, they didn't understand. How come this person who's healthy, who's fine, who doesn't have any obvious schizophrenic logical problem, he's not illogical, he's not obviously schizophrenic or delusional or psychotic, how come he wants his arm amputated, for God's sake? It doesn't make sense; it makes no sense whatsoever.

2.9 So, first, they thought, "Okay, these guys must be crazy. This is the only explanation. They must be insane." This is the knee-jerk reaction of a lot of clinicians. They say, "These guys are crazy," but that's usually because they're not smart enough to figure out what's going on in the brain; what's actually happening.

2.10 So, based on the information that you have now, we know that there's a part of the brain that has to do with feeling a sense of a body. Feeling it, okay? That's the postcentral gyrus in the parietal lobe. As we said, there's a map of the entire body. So, when I touch a leg, when I touch a Baland's knee, there's a region up there mapped like a little human, homunculus it's called. It activates up there. If you record the neurons in that region, they'll become active when I do this. Okay.

2.11 Behind it, there's a superior parietal lobule that's involved in the more abstract sense of Baland, the more abstract feeling, like, I feel anchored in this body. So, the superior parietal lobule (SPL) is an abstract sense of self. The postcentral gyrus, a strip in front of it, is involved in the actual sensory feeling.

2.12 Now, what do you think? What do you think might be going on? If you are the scientist studying these patients, and trying to understand, where would you look? Perhaps you might look in the postcentral gyrus to see if that arm is maybe missing from their body image as the first step. Does that make sense? Would you do that? Well, my colleagues did that. They actually looked in the brain of the somatosensory area: S1, S2, S3. These regions have fancy names; I'm just trying to impress you with these S's. So, they looked there, the brain scans, and they touched the arm, and it was normal, normal activity, nothing going on, nothing interesting. Completely normal. So, my God, they're completely fine.

2.13 But then, it occurred to one of the scientists to actually look at the superior parietal lobule with a more vivid, abstract sense of where self is, and they looked over there, and lo and behold, the arm was missing in that region. So, the arm representation was missing in that region. Interesting.

- 2.14** So, the somatosensory cortex over there, the superior parietal lobule, works in congruence; they work together to create a sense of the whole self. There are a lot of these wires, these axons, going back and forth over here, and that region is missing. So, they feel like, "The arm doesn't belong to me." In fact, these patients—the condition is called apotemnophilia or xenomelia—will actually say, "It feels too much like a part of me. It's intrusive; it's too much. It's kind of in the way."
- 2.15** Now, a lot of these patients will actually have it amputated. They will go ahead with the amputation. In fact, it turns out people that kill themselves, and I'm going completely off topic, but people that kill themselves, when they jump off these bridges and kill themselves, a subset doesn't die. A subset survives. When they jump, they get paralysis, hemiplegia, or whatever. They can't move, but they survive. And when you ask them at that point, "What were you thinking when you were jumping?" After jumping, a lot of them will say, "I regretted it. At that moment, I had so much regret that I just felt like I made the worst decision of my life."
- 2.16** That's pretty interesting from a psychological perspective. Before they jump, they just want to end their life, but when they jump, during the jump, they regret it completely. Now, that logic, let's apply it here. Might it work? It kind of makes sense that it might be the same thing; they regret it afterward. But it turns out they're happy, they're delighted, even though that doesn't make sense, but for the rest of their life, they go, "This was great. I got rid of the hand." So, apotemnophilia. A very strange condition, very bizarre.
- 2.17** But they tend to have some other quirks, too, which I personally find very interesting. They tend to have this weird sexual attraction to the people who are missing that arm that they want to have amputated. So, we have an innate attraction to the human form and not, say, a chair or a camera or a sheet, at least not most of us. We tend to have—that was a joke. When I tell jokes, by the way, you have to laugh; otherwise, they'll get worse. Okay? So, you go from this phenomenon in neurology, and then you go all the way to attraction and how attraction comes about, potentially, in the brain. Why humans are attracted to the human shape. These structures might explain that.
- 2.18** You have these conditions—they show us a lot about body image and how the self is constructed. It shows us that the self is fluid in many ways, the rubber hand showed us that we may feel one with a table, or we may have our body image centers disturbed, so we feel our arms are floating somewhere in Italy, right? Or you may want to cut it off because some of these centers are not working, so you want to get rid of it. So, the human sense, the sense of self, is very malleable; it is fluid, and it can change in many ways.
- 2.19** We wanted to study this. We wanted to understand, "What are the limits of the body image? What are the limits of self-other distinction?" So, it turns out there's a region in the brain called the temporal parietal junction. I know I keep throwing names at you, but hopefully, you'll get some of these—TPJ. It's a region, again, in the parietal lobes,

the regions that have to do with body image, the sense of self, the sense of space. There's a region over there called the TPJ, the temporal parietal junction.

- 2.20** It's a very interesting structure. You know why? It is strategically located to integrate various sensory modalities. It's right there at the cross-section between vision, touch, hearing, and so forth—a sense of balance. So, there's a balance center in the brain. So, it gets information from various sensory modalities and puts it together to help you create a sense of self, and it works very closely with the superior parietal lobule. In fact, it's very close to that region. So, there is that.
- 2.21** If the self-other distinction, the distinction I have between me and Jose—I know I'm not floating into you, hopefully, right? I know I feel here, crisply here. So, there's a barrier. It's a temporoparietal junction. If I have a stroke over there, damage, there's a sense in which I actually feel like I'm floating into you. There's a sense of that temporoparietal junction. The reason—so, the self-other distinction, if you have a stroke, it can go awry. So, you actually feel like you're floating into people. The frontal lobes are very much the place—the prefrontal lobes are very much the place where a lot of this inhibition occurs, the brakes on the brain that keep things in check, the prefrontal.
- 2.22** So, let me give you a good example. In your dreams, when you dream, do you sometimes feel yourself occupying somebody else's body or that you're looking at yourself from the outside, or is it this kind of vantage point? I'm looking at myself here. The answer can vary. Sometimes, you see yourself from this perspective, like this. I see myself like this. Other times, I see myself from the outside. Sometimes, I might occupy somebody else's body during dreams. So, I feel like I'm occupying Napoleon's body, Gaddafi's body, or some person's body. So, I may have that.
- 2.23** And the reason for that is, guess what: the TPJ deactivates during dreams. It shuts down during REM dreams, and we'll get a bit to dreams later on. During REM dreams, the TPJ shuts down; it becomes less active. So, the crisp barrier between me and you breaks down, and I may feel like I'm occupying your body or a bird's body or something like that.
- 2.24** The frontal lobes, the prefrontals, during the dreaming state, the REM stage of sleep, also turn down. You see, the brakes on the brain are not there, or at least they're not working as nicely as we want them to. And so, you feel like you don't have that ability to control your body image, so it's anchored firmly inside of you. Does that make sense? Very good. TPJ, superior parietal lobule, postcentral gyrus—a lot of fancy names, but now we have all the interesting functions, right? Very good.
- 2.25** Now, we did some studies, and I went off a tangent here, and sometimes I may do that. You've sensed that, perhaps. But we wanted to see if you have somebody with OCD symptoms—I like OCD; it's a very interesting condition, very fascinating—you bring them in, and you sit them in a chair.

- 2.26** You have this patient, Joe. Joe is sitting there; Baland is sitting here, and Baland is the experimenter. Baland has this disgusting item in front of them, let's say, fake feces. Excuse me, but fake feces, you have it here, and the person is over there looking at me, and I touch it. The question is, "Will he feel disgusted in his hands over there?" In other words, can we override logical inferences that I know that my body's over here, and you know your body's over there? Can we override that logical, cognitive, impenetrable barrier? Is that possible? That's the first question.
- 2.27** We did it. Touch, touch, touch, touch, touch, touch. Over there. Lo and behold, the guy actually feels contaminated in his hands, disgusted in his hands. In patients with OCD—we did the study—they have an urge to go wash their hands. Like, "My God, I want to wash them. I know it doesn't make sense, this thing, but I'm watching you touch that. But I feel like I want to go wash my hands." Interestingly, we actually found that disgust levels were nearly or even as much active when you were just watching somebody else. Very, very interesting. Surprised?
- 2.28** But then, we said, "Let's do something else." Now, patient Joe touched the contaminant; touched the fake feces over here. Touched it. So, Joe touched it. Touch, touch, touch, touch, touch. Now he's disgusted, he's anxious. Not good. Then, Baland over here washes his hands. Guess what happens? He feels disgusted, anxious. Then, he watches me wash my hands. You know what he said? The patient said, "Baland, wash more between your fingers over here. Can you scrub a bit more over here?" He was telling me how to wash my bloody hands now. So, he was that affected by me washing my hands that he felt relief in his hands. That was quite interesting, wasn't it?
- 2.29** Vicariously, he was feeling my emotions and sensations as if it was his emotions and sensations. Very interesting. And a lot of these regions involved in this— one region is called the insula. It's the disgust part of the brain. The disgust part of the brain is the insula. It's a very nice structure; it's in the temporal lobes; it's tucked in there, behind them. And the insular cortex would be on the outside, of course, on the cortical side of the cortex.
- 2.30** Now, this part has a map of all your organs and your internal states. It has an entire map. So, here's a map not only for the body and the brain but also for your organs in the insula. And a lot of the receptors, the pain receptors, the touch and so forth, temperature, it's also in the insula; it's also in that part of the brain. So, one would predict from this experiment that when they touched that contaminant, the insula would activate when the touching is occurring, and they feel disgusted, but the insula becomes less active if you look at brain scans and scan their brains while doing this activity. Does this make sense also?
- 2.31** So, the insula is the disgust part of the brain, but also temperature and pain. It has a lot of wires to the limbic center, a lot of wires, a lot of communication going on, so it can tell your emotional part of the brain, the limbic regions, "I'm feeling disgusted." Now, you should have a sense of a negative feeling, an aversive feeling. Or, "I'm feeling pain

right now." The insula would monitor that with the anterior cingulate, another region there; it would monitor the pain.

- 2.32** So first, it goes to the somatosensory cortex, the physical part of the brain, that region. Okay, I'm not going too fast here, but it's going to the somatosensory region, going to the insula, and the insula would communicate to the emotional part of the brain. Now, you should have an emotional response to that pain, not merely a physical response, but there is a lot of emotionality and psychological factors to pain, which I mentioned earlier, and we'll get into it in depth. There are a lot of psychological components to pain. Very good.
- 2.33** So, we did this experiment. I talked about brain imaging, that you can image the brain using, obviously—and this would be a very good experiment to do. And I should mention some of the techniques used to study the brain.
- 2.34** So, OCD. On the topic of OCD— I find it so fascinating because it explains the general human brain. Just like sleep paralysis, it's like a window to understanding the overall human brain, like disgust and a lot of factors, like why we are so vigilant at times and so on. Now, in OCD, you have a region called the orbitofrontal cortex, and the orbitofrontal cortex helps with filtering sensory information in the environment. Is this dangerous? Should I be cautious? Should I stay vigilant, or can I tone it down? What should happen now?
- 2.35** The orbitofrontal cortex is kind of hyperactive, as you might guess in OCD, meaning the neurons over there are like a pinball machine, firing everywhere, and they communicate—the structure called the cingulate cortex of the anterior cingulate, a vigilant structure, part of the brain that has to do with vigilance, but also some kinds of emotions. They communicate with the movement parts of the brain we talked about earlier, the basal ganglia, particularly the caudate stratum, these regions. Because in OCD, you have a lot of automatic movements. Like, "I'm washing my hands. I know it doesn't make sense, but I'm washing them for hours and hours and hours." It's another example of how that automatic movement part of the brain goes awry. You have too much movement, automatic movement like that. Orbitofrontal cortex.
- 2.36** There was this guy, this college student in the US. He had severe OCD. Very debilitating, very severe. So, one day, he took a gun and said, "I'm gonna end my life right now. I don't want to live." He took his gun and shot through his head, right here, to kill himself. Guess what happened? He shot through his orbitofrontal cortex and survived. He did a brain surgery on himself without knowing. He created a lesion in the brain, and his OCD was cured. Can you believe this? It's interesting, right?
- 2.37** So, this shows us that brain lesions are often used to destroy pieces of brain tissue to get some of the functions that the hyperactive region is creating, so to speak. Is that clear? It's interesting, isn't it? So, if you have OCD, you actually do ablate, destroy some of these regions, and you get a positive result at times. But, of course, it's extremely

invasive doing brain surgery and getting all inside, because we said the basal ganglion orbital—you have to get a bit inside. It's quite aversive, but in severe patients, it's done, for example.

- 2.38** But can you do the opposite, then? Could you activate neurons over there? The neurons are too silent; they're like sleeping or dead, even. Could you maybe implant an electrode there and then make the neurons flare up again and become active? Is that possible? You do that sometimes, the implants. You might have a little electrode put in the brain, and it revives the neurons.
- 2.39** So, what condition have you looked at in this course—it's like an exam here—which condition have we looked at in this course that might need some brain revival in a certain structure that has to do with movement? Parkinson's. Right, Parkinson's. Because the cells in the substantia nigra are not working as they should. They're dead over there and cannot invigorate the basal ganglia, so you get stiff movements. So, you plant this electrode inside the basal ganglia, in the striatum, in the substantia nigra, possibly in the basal ganglia, if you want that to be active. I'm not a brain surgeon, but you could probably do both. And then, you see the dopaminergic neurons become active. Is that clear?
- 2.40** Two opposite ways of doing it. One: lesion. The other what? An electrode. You need batteries and stuff for it, probably. Maybe they have newer models that are less heavy, but yeah, these are two methods.
- 2.41** On the whole, dopamine. You might have a question. You might say, "Okay, well, if dopamine is missing in things like Parkinson's, can't you just give them a dopamine pill, for God's sake, and be over with it?" What do you think? Perhaps, right? Give them the dopamine pill, and it's going to, maybe, give them that dopamine they're missing. There are two objecting reasons.
- 2.42** First, you can't just—dopamine doesn't cross the blood-brain barrier. The blood-brain barrier is that the blood vessels in the brain have a very thick and dense wall, so a lot of molecules and invasive stuff won't actually go into the brain and destroy it. So, it's like the brain went, "Look, I want to make a compromise here. Having dense blood vessels is a hard thing; it's very difficult to have; it's a complex thing, but I'm going to have this structure because it's very costly for me to get brain damage by swallowing some molecule, like a large protein. So, I have these very dense walls in the blood vessels called the blood-brain barrier."
- 2.43** The brain, you got to protect it. If you have, let's say, a Rolex watch at home. I don't, but let's say you have a Rolex watch. And that Rolex watch, you want to put it in a safe. So, the brain also. You have the blood-brain barrier to control it from having all these nasty molecules contaminate the brain. By the way, you also have three meninges that help you; the membranes protect the brain should it get, like, some damage.

- 2.44** Now, it turns out dopamine will not cross the blood-brain barrier. So, you take it, but the brain rejects it; it doesn't get into the brain. So, what do you do instead? You take the precursor of dopamine, L-Dopa. You take that, and that will cross and, eventually, synthesize into dopamine. So, you could do that. You could take dopamine. Yes.
- 2.45** But the problem with a lot of the pharmaceuticals, the drugs—guess what the problem is. When you take a drug like SSRI or serotonin or dopamine, you kind of put it all over the brain. It's like, you put it, you spill it all over. You can't really target that region. That's why these days, we're working on more targeted therapies where you can get to particular structures. There are certain things coming, but it's a very early stage.
- 2.46** You can manipulate the cortex in the outer region by a stimulation called transcranial magnetic stimulation, and I'll get to that later on. That's very interesting. You zap the brain with an electrical current over here, and then it's high frequencies, and then, the brain becomes more active, the cells, and produces the right kinds of chemicals. But that's right, you can't. You get a lot of symptoms from just flooding your system with something like dopamine, but it works—L-Dopa, sorry, L-Dopa that will turn into dopamine—but it's still the main treatment for Parkinson's.
- 2.47** Unit recordings are very interesting. Some of the very early studies on animals, for example, where they discovered a lot of things, were recording individual neurons and their activity. So, that was very, very cool because now, if you had a chimp—the ethical guidelines, maybe unfortunately, but it is what it is, are less strict when it comes to animals. So, they open the skull of an animal, and they will measure individual neurons. They would find things like a map, a lot of things on the body based on which neurons would become active. You touch here, here, single neurons. It's called a unit recording or multiple—multiple, what is it called?—a multiple unit recording where you record thousands of neurons. Another option that takes less time, obviously. The other one is hours and hours to map if you want to map regions. So, there are these methods.
- 2.48** There's the EEG, of course. You may have heard of EEG; I forgot the whole name. Is it here? Electroencephalographic recording. I always miss this one. EEG. This is a standard one; it's used so much. You have this net-like thing on the brain, then you put the gel on the various spots, and then you have two regions, and you measure the electrical potential—electric potential—between two regions to know which regions are active and to what extent. So, it measures various brain waves: alpha, delta, theta. You don't have to know these names, but it gives you a peak inside the brain, but only the cortex, the outer layer. And it's very noisy as well, so it doesn't give you much information about various regions, but it's a good tool to have. So, that's used.
- 2.49** And then, you have the CT scan. That's another thing you can use, but the CT scan is more like an x-ray. So, it gives you images of the brain, exactly eight or nine images, put them together, and that's it.

- 2.50** Of course, the MRI scan. If you don't know this, the MRI, the magnetic resonance imaging, measures the waves that hydrogen atoms emit. It gives you a pretty picture of how the brain looks.
- 2.51** The PET scan I actually find interesting. You can look at receptor activity, the activity of individual receptors. And now you say, "What are receptors? We didn't get into that Baland!" But it turns out neurotransmitters, when they communicate, can also activate different receptors. Receptors are like parking spots on the dendritic spine, on the dendrite on the other side, or like doorway and keyholes, where you talk to the cell. You inject a glucose-like substance into the body. The brain likes it. It thinks it's glucose, but it can't really metabolize it. So, when you scan the brain, you'll be able to see the activity of that particular region. When you do something like reading or whatever activity you're doing looking at images, you can see that now because it sort of stains the cell. So, that's a PET scan.
- 2.52** And then, probably the most interesting, not interesting, I don't find them particularly —the functional MRI is where you measure the flow of oxygen and blood, activating various areas, increasing them when you do a certain activity. This is the most popular one in neuroscience. When you do a lot of studies, when, "Oh, look at this pretty picture," or you see this apple, you see a picture of your romantic partner, and this part of the region is active. So that's a lot of the fMRI studies (functional MRI), where you see a lot of functions mapped in the brain. They are important. You want to know what parts of the brain are lighting up. So, very good.
- 2.53** So, these are the various regions and the various methods, excuse me, to study the brain.
- 2.54** Studying sleep paralysis got me into things like vision. It got me interested in vision. Why do we hallucinate? What's going on in the brain? How does seeing come about? And it's one of the very interesting areas, I think, of brain science. So you, of course, have the visual cortex over here. The visual cortex is large. The occipital lobes are huge, huge compared to the other lobes in the brain. The occipital lobes are very big, and that's because our survival depends to a large degree on vision, on being able to see.
- 2.55** And so, when you see something initially, first of all, a question always comes up. It's called the homunculus fallacy, this idea that if you see things in the back of your brain, then surely there must be a little man over there watching, like a screen or something like that. And then, inside that guy, there must be another guy watching the screen inside his brain. Infinity occurs, right? But, of course, it's a fallacy. There's no little man in the brain, and when we look at the world, what we see is actually not the world but symbolic descriptions. There are neurons talking and making a representation of what's out there in the world.

- 2.56** So, you get light through the retina. It's the retina, the back of your eye, it travels through the optical nerve, you have the lateral geniculate nucleus, and you have the thalamus, of course. The thalamus is a stop station, right? Going to the V1, the visual cortex, called V1, where you make sense of simple objects, simple—not even objects, but lines and dots and simple, very, very simple percepts of vision. All right, visual cortex. Now, this is the main pathway. This is the main visual pathway, but there are a few freeways, high-speed highways.
- 2.57** One is the one for sleep. We talked about this. I mentioned it. It goes from seeing things, from vision, from the retina, but then it bypasses a lot of these regions and doesn't go to the visual recognition area of the brain. It goes to the hypothalamus, the sleep area of the brain, which we talked about, or the hormone factory of the brain, the hypothalamus. So, it goes directly there to the suprachiasmatic nucleus of the hypothalamus. This is a fast pathway.
- 2.58** Then, there's another pathway, and this is interesting. This is the ancient pathway of object recognition. It goes from the retina, the back of your eye, sits in the orbit, the round thing. It goes in the retina, and then it goes to the superior colliculus pulvinar and the parietal lobes. I'm just throwing fancy names at you, but it's going to a lower part of the brain and then going to the parietal lobes. It bypasses, again, the visual cortex, the vision area of the brain; it bypasses all that. It's very quick. Let me illustrate this a bit better.
- 2.59** There was this Oxford neuroscientist. He had this patient with him. He was studying this patient. He was completely blind; he couldn't see. He couldn't see nothing. Completely blind. This patient was sitting with the scientist, and the scientist said, "Well, Joe," I'll use a different name now. "Bob, show me where this object is." And then, he'd flash the object, and Bob would go, "Here," and he would say, "My God, you're right." Might the ancient pathway in the brain be preserved even though his visual cortex is damaged? He might be blind in his visual cortex, so he cannot see anything, but this ancient pathway, which bypasses—it's the ancient fast pathway that goes through the spatial navigation centers in the brain, the parietal lobes, in a quicker manner—might that be preserved, that pathway, but he's still blind? That's exactly what's going on. It has the oxymoron name called blindsight, this condition. Blindsight. So, these are the two super ancient fast highways.
- 2.60** So, vision comes to the V1, over here, to the occipital lobes. Early processing occurs. It's like Mount Everest, like climbing Mount Everest, with each step in the visual hierarchy, with each step you go up, climb the ladder, it becomes more and more sophisticated. When you first send information there, it's sent further on, for example, to the regions in the parietal lobes. This is called the how pathway. There's a region that's called the MT region. MT is important for detecting movement. So, the brain is here. Vision comes in, it goes through the thalamus, it goes to the back of the brain, and from the back of the brain, it splits. It goes to the MT, it goes to the parietal lobes, and you make

a sense of where you are in space. Now, this is, of course, different from the ancient pathways, but it's still unconscious. When I walk around in space, I don't bump into things. I don't think about it really. It's just me seeing. That's that parietal lobe.

- 2.61** Other information is sent the other way through the temporal lobes. That's the side of the brain. It goes to the temporal lobes. Parietal, up here. It goes to the temporal, and in the temporal, it goes to something called the fusiform gyrus. Gyrus is a little fatty thing we talked about. It goes there to one of the gyri. It's called the fusiform gyrus. The fusiform gyrus is interesting.
- 2.62** Now, we get into more sophistication in the network. Shapes of things. Shapes of letters are there. Dry discriminations of faces: I can discriminate Jose's face from Michael's face. Dry discrimination, nothing emotional, no meaning, or anything like that. That's later. But I do a dry classification of objects: car A from car B, Jose from Michael, Alejandro from Joe, and so forth. That occurs in the fusiform gyrus.
- 2.63** Facial recognition is very much in that part of the brain, and if you have a stroke over there, guess what happens? You no longer can recognize faces, but dry classification of objects is not enough. We need to know what these objects mean to us. Have you seen them before? And so forth.
- 2.64** So, information goes to the Wernicke area, for example. Do you remember Wernicke from before when we talked about language? You have a stroke in the Wernicke area; your semantics are out of sync. You don't know how to understand language. Well, the Wernicke is also very much involved in semantics and understanding the world around you. A greater meaning. Now, at this point, it's very important also to have an emotional component.
- 2.65** After all, the purpose of us living is to stay alive so we can send our genes to the next generation, so we can procreate and all that. So, you want to know, "Is this a friend or a foe? Amari, is it a friend or a foe?" So, I look at your face, I look at you, and I bring in the limbic structures, the amygdala, the limbic emotional centers. I communicate with that, that's next in the network, and I say, "My God, this is dangerous!" This is dangerous now, so it becomes active, and if it's very, very active, it alerts the hypothalamus, the hormone part of the brain, and the flight and fight reaction part of the brain. So, it sends out the right hormones to have a reaction: fight or flight, freeze if you need to, whatever. So there's this hierarchy.
- 2.66** You bring in the hippocampus. We haven't talked about that, but we'll get into the hippocampus, trust me. The memory part of the brain. So, you bring in old memories. What can I do? What does this mean to me? Not only that, but what is there of memories that I can use right now to make me survive this encounter? Does that even make sense now? I'm supposed to see an object, recognize it, give it depth, give it color, give it meaning, give it memory, give it emotion, and then get the hell out of

here. That's the right order, right? That's what's supposed to happen. Yet, there are more connections going in the other way. This way.

- 2.67** Well, this might mean that seeing is more conceptual than we think. Then perhaps, when I see something, it may go the other way around. Well, I see something, but then information is quickly sent to the meaning part of the brain, and that meaning part of the brain goes back and tells me, "Well, you're supposed to see this. This is how these things looked in the past. I've seen people that look like this in the past."
- 2.68** Vision is very efficient, and it has to be very quick and precise at all points to facilitate your survival. If it's slow, if it's just going from one step to the other, it wouldn't work. You would die. Exactly. So, it has to be efficient, even though it's kind of lazy also. It takes shortcuts whenever it can. So, it always is in this balance of, "What do I think is out there, and what am I actually sensing is out there?" and then, at each stage, it's iterating the information and double-checking, "Oh, this is probably this." and then you go, "Oh, it looks like this. This is what the sensory information is telling me, and I know from conceptual history that this might look like this." And then, it's matching, matching, matching at each step, matching, matching to find the best solution and the quickest solution. Does that make sense? Am I being clear?
- 2.69** So, as much top-down, it's called top-down, penetration goes up from the higher regions to the simple regions, as there are information wires traveling from lower to up. So, this is a lot of what vision is about, and you won't really appreciate this until you meet a patient with some damage to his optic nerve, maybe a scotoma, a blind spot in his visual cortex. So, the sensory information that is coming through his eyes to his visual cortex is messed up. Or Charles Bonnet syndrome, patients with eyes that are just degenerating. Sometimes in older people, you find this. And so, the sensory information coming through their eyes is messed up. So, they can't really rely on it very much. So, they tend to over-rely on the conceptual stuff, on the higher regions from the memory regions, the meaning regions. There's over-reliance.
- 2.70** In fact, what happens is a lot of these guys will tell you, "Well, I'm sitting with you," and then they might say, "There's a monkey in your lab, doctor, or am I hallucinating cartoon characters? They're right there." It's because their brain is filling in too much. It doesn't know there's this gap; there's this hole in the whole visual process, and my sensory information is lacking. And we said earlier on, the brain hates what? Incongruencies.
- 2.71** The visual process is an extremely active process. Have you seen some of these bistable illusions, where you see, for example, you see—there's a picture of an old lady, like an old hag, like a witch, but then you look at it again, and then, oh, my God, it's actually a young girl. Have you seen those? And then you look at it again, and it flips on you. Oh, it's actually the old lady. You can't have both at the same time, and the reason here is that vision is very much an active process. This is one very, very, I think, elegant

illustration of that point. The fact that you're constantly creating the world, you're constantly imposing what you think is out there.

- 2.72** Now, there's a final pathway you must know. It's a very simple pathway, but it goes from a center that is involved in—well, it goes from the visual cortex. So, information comes in through the eyes, optic nerve, boom, it travels to the visual cortex, it goes to the face area or the recognition area, and then, before spreading and having that diffused effect on other regions, it goes directly to the amygdala, the emotional part of the brain, limbic. Why might this be important, this super high fast freeway pathway? Survival, emotions.
- 2.73** In certain situations, literally, you can project things on a screen and you look at it, and you will know that it's dangerous before you actually recognize the object. That is possible. You can do reaction times, and they'll look at things fast, and you can measure their sweating, their autonomic reactions. They will start sweating, but they don't know why they haven't recognized that. Is that fascinating?
- 2.74** Well, we can talk a little bit about the vestibular sense, the sense of balance, because this is a very, very interesting region of the brain. The sense of balance is a curious one. So, you know that we have five senses, as you guys know, but there's this sense called the sense of balance, the vestibular sense. And I thought, "What sense of balance? That's so boring. Like, what the heck? Have a sense of balance? Balance—do we have a vestibular center? Well, that sounds kind of boring."
- 2.75** It's inside your ears; it has this fluid with some hairs that move in these three semicircular canals. When you move, either when you move yourself back and forth or when you detect movement in 3D on the vertical plane, this thing will monitor that, and then, you will feel like a sense of balance is being registered. Then, you send information to the vestibular nucleus, and that has a lot of pathways going to things like vision and the motor system. All fancy stuff. Vestibular system.
- 2.76** It's actually quite important because if you have destruction to your vestibular system, first of all, your sense of balance goes completely off. You will feel like the whole world is moving. So, this is an interesting one. Also, to create a sense of body image. A lot of the studies we did on sleep paralysis and the research and what we came to conclude was that when you are sleeping and paralyzed in sleep paralysis, as you are lying there, what is happening, in fact, is that your motor cortex is sending signals to your body to move, to get out of there. Motor cortex up here.
- 2.77** It has the primary area of the motor cortex, and then it has the premotor area, or the area that's called sometimes the supplementary motor area that's involved in planning the movement before it occurs. Now, during sleep paralysis, for example, as you are preparing to—you want to get out of there. That's the first thing, right? So you're lying there, you're paralyzed in your sleep, you're sleeping, but you want to get out of there. You can sense a demon in your room. You send commands to your body to move.

"Move, move!" Is there feedback coming back from the paralyzed body? No. This feedback, by the way, is called proprioception. The feedback that comes back from the body. It's called proprioception. It goes to the parietal lobes and the cerebellum partly, and it helps you create a sense of self depending on the movement. So that's that part.

- 2.78** Now, because that center is off and because you're not getting that vestibular sense, and the vestibular neurons cannot do their job, that is the reason why the sense of self, the body image, becomes disturbed. And as it becomes disturbed, in this case, you will have a lot of bodily hallucinations. So, in order to anchor the body and feel completely normal in your movement, you need the vestibular sense system and the information you get from that. The body's movement, at any time, sends information to the vestibular system, allowing you to feel anchored in your body.
- 2.79** We've covered a bit of ground here, so I think you're ready to get the grand finale. You know, "What the hell is that ghost all about, then?" Okay, we have the vestibular system that helps us with a sense of balance and a sense of body image. The command is going, no feedback is coming back from the body, so you go—my body image becomes disturbed. I have bodily hallucinations. Why do I have bodily hallucinations? Because the brain hates incongruencies. So, it fills in the blanks for you. We talked about it. If I can't create my body based on the information coming back from my senses, well, the heck with it. I'm just going to create my own body, and that will be Baland's legs flying up in the air.
- 2.80** Occasionally, I might have an out-of-body experience because my sense of a body, just as my legs can fly in the air and my arms can fly in the air, well, the whole thing, the whole bloody thing, can also be projected out there, and I can have an out-of-body experience. But, of course, out-of-body experience is just one manifestation of what's going on because of all the fear centers in the brain being active: the panic centers of the brain, the limbic—hyperactive.
- 2.81** The prefrontal lobes, by the way, are supposed to inhibit this activity. Inhibit, inhibit. Stop, stop. Don't project any ghosts. See the world as it is. Inhibit, inhibit. Prefrontal lobes during dreams, during REM sleep, are dialed down. It's down. During sleep paralysis, a dream with your eyes open, it's down. So, when you see that shadow-like being, which is a shadow of yourself, we've argued, it's easy to activate a lot of these centers in the brain where you have to do with color, you have to do with memories of Freddy Krueger, and what people are telling you. You project that into the frame, and you see that. You see that over there.
- 2.82** Now, interestingly, a lot of the cases, 50, 60, 80% sometimes, will tell you, "Oh, Baland, but I only saw the shape of a ghost. That was just a shape; it was just a frame." I said, "Yes, of course, because your brain didn't activate the whole conceptual area. It just went with some of the earlier regions in the brain." It didn't go all the way to the temporal regions and the meaning and filling in, and 3D. It didn't bother; it said, "Look, I'm seeing—I'm sensing a ghost. I'm seeing—I let me just create a frame for it, and then

I can use my limited processing power on getting the hell out of here or thinking about navigating this room."

- 2.83** Where is the window? Where is the door? Vision and computing power at this moment is very costly. It's very precious. My attention is extremely scarce at this moment, so might I want to use that for a better purpose than hallucinating the whole thing with the hat and the bloody thing, Freddy Krueger or whatever. That doesn't make sense at that point. Let me just use my limited attentional abilities for something more meaningful, like getting out of here. And they sometimes say, "Yeah, I knew where the room window was, and I knew where the door was." Well, yeah, you want to survive. It makes a lot of sense.
- 2.84** So, here we go from the nitty gritty of neuroscience all the way to why we see ghosts, all in one preparation. We'll get to attention a bit. Attention is an interesting one because it kind of aligns with these systems. You have visual attention, auditory attention, but attention is also kind of a part of this. And you're in a situation where you see a ghost. Obviously, your attention is completely hijacked by that. But I'll mention briefly with attention.
- 2.85** There are two forms of attentional processes, two forms of attention: endogenous or exogenous. Endogenous attention is the kind of attention that comes from the inside, and there's exogenous attention, which comes from the outside, also sometimes called top-down or bottom-up.
- 2.86** A bottom-up is like this glass over here falls bottom up. It goes from the bottom of the circuits, and let's say the visual attention, bottom, and then up. If I'm looking for these notes, attention goes top-down. Top-down, bottom-up. Somebody throws a spear at me = exogenous.
- 2.87** So, attentional circuits. Why does attention—what is the purpose of attention? Why is it so important? I think this ghost illustration shows us that attention is so crucial for your survival. So, having those circuits active by having an attentional system is quite important. But I'll get to attention; I have a couple of more points on attention because there are some interesting points on attention.
- 2.88** There's something called the cocktail party effect, which is quite, very interesting. You can sit in a room, you're sitting there, and you're having a conversation with this chap, and you just feel like, "Oh, my God, this is a very good conversation," and then somebody says your name over there, and your attention goes to it immediately, even though you were just focusing on this guy. It shows you that the brain works in layers. I was actually getting all that information at the same time taking it in, but my brain didn't consciously know. So, I find that very interesting, that your brain can inhibit information even though it's actually taking that information in.
- 2.89** Thank you.

Lecture 3: Principles of Neuroplasticity

At a glance — Principles of neuroplasticity.

- 3.1 I think I might want to start off with a little bit of review. I think cementing some of these concepts in your brain would be very good, so you have them as concepts that are very clear and crystal in your brains or in minds, whatever.
- 3.2 So, we talked about the cortex, frontal lobes, the four lobes.
- 3.3 So, the prefrontal, the part of the brain that's very well developed in us humans, and has a lot of these bricks on it, on the emotional parts of the brain. It's good for inhibiting your emotions and impulses. It gives you that higher feeling of being human when it interacts with your emotions in the sense of self in the parietal region, letting you know that you are Amati and you have a sense of identity and a sense of self that's uniquely you. It's your identity. So, the prefrontal lobes have that function.
- 3.4 Sense of logic, sense of agency. I feel I'm here at the Peterson Academy. I'm anchored in this body, there's a room, here's some whiteness; I'm here. And that feeling also comes about in the prefrontal lobes, in particular, the dorsolateral prefrontal—I keep throwing out this name, but dorsolateral prefrontal cortex, or simply DLPFC, are regions on the sides of the prefrontal that are very, very important for the emotional components. So, they're very much important for the emotional components.
- 3.5 Of course, emotions are in the limbic, in the limbic part of the brain, and for a long time, they said, "These two shall never meet. The twins shall never meet; they are two separate systems." Emotions and intellect/thought. Well, it turns out that, in fact, a lot of emotions are regulated from the prefrontal regions that speak to the emotional parts of the brain. So, it has that component. It can dampen emotion. It can fire up emotions and make you more excited. So, people with depression, for example, in this part of the region of the brain, there's not a lot of activity. It's dead over there. It's slow; the neurons are not that active. So, you have that. This is the prefrontal cortex.
- 3.6 The motor system is in the prefrontal cortex. We said there is a supplementary motor area that helps you. When you make a movement, for example, you plan, "I want to move my hand from A to B." So, there's a first, a planning, saying, "Okay, go ahead and execute that." And then, there's an actual signal from the motor cortex, which sits in front of it. This is also a map of the human body, of the arm area, saying, "Now move!" and as you move that arm, there's feedback coming back from the moving body; we call that proprioception. From skin receptors, from muscles, from joints, information comes back telling you how to move your body, where your body is in space, and it also sends commands to the cerebellum, that movement part of the brain, in the lower part of the brain and the parietal lobe, creating your whole body image as you actually move.

- 3.7** And again, like before, there's this iterative process where when you move your hand, as you move it, and this is quite elegant, actually, as you move it, move that hand, there's the ghost feedback, "Move!" You move it, and at the same time, at each little movement, there's feedback going back, "Oh, good, you moved it right." You didn't move it too much; you didn't move it too little; it's the right movement. So, feedback, this iterative process in the whole motor system, just like the vision system, too, so you know how to move. So, this is quite important, the motor system.
- 3.8** Then, there are the sulcus, the sulci, the little ridges, the little furrows, and then the central sulcus in the middle here. Then, you come to the post central gyrus. The precentral gyrus is the motor strip, the motor region. The post central gyrus is the sensory map; your whole body is mapped onto the touch region of the brain, the somatosensory region of the brain. Over there, I touch your hand, I touch your leg, there's a specific—there are nerve cells over there, some for pain, some for temperature, some for touch; we have different types of receptors. They will activate in that region of the brain, in the somatosensory area. So, you have that, the parietal region involved in spatial navigation, multimedia studio over there, sense of self.
- 3.9** And then, you go further back; you have the occipital lobes: vision, sense of touch. It's huge because vision trumps other areas in size because it's so important for survival. So, you have that vision part of the brain. On that whole vision part of the brain—this is going off tangent again, but that's my specialty. Have you noticed when you go into—I'm wearing all black right now, and it was semi on purpose, but not really; I just wore these clothes. But notice there's a contrast with this red, I think it's a horse. This is a contrast. If I was wearing all black, it wouldn't be as nice as if I had a little bit of a redness in there. If I wear a black suit, I might want to have a red tie, correct? Have you noticed that?
- 3.10** But why do we, the human brain, the human visual system, prefer that? Why do I like Michael's suit with that bluish thing here? Have you thought about that? Why do I find it dull if it's all black? Any thoughts? The law of contrast. Absolutely, contrast effect because when we were living in Savannah, when you were looking at the bush over there, and you spotted a lion, let's say, or you spotted a gazelle or something like that, it had this color upon another color. So, it had this contrast effect there, and your visual system, which is also wired up with your dopamine system, gives you a sense of pleasure when you see that contrast. It means fear, or it means a new mate, or it means a new meal. So, it has that; it likes contrast. Interesting.
- 3.11** Another example is, how come you generally like groupings as well? If you wear clothing—if I was wearing red shoes, blue pants, they are kind of blue, but blue pants, and this was purple or gray. Everything was colors all over. Would you prefer that? You wouldn't prefer it? Maybe in California, they like it. Well, the brain likes grouping. It likes grouping things together because, again, seeing the lion behind the bushes, we see those splotches. We don't see the whole lion; we see patches of a lion, a lion

behind a bush, and we go, "Oh, here, here, here. Oh, my God, it's a lion." That 'aha' there gives you a dopamine boost.

- 3.12** Dopamine is not only involved in the sense of movement, as we talked about earlier in Parkinson's but is also involved in pleasure. You have pleasure centers, and there's a pleasure center that thrives off this dopamine. There are dopamine centers here. So, yeah, that's the law of grouping, another law.
- 3.13** Here's another one. Now I'm going really off-tangent. This is bizarre. How come we like Picassos? How come we like paintings? Are we just fooling ourselves to seem cultured and, like, "I like the Picasso because I am so cultured," or is it actually something in your brain, some neural substrate, some circuitry allowing you to be interested in the Picasso? What do you think? What's that? Well, they have harmony and symmetry. That's a great point.
- 3.14** Yes, symmetry is another rule. I didn't think of that, but you reminded me. So, most objects out in nature, like a lion, a giraffe, or whatever, have symmetrical forms. Humans are symmetrical. If you look at a tree, it's not very symmetrical. So, we have an innate liking towards symmetry, and some of these artists will deploy that. They'll deploy the laws of grouping, the laws of symmetry, the Taj Mahal. The Taj Mahal is gorgeous; it's symmetrical. Right, that's true.
- 3.15** But let me go a step further. I wanted to take it a little bit more to the next level. So, now there's this guy, Nikolaas Tinbergen. This guy was fascinating because he was asking—you know these small seagull chicks, little baby birds, seagulls, chicks, whenever they see their mother, the seagull mother, and her beak with this big spot on it, they go wild, and just peg for food. They want food. They open their mouth and just go wild. And this guy was asking, he was saying, "Why is this the case? How do these small little chicks, or rather their neurons, recognize this beak and this spot on it?"
- 3.16** They didn't know what was going on there in the neurons. So, he took—he chopped off the head, or he got a beak of a dead seagull, and he waved it in front of those chicks. Guess what happened? Still, they begged for food. They got excited; they still got excited, begging for food. It didn't matter that it was an evil ethnologist begging a disembodied seagull. It didn't matter. They wanted their food and were begging for food. Next, what happened? The guy was smart. He said, "Let me take this, this little thing here, cardboard from anywhere, from Walmart, cardboard here, and then put a dot on it and just wave it. Will they know?" The chicks down there were still begging for food as if it's the mother. They think the cardboard is now the mother. They were begging for food; it didn't matter.
- 3.17** Next, here's the exciting part, at least for me. This is the exciting part for me. He took the cardboard and made three stripes on it, and did like this. The chicks went insane begging for food. They went out of their mind, begging for food. Pip, pip, pip, pip. They just went crazy. He had, incidentally, created a super beak by putting three stripes on it,

and he didn't know why. Why would these birds here go crazy when they see this cardboard with three lines on it?

- 3.18** But somehow, Tinbergen had tapped into the neural algorithms of these chicks, these birds, and how their system works. Now, nobody knows for sure why the three stripes made them go berserk and just go crazy and beg for food, but it could have something to do with the outline and the level of the outline. The brain says, "Look, I want to survive, and the quickest way for me to survive is whenever I see something long like this in nature, elongated and some outlines like this, it means mother, it means survival, it means food." So, it merely reacts to that. It doesn't care, whatever it is, but the neurons just want to shortcut everything and just go directly to begging for food.
- 3.19** What some people have argued, a colleague of mine has argued in this case, is that when you look at a Picasso, we look at a Picasso, we go crazy. Not crazy, but we find it very inspiring. We find it beautiful, perhaps. It's tapping into some neural substrate, just like those little chicks, and those chicks, imagine this. Those chicks, if they had a little society, they had a little society on their own, a little civilization on their own, you know what they would do? They would hang these cardboard with three stripes on them, and they would worship them, call them a Picasso, and not know why. You see? They wouldn't know that this is their mother. It doesn't look like their mother at all, right? But this is the point I'm getting at, that there might be algorithms in our brain, too, that allow us sort of to find these Picassos attractive, find them beautiful, and we don't know why. Just like the chicks don't know why the cardboard with the three stripes. Does that make sense?
- 3.20** If that's the case, though, then Michael might say, "Baland, hold up. Not all people love these Picassos; not all find them attractive. Wouldn't you say that? Good. You would say that, right? Well, perhaps, and here's the argument that's been made for this. I didn't make it, but the argument that has been made here is that, well, in fact, all of us do like these Picassos, the Taj Mahals, and these beautiful paintings of Michelangelo. We do love them, but for cultural reasons, the prefrontal cortex has inhibitory structures from social standards. We've been raised in a certain way that is telling our limbic, emotional part of the brain, "Hold on buddy. You don't like this. You're just not supposed to like this."
- 3.21** So, perhaps if you measure his emotional state using a way to measure his skin, called the GSR, galvanic skin response, or skin conductance, you measure his sweating, he can't really fool you with his sweating. You see what I'm saying? You can't really use his frontal lobes for that, so you will still know if he actually likes it or not. You could test it in that way. So, this is a way of thinking about vision, visual neurons, how that occurs in the brain, and then thinking about how Picasso might be beautiful.
- 3.22** On the whole Michelangelo thing, have you been to the 16th chapel in Rome? You've been there? It's beautiful. It's beautiful. When you go there and just admire these pictures, these images, these paintings, you feel like you're in another universe. You

feel like you're not really here on Earth, but you're looking at something that's just from a different planet. All the details, the colors, these guys were tapping into neurochemistry that makes the dopamine system just flare up, that's for sure.

- 3.23** It's said that Michelangelo, when he was painting the big one in the Sistine Chapel, the one with the Jesus character and all that, was actually lying. He was on a scaffold for 22 months. For 22 months, he was lying there and painting there, just painting, painting, painting, painting. You know that? That's kind of crazy. Guess what happened when he came down. For months, it's said that he could not look at images or a design except if it was upside down. His brain just kind of flipped in that way.
- 3.24** There are certain goggles that you can wear where you see the world upside down. In a short time, your brain will adapt to that. You take off the goggles, and lo and behold, the world is upside down. This takes us to the next thing that we've been chatting about today. We'll go into it a little bit: neuroplasticity.
- 3.25** The brain can change its shape. So, even though the brain has a certain structure, and we were born with this, a lot of change can occur in the brain. So, just on the whole language thing. So, language is in the left hemisphere. Language arises in the left hemisphere. For example, we talked about the two modules. This is a good review. Wernicke: semantics; Broca: syntax, speech. Arcuate fasciculus: the fibers. The left hemisphere—it's a specialty of the left hemisphere. The right hemisphere doesn't care about language.
- 3.26** Now, if you have damage to the left hemisphere, guess what? Some patients can sing. Now, they can sing if they want, but when they try to speak, they speak nothing depending on where the damage is, of course, but it's gibberish. But when they sing, for example, completely eloquent, eloquent songs. It's a different circuitry in this case, again, showing us the difference between the two regions.
- 3.27** Now, in children, we talked about this early on a little bit: the right hemisphere is very active early on until the age of two or three. If you do scans, the left hemisphere is not very well developed yet. The right hemisphere is doing all the things. The right hemisphere is very emotional. A lot of the emotions are being processed over there. Of course, the emotions themselves aren't limbic centers, but there's a lot of regulation on the right side, as the lady that tried to strangle herself demonstrated. And if you do stimulate the brain, if somebody has anxiety or certain forms of psychiatric conditions, you will dampen the activity of the right hemisphere. You would say, "Cool down, buddy," making the neurons less active using TMS, transcranial magnetic stimulation, for example. Very good. So, we covered that.
- 3.28** By the way, the baby's brain. My God. So, the brain of a child, you know, the specialty of the different regions are very, all over the place. Babies' brains aren't very differentiated, so modular structures are blending into each other a lot, a lot of the structures. Meaning, for example, language, which is not very developed in kids, of

course, until year two or three, occurs mostly in both hemispheres, when it occurs, very little of that. So, it's at the ages of two or three that we begin to see a more clear structural difference between different modules where the left is language, the right is spatial. The occipital lobes—these things become more specialized later on.

- 3.29** In fact, baby's brain, it turns out there are 50% more synapses in the baby's brain. That's interesting. That's a lot. I mean, it's just hyper-connected in many ways, the baby's brain. I mean, so many connections all over the place. And there's a particular gene. Again, this is when we get into the topic of genius. Can you sculpt—can you shape the human brain just like Michelangelo and Leonardo was doing sculptures? Can you shape it? But it turns out there's a gene called the pruning gene, and this pruning gene cuts down on connections depending on whether we use them or not. Interesting. Use it or lose it. It's a principle of the human brain that's intrinsic, an intrinsic rule, an intrinsic algorithm. Use it or lose it.
- 3.30** So, the baby's brain is hyper-connected between different modalities. Everything is flirting with everything, and then, as time goes on, you use certain structures. This pruning gene allows you to prune down and remove the redundancy, so you end up with the perfect sculpture. Is that kind of clear? So, you have that.
- 3.31** Of course, in some people, it doesn't occur very well. Have you heard of synesthesia? There's a condition called synesthesia where people—it's been argued that this pruning gene, this gene right here, that helps us prune down the excess connection and the excess cross-activation between different regions, is just not working right. There's some abnormality, mutation, to that gene, and therefore there are excess connections. So, when I hear something, I see a color, for example, or I see a letter, I see this Jose over here, I see Jose in blue, the j in blue, the o in green, and so forth. This is a real sensory phenomenon because the sensory modules and the brain, the vision and hearing, might be hyper-connected in this fashion.
- 3.32** Now, an interesting character, Dr. Wilder Penfield. He was interesting, this guy. He was a surgeon in the 1930s, I believe. He was doing surgeries, brain surgeries on these people, because he wanted to see what part of the tissue was healthy or not, of the brain tissue. So, he opened the skull, and these were human beings, not chimps, nothing like that. Human beings. He opened the skull, and then he would look, using an electrode to stimulate parts of the brain to see, "Is this tissue cancerous? Is it healthy? What's going on?" So, he was doing that. And while he was doing that, the arm started to move, jerky arm movements. And then, further down the the leg was moving. Opposite, opposite, opposite.
- 3.33** Actually, he did start with this region further back. Remember that? What is it called? Do you remember that name? Somatosensory region. He did start with that, actually. So first, he went over there, and the patient would go, "Oh, I feel it in my fingers, doctor." Or he might say, "Oh, doctor, I feel it in my leg, doctor." And that's how they

discovered that there's actually a human map, topographical organization. A fancy name for that. It looks like a human being up there in the somatosensory region.

- 3.34** But not completely, this guy, Penfield, discovered it because certain regions were more represented, more space. So, it was like a human being, like Baland over here, but if you're—okay, I am the homunculus in the somatosensory cortex right now. Imagine that. Now, the tongue would be much bigger than the actual tongue. It's like a huge tongue. And then, other parts, the genitalia, for example, might be extra large. The feet were quite large, and so forth. Now, guess why would the tongue be bigger, for example? More connections?
- 3.35** Well, it turns out it's very sensitive. There are a lot of receptors on it. You know, the tongue is very sensitive. There are over 48 different taste receptors, taste buds on the tongue. I believe it's like 14 on the tip of the tongue alone, and about 10,000 to, I think, 50,000 wires going from the tongue. When you have some molecules, like a chemical, the tongue is—it reacts to chemical molecules, and then it goes to something called the solitary nucleus, and there it goes through the gustatory, primary, secondary, gustatory cortex. Of course, it passes through the thalamus, the stop station. Now the review point. The solitary nucleus, then it's going to the primary and secondary gustatory cortex. I just wanted to put that in there, so it's recovered, that sensory organ, you see.
- 3.36** The tongue is interesting because the tongue is actually an inner organ that's on the outside of the body. It doesn't have all the dead skin cells. So, a lot of scientists these days, if they want to get directly to the brain, they use the tongue. They activate the tongue. There have been some, not all sciences, but some have done that. It's very interesting, I feel. But, it's over-represented in that region because it's very sensitive. So, you need a lot of cortical real estate for it, more space.
- 3.37** Did you know that animals have different sensitivities of receptors? When I was in Turkey recently, I was in Istanbul and Antalya, and I saw these little cats. There are cats everywhere over there. They're so cute. So I'd go to the cat, and I would go like this on its head. It's very soothing, by the way. It feels very nice, just the head. You massage its head. You feel good. It feels good, right? It turns out that in animals, their head, for example, in a cat or in a rabbit, their head or their teeth might be overrepresented. They're more sensitive than the rest of their body. It's interesting, isn't it? So, I was using that knowledge of neuroscience to appease and make the cat happy. Kind of cool. Well, it's overrepresented.
- 3.38** The next question that Talia wants to ask; the physician here wants to know how come—probably she already knows, but she wants to test my knowledge over here. She wants to know, "How come the places in the brain—how come the body parts are organized in this manner? Why is this the case?" It's a good question. Well, it turns out there's a law called the Hebb's law. And Hebb, nobody argued, he said that neurons that fire together wire together. Actually, he didn't say that expression, but that's the law. Meaning, when we use certain neurons at the same time, in time, or right after

each other, just close to each other, we tend to have the neurons in that region fire at the same time, and then, they also tend to wire with each other. And if you want to wire, then it makes sense to wire with neurons that are close in proximity, so you avoid some of the synapses. You want to skip synapses, so it makes sense to have things next to each other. Does that make sense?

3.39 So, you want to skip synapses and have things close to each other. And since—and now, here's the point; here's the punch line of this whole rambling here—whenever you use your thumb, you often use your, this finger, the index, right? Is that right? You use this finger with it, and then you tend to use this finger next. Okay, Baland here is reaching out for this cup. Notice: thumb, the first point of the somatosensory area. This finger, okay? You see that? So, because we tend to use these together, they're close to each other on the map. And then, we usually come with this third, fourth. You see what I'm saying? Does that make sense? We tend to use these body parts together at the same time, and therefore, they are wired next to each other in the brain. So, that's another explanation. That's an explanation for the proximity and the map, the topographical map.

3.40 Now, we said synapses. If they want to communicate better, they want to skip some of these synapses. The less synapses, the better. A great example of that is a sense of smell. The sense of smell: inborn chemicals, you inhale them. It goes through your olfactory bulb, and from there, it travels; it goes to the piriform cortex—that's the fancy name for the primary cortex, the primary cortex area of this brain that processes the sense of smell. Then, it goes straight to the limbic, the emotional part of the brain. It skips the thalamus. What's going on over here? I said earlier that all sensory information coming through the brain, coming inside, like hearing, taste, vision, and all that stuff, has to go through the thalamus. But here, I made an exception because it turns out that smelling, once upon a time, was crucial for survival.

3.41 Is this food rotten? What is this? Can I eat this? Let's skip the thalamus. Let's go directly to the limbic. Let's skip that bloody synapse and go directly to the limbic, the emotional part of the brain. Isn't that cool? That's very cool. And then, it actually goes to the amygdala first, which is, of course, the limbic system, a part of it. But then, at that point, it sends a projection to the orbitofrontal cortex, which is involved in OCD, near the eyes. And here, you have a cognitive, intellectual judgment of that smell. "I don't like that smell. I certainly don't like that emotional, intellectual component of smelling." But you also have another projection going from the amygdala to other diffused limbic centers to give you that emotional, visceral response to smell.

3.42 And, of course, taste and smell often work together in a congruous manner, and they produce something called flavor. When you eat, you taste something, but if you have the COVID virus or something like that, and you can't smell, you notice the food doesn't taste as good. Or maybe it's just a regular cold. Is that right? Have you noticed that?

Yea, absolutely, that's very true. So, there you see. This is the reason that there's this organization going on.

- 3.43** Have you heard about David Hubel and Torsten Wiesel? These guys are interesting characters. They won the Nobel Prize for some of the early studies in vision. So, what they would do, they would have a cat, and they would sew, have the cat's eye blindfolded, just sewed it together, like this. So, it couldn't see, and then they would measure the visual cortex, the neurons of its brain, in the occipital lobe, using microelectrodes. So, they measured the activity of the neurons as it's using its eyes.
- 3.44** So, first of all, they discovered some simple things like that when it looks at a certain line, for example, certain neurons would activate in the visual cortex. Trrrr. That's how it might sound when you listen to it. Trrrr. You'd see a dot, see simple stimuli like this as a corresponding region in the visual cortex becoming activated. In some parts of this experiment, they had the blindfolded—the cat couldn't see on the one eye. And then, what they noticed was, they had this done so it couldn't see with its left eye, say, from week three to week eight of its life. And then, they opened the eye. Guess what happened to its eye? So, you have a baby kitten there and from week three to week eight of its life, with no visual input. It was blind in one eye. It was blind in one eye for life. The poor kitten couldn't see anymore. That was the first observation.
- 3.45** The next observation: it turned out that the visual cortex on the other side of the brain that was processing vision from the other eye, invaded that part of the brain. It invaded it. It said, "Look, if you're not using that cortical real estate, if you are not using that eye and that place in the brain, I won't let that go to waste. I will invade it like an invading army, and I'll have double sensitivity in that eye. I see the world much better." It's cool, isn't it? So, it did that, and this led to some great work in plasticity.
- 3.46** You know, these guys, Hubel and Wiesel over there, they were doing these experiments. But one guy who was doing some of these similar experiments was a very fascinating character, named Horace Barlow. Horace Barlow was Charles Darwin's great-great grandson, I believe, a great scientist. He was in the names for the Nobel Prize with these guys, but he didn't get it. So, I think he was sad about that. When I spoke to him, he wasn't happy.
- 3.47** One day in Cambridge, in the UK, when I was giving a lecture back when I was a student there, I was giving a lecture, and this old guy next to me looked very, very old. I was talking about my research, my OCD studies, and various things, and he looked at me, and I looked at him, and he said—this is after the talk—he said, "By the way, do you want to have some coffee with me? You sit down, and we can have a chat." I thought, who is this guy? "Sure, what's your name?" He said, "Dave Barlow, David Barlow. In fact, you can come to my house tomorrow." I didn't know who he was initially. Then, I figured out, I found out that he was this great scientist.

- 3.48** One of the great things about being in Cambridge is that you have a lot of opportunities to meet great scientists like that. I would go on my bike sometimes, like heading to the lab in a rush to get my experiments, and then I would crush, like crush into this person, and I looked at him, "Oh, my God, this guy has a Nobel prize in physics," for example. It happened a couple of times. So, you see these characters quite a bit.
- 3.49** But then, yes, I went to Barlow's house, and he knew my mentor, who also went to the same college over in Cambridge, and we had a great time. We were just chatting, and it was quite fascinating. He told me a very interesting story about the guy who discovered DNA. And this is interesting. You may have heard of— what's his name? Now, it's escaping me. Not Watson. Crick. Crick discovered the double helix structure of the DNA. Now, this guy actually went to Barlow, because Barlow was doing vision research at the time. So, he went to Barlow, and he said to Barlow—and this was before he discovered DNA. He said, "I want to go and study brain science and vision." And then, Barlow said to him, "Don't do that. Don't study brain science and vision. It's too early days. You won't find anything interesting." So, he dissuaded him, and so he went off and discovered DNA. And so he said to me, "I should have gotten the Nobel Prize along with him for the discovery of DNA. I think he had a Nobel Prize complex a little bit, but he's a great character. It was a nice story. It was a nice experience.
- 3.50** So, visual input is important, but then, it brings us to the whole thing. This is called the critical period, the time when the brain is especially plastic in a kitten, in a little baby cat, obviously, is from three to eight weeks, right? Three to eight weeks, but is there a similar period in humans? Is there a similar time when the brain is more plastic, more able to change? Possibly, right? Could be. Well, it turns out there is. It seems to be.
- 3.51** Have you noticed some people that acquire an accent? They weren't exposed to the language early on, so they didn't really get the language fully, like my English. It's okay, but you can sense, on the edges, if you really listen, there's an accent there. I came to the US when I was way over my critical period as a child. I was born in Bulgaria. I'm Kurdish, originally from Iraq. My parents came to Copenhagen as refugees. I was six months old; I was a little infant in Copenhagen over there. So, my Danish is very good. I dream in Danish, a mix of Danish and English a little bit, but Danish is my main language. Can you hear that I wasn't born in Denmark? No, but my English, you can sense.
- 3.52** So, the critical period for language, for example, where the brain is extremely plastic and allows you to incorporate a new language, is around the age of eight or twelve, meaning after that time, it's too late. So, critical periods for different skills. Not all skills have the same critical period, but for language, it's about that age. At that time, the brain is ultra-super plastic, and there's a lot of possibility for change. The brain hasn't been pruned away yet completely, so you can allow certain synapses to exist. In fact, when you learn a second language later on, it's in a different part of the brain than the

first language, if it's later on. But if it's early, then it will be the same language as, let's say, your original mother tongue. Is that clear? Very good.

3.53 So, you might ask, "Why is the brain of a child, of a kid, a baby, so damn plastic? What's going on?" The answer lies in the attentional circuits. Attention, attention systems. Acetylcholine—remember that neurotransmitter that was in its own class called acetylcholine? Again, this is a very simple way of doing it, but acetylcholine is that chemical, the neurotransmitter, that's important for having focused attention. Nucleus basalis—remember I mentioned that structure in the brain? The nucleus basalis is where the neurons that produce acetylcholine are. So, you need focused attention for neuroplasticity; it's one of the principles and one of the rules. Focused attention, acetylcholine.

3.54 Now, a baby, a little kid—when it's like a baby, it's looking around, does it know what's important and what to pay attention to and what to use its attentional system and acetylcholine for? Does it know that, or it doesn't know? It doesn't know. It's all over. It doesn't know. That's true. It doesn't know, and for its own survival, its brain is wired in a way where it pays attention to everything, all stimuli, everything in the environment will trigger acetylcholine. It's hyper-attentive, and that hyper-attentiveness allows it to tune in on different things and wire that stuff into the brain in a plastic way. Is that clear? So, that is a principle. Attentional system.

3.55 Now, the acetylcholine system, the attention system, in a baby is up. Plastic rewiring. But then, at eight, 10, 12, there's a massive downturn of that system; it turns down. Acetylcholine doesn't have the same function anymore; it doesn't get excited all the time. Is that clear? Acetylcholine. That's why I find acetylcholine so interesting. So, if you as an adult, want to learn something new, you have to have that attention, focused attention, on that new skill.

3.56 Another time when we do have acetylcholine is when we experience novelty. When I go outside, for example, and I'm out here in Miami, and then I see a lion, for example, will I have the acetylcholine attention system on? Does that make sense? Because I want to know how that lion got here and where I'm going. So you have a lot of attention. The brain of a baby is like that all the time, but we have it when we are exposed to novelty. So, if you want to have a lot of neuroplasticity, expose your brain to novelty. Traveling, for example, is a great way. When you travel, you have a lot of novelty. Everything is new, and so your brain says, "Look, I need to know where the roads are; I need to know where space is, like different spatial locations, because it allows me to survive. Or it allowed the guy in the savannah to survive in similar situations. Is that clear? That's interesting. Acetylcholine.

3.57 Another key point for neuroplasticity is, it's not enough to have just attention. The attention system allows you to have certain proteins released in the brain that allow for neuroplasticity. That actually allows for neurons over here and neurons over here to create new synapses. You need some protein to allow that growth to occur.

- 3.58** There was this lady, and I found her so fascinating: Rita Montalcini. She was this lady; she lived during the Second World War. She lived over in Italy, in fascist Italy at the time, and there was a lot of—she was actually Jewish of origin, and so she couldn't really do her science. She couldn't do science. All the labs for Jews were closed down, and so she had to go into her bedroom and do experiments over there. She would do various experiments in her bedroom. That was pretty crazy. Then, she was looking at why neurons and nerve fibers tend to grow in the presence of tumors when there are tumors around the brain. Well, she later discovered—she eventually left Italy, and I think she went to Austria, and she joined—what's his name?—Viktor Hamburger and Stanley Cohen, I believe, and some other people. They won a Nobel Prize together for great work.
- 3.59** They discovered that there is something called the nerve growth factor, NGF, which is released in the presence of tumors, and this allows for more plasticity to occur, for the synapses and these wires to grow more rapidly. This led to the discovery of something called the brain-derived neurotrophic factor, BDNF. Another interesting fact is that these growth factors, once you release them, you have allowed plasticity to occur. So, growth factors, you need growth factors, proteins in the brain, being released. So, this acetylcholine system, this attentional system, is a way to trigger that BDNF.
- 3.60** There are other ways, too. Some of these drugs you use when you have people depressed; they take antidepressants. How do you think antidepressants work, for example? For a long time, it was thought, well, antidepressants increase, like serotonin, for example. This might be a way to increase serotonin, and that is the way that it works. Serotonin is an inhibitory neurotransmitter, and then you may ramp up some GABA as a side effect of that. GABA is also inhibitory, so you inhibit the emotional part of the brain. But it turns out, actually, that one way that these antidepressant pills work is by allowing you to have more BDNF, growth factors. When you have these growth factors in your brain, which are released by things like attention as well, it allows you then to have more plasticity. So, growth factors.
- 3.61** One of the ways, something that they do these growth factors is they allow for this myelin—do you remember myelin? Myelin is the fatty-like substance on the axon. So, remember we said a cell has a soma (cell body), has the axon (extension), which sends the signal, and then you have the dendrite, which is usually smaller. It gets the information over here in the synapse and then sends it to the other cell. So, we said that. And then, on the axon, you have this fatty-like substance over here, and that allows for rapid communication. So, signals are sent quicker and faster. Some of the things, paying attention and having a lot of growth factors in the brain can help you with some of these fatty substances so neurons become quicker and more effective.
- 3.62** But on that point, when you learn a new skill—so, I want to learn, what am I really bad at? Tennis. I'm very bad at tennis. So, when I learn tennis, for example, initially, I pay a lot of attention. Isn't that the case? That would be the case if I was to learn tennis. I

learn how to use my hand and all that. But then, as time passes, I will use, as the brain plastically rewired itself, that part of the brain that has to do with playing tennis, say, the hand area, becomes larger and larger. What happens is, eventually, of course, you would run out of space. Isn't that true? If I was just playing tennis 24 hours a day for years, eventually, this would grow big, but it would grow out of space. It couldn't just keep growing. So, what occurs at that point is that neurons become more effective.

- 3.63** So, neurons not only take up more space in a plastic way by these growth factors overtaking more and more of the cortex, the cortical regions, but they also become more effective at some point. At that point, I need less and less of neurons for the activity because they're more efficient, and I know it very well. Now, I'm like Roger Federer or something. I'm very good. So, this is kind of how it organizes itself in that way.
- 3.64** But there's another side, the flip coin to all of this. Do you know what that is? The flip side is that if I was very good at playing some other, like the piano or something like that, and I stopped completely for a very long time and just played tennis, I would gradually become worse at that skill because there's a competitive nature of plasticity of the brain. It's like competing for cortical real estate all the time. It wants to have these regions and use them for whatever they're being used for. There's not an infinite space in the brain. So, you see that? So, it's also important to 'use it or lose it.' If you don't use it, then it will lose it to the new thing that is so important that you're using all these hours on.
- 3.65** But it has to be a novel thing; otherwise, it's not gonna have a change. If it's the same old thing— I play soccer, well, I did when I was younger. I still like it, but if I was playing soccer now, which I've done for many years, I wouldn't have much plasticity. I mean, no reason to plastically rewire my brain because it's already there, the modules. There's no focused attention when I'm kicking the ball or dribbling like Messi or something. There's no need for it because it's all there, but if I, as I said, went to learn ballet or if I was to learn some new, interesting thing, then I would need a lot of plasticity. This is one of the key things of plasticity.
- 3.66** Dopamine is also another source of plasticity. Acetylcholine is one branch. The other branch is the dopamine system. Dopamine. Pleasure. Pleasure system. So, here we can talk about the pleasure system, so you know what kind of pleasures the brain processes.
- 3.67** So, there are two forms of pleasure, as we said. There's pleasure when I'm looking forward to that great meal. There's this great meal in front of me. The meal is a cake; it's a beautiful, gorgeous cake. I want to eat it. I want to devour it. At that point, I feel excited. My tension is high; I'm anticipating eating that cake. I'm excited; I'm in intense mode. This is the dopaminergic system. This is a system that relies on dopamine in the mesolimbic pathway of the brain, in the limbic structures, and it's very different from other types of pleasure.

- 3.68** A part of the brain called the septum—this is interesting. I said earlier on that brain regions like to inhibit each other. For example, the right hemisphere might inhibit the body on the left, and the left might inhibit the body on the right because they have different functions and they want to keep their friend in check. Same within the brain, different regions like to keep their friends or their opposites from overdoing stuff, so they keep them in check. They inhibit; it's a cross-inhibitory activation in the brain. It turns out that this region of the brain called the septum, dances with the activity of these dopamine receptors.
- 3.69** So, they did a study where they planted an electrode in the brains of people to activate the septum, and these guys were happy. They were so excited all the time. They were talking about how it feels like having an orgasm and crazy stuff from having the septum being on all the time. It was crazy. One guy pleaded not to stop the experiment. He just wanted to keep going. This is the septum region of the brain—sorry, the septum of the brain, which is part of the emotional part of the brain. The limbic center. It's a center. This septum right here is known for inhibiting, for example, the activity of the amygdala and the fear-processing parts of the brain. So, they compete over there.
- 3.70** So, dopamine you really need that for plasticity as well. So, having novelty is one way to trigger dopamine. When you focus, you have acetylcholine. When you focus, you also have dopamine, but dopamine has that pleasure component as well. It's kind of refreshing being in a new environment, for example, and you turn on the dopaminergic system.
- 3.71** It turns out that by walking, you have a lot of dopaminergic neurons being active, and you create new synapses by walking a lot at a fast pace. And the reason for this is something called—I think it's called the anticipatory proliferation, and it has to do with when you walk to a new place, and you keep walking, your dopamine cells start firing because you're preparing for a new environment. So, you are laying down—you're ramping up the dopamine system to be ready for new plasticity. So, you have to learn new things in the new environment. So, merely walking is a good gateway for that.
- 3.72** So, dopamine pleasure system, excitement; you feel so happy, but not really happy, though. Not really happy. It's not the calm, satisfying sense of happiness I have when I consume a good meal, and I come to sit there and relax, and I feel calm. I have my niece with me, and I kiss her cheeks and play with her. It's a different type of happiness. Do you agree? That's the consumatory pleasure system, and it's a different system altogether. It's based on the endorphins we talked about earlier a little bit, the neuropeptides, that allow you to feel more calm, more relaxed, and that's a different pleasure system. So, you have these two opposite systems over here.
- 3.73** When you fall in love initially, the first year or six months, what kind of pleasure system is it? Dopaminergic. Yes. Anticipatory, appetitive, called appetitive pleasure system. Absolutely. Everything is novel; everything is new. What's next? "Oh, I'm excited. Did she call? Did she text?" You check it three times. If you have OCD, maybe 100 times,

you check. All right, so this is what? The dopaminergic system. Later on, one year, two years, what happens? Different kind of system, right? Serotonergic, well, not serotonergic. Serotonin is more involved here, but it's mainly the endorphins. You feel a sense of calmness, a sense of pleasure. Now you've grown, it's a different kind of love. Is that right? Oh, well, you got it there. Exactly. So that's these two pleasure systems. So, dopamine you also need for a sense of neuroplastic wiring in the brain. Am I making it clear? Am I making sense so far? Everything's clear? Very good.

- 3.74** I always double-check. I want to make sure that I don't end up like my old professor when I was in high school. I hope he's not going to be watching this, but he was just saying random Latin names, and I felt so lost. Honestly, I don't know how I passed that biopsychology neuroscience course. I don't know how I passed it.
- 3.75** In fact, when I later went to—I was not very into biopsychology or neuroscience or anything. I went to California, and I was doing part of my undergraduate at UC San Diego over there. I was taking the class of this chap, this professor called V. S. Ramachandran, and we'll get to his work. He's a legendary figure in neuroscience, and his class was so interesting. He was talking about all kinds of mysterious conditions, why we fall in love, why we are conscious rather than nonconscious, and all the stuff that is sort of on the fringes over here; he was talking about that. I said, "My God, is that neuroscience? That's very, very cool. That's very interesting. It's very enriching. Now I know why I feel those feelings when I fall in love. Or now I know why I love that cake." I got a different impression of all this. So, this is how mentors can influence you. I ended up working with him for many years. For over a decade, we worked and did a lot of research. Some of the OCD work was together with him, and it was quite fascinating.
- 3.76** During the next lecture, we'll continue down this course, talking about plasticity and answering the question, "Is plasticity, this rewiring, possible in adults, in grownups? And to what extent? What are the limits? And can we all take advantage of this and use it to shape our lives in a more direct way?" So, we'll get into that tomorrow, and that will be interesting. We'll talk about some interventions and treatments, neuroplastic interventions. I created an invention, too, for sleep paralysis, a treatment that is used and also has a neuroplastic component to it, but that will be perhaps on the third day that we'll get into that one.
- 3.77** I hope you enjoyed it. So, thank you very much.

Lecture 4: Neuroplasticity and Therapy

At a glance — Neuroplasticity and therapy.

- 4.1 All right. Thank you very much for coming today. Yesterday, we ended off on the whole plasticity thing, right? We talked about plasticity. We talked about plasticity in kids and babies. Then, we went to the critical period. Remember that? So, during the critical period, the brain is plastic, and the critical period varies for brain systems. So, language was from what age? What was the critical period for language? Remember? You got it. So, about eight to 12 is that period, right? What about other brain systems? Different, right? Different periods for different brain systems. So, we have that; there is that flexibility, that time when we can shape our brains and make a lot of use of it and mold it in profound ways.
- 4.2 Here's the thing, though. The question is, "What about in adults? Is it possible to have plasticity in the adult brain?" That is the question that has been on the neuroscientists' minds. For a long time, the answer was, "No, that's impossible. We can't have that. It's only in babies, it's only in young people; you won't see it in somebody who is like 30 or 40. Impossible."
- 4.3 Now, this guy came along; his name is Michael Merzenich. He did some studies on monkeys, very interesting studies to look at plasticity. These were adult monkeys. What he did, first of all, he had the nerves going from the arm, the sensory nerves, and he cut the middle nerve here, right here, he cut it. Then, a few months later, he looked to see if there were any changes in the brain. So, they touched the arm over here, and they looked at the area of the brain that corresponds to that area. Remember we talked about the somatosensory area, that strip of cortex? Over there, when they touched it after a few months, guess what happened. Anything? Anything here if you cut the nerve? Anything? No.
- 4.4 So, basically, it's like somebody who had a stroke. Up here, you touch it, nothing, no feeling. The monkey has no feeling. Of course, the monkey can't speak, but you have these unit recordings. You record the activity of the neurons up here, and there's no activity for this region. But something interesting happened. It turned out that the area for the sides of the hand invaded, in a matter of months, invaded this area.
- 4.5 Those neurons that previously were only for the sides of the hand would now light up. Is that clear? Invasion: armies invade, use it or lose it. So, this part of the hand no longer was doing its job. Now, in another study, they actually cut off the middle finger, to be sure. Again, they saw an invasion. The nerves were cut off here, and you saw the brain area for the other fingers invading. So, this was the first step, showing us that plasticity is possible in adult monkeys. So, this was a good first step.
- 4.6 Now, interestingly, this guy, Michael Merzenich, he was also finding that in some cases, when you would map the brain of these animals, randomly, there would be some

changes. Sometimes, the map for the hand would be over here, in the somatosensory area, but over here. But then, a month later, he would map again, and it would randomly change a little bit. Again, giving us clues that sometimes these maps just randomly change. No need to cut off the nerve sometimes; there is this change all the time occurring in all of us, potentially, at least in animals at this point. So, this was the first step, a very interesting finding.

4.7 Now, later on, they wanted to—t his is a very interesting story, actually. The question is — and this is the work of a guy called Edward Taube. Very interesting stuff. There were these monkeys, and he cut all the nerves going from the hand to the somatosensory region. He cut off all the nerves. Now, the nerves over here, when you cut them, guess what it's called? It's called the deafferentation. This is a technical term, deafferentation, when you cut the nerves. When signals go from the body and out, it's called an efferent signal. When the signals go from the body to the brain, it's called an afferent signal. It's a technical term, so you know this. Efferent, like exit, and then afferent.

4.8 Now, when you cut the nerves, it's called deafferentation. "Look, Ismail, they cut all the nerves." But there are some militant-type activists out there—well, I think they're militant. What they did was they infiltrated this scientist's lab. So, they went to him and said, "Look, doctor, I want to be a research assistant in your lab. I want to join your lab. I love animal research. I love all the stuff you do. Can I be a part of this whole thing?" And the doctor was, "Sure! Just join the lab. Come in," and he joined the lab. And as he joined the lab, one day, the doctor was, I guess, on vacation or something like that, and what he did was he— this is on camera, so I have to be careful because this was a big deal, actually, at that time. The FBI got involved and stuff.

4.9 He took pictures of the animals. It seemed like he maybe staged it or something like that; that's what some people argued. But it was made to look like the animals were suffering, and they were starving, and they had all their nerves cut off, and it looked like they were just dying there. So, he was taking all these pictures, and, eventually, what they did, they took these monkeys and removed them from the lab. So, they were gone, and so was this guy's career. For years and years, he couldn't do research. Horrible.

4.10 So, years went by, and this guy went pretty much to poverty. He couldn't do anything. He couldn't work, and the whole science community was against him at the time. Well, not the scientists, but more the funding in university and stuff like that. So, he couldn't really do anything, so he went into poverty. And yeah, it was a big deal. But guess what? He was a smart guy, and so what he did was he got evidence and things like that and actually showed that much of it was staged. Some of the photos seemed to be staged and things like that.

4.11 Twelve years went by. You know what? The monkeys were brought here to Florida. This is interesting. They were brought here to Florida. And so at that point, because 12 years had gone by and it was clear that he was innocent, perhaps, what they did, they

were able to get some of the monkeys back. Then, they got secret funding because they couldn't get governmental funding. That would be very difficult at this point because there was so much scrutiny from the media and all that stuff. So, they got government— private funding to fund brain surgeries on these monkeys to look in their brain to see what had happened to these maps in the brain now that all the nerves were gone.

4.12 Mind you, all the nerves from the hand, from this left hand, were gone. It was all gone, and 12 years had gone by. Not a month, 12 years. So, this was with a guy called Tim Pons. So, they looked inside the brain, and at that point, we knew that brain plasticity changes in the cortical areas about 1-2 millimeters and only in the regions that were neighbors. Only neighboring regions, like the premotor area and the motor area, might switch careers a little bit, but not far-flung regions. They wouldn't, but then, when they looked here in the brain of these monkeys, it turned out 12 millimeters of change had occurred. In some cases, far-flung regions had invaded parts of the brain. This was the first clue, and this was a seminal experiment. It made shockwaves in the scientific community, the fact that 12 millimeters of change had occurred. This was the first really interesting study.

4.13 There's actually a fun point about this whole study because back in the day, for a very long time, it was thought that you actually need your sensory receptors in order to move your hand. It was thought like that. This was because people would have strokes, and then they would have—the sensory area would be gone, but then, sometimes, they really wouldn't move it. So, it was just thought, well, sensory area, you need that for movement. Is that clear, Moni? That was the idea.

4.14 So, these researchers challenged this. They wanted to see, "Do you actually need your sensory receptors for movement?" It makes no sense, really. You have a motor cortex. So, what happened here was they noticed, for example, when monkeys had all their—for example, this monkey, all the sensory neurons were gone, no feeling in the hand whatsoever. There was no feeling in the hand. But then, what would happen, this chap, this researcher said, "Look, what if I take a sling or a cast of some sort and the good hand, I wrap it around, so it can't use it?"

4.15 So, he was forced to use the hand, which was not good. So, he did that. It was weird, but let me just try this; why not? So, he did that. So, the monkeys, for a month— a month went by, then it became two months, and they started to use the hand more and more and more and more, and it turned out they were just fine. You see, before they had the opportunity of using this hand. This hand was good, so they could just use it to grab objects, bananas, peanuts, fruit juice, all the things they love. They can take that. Interesting.

4.16 So, actually, what he did in a very interesting experiment, you know what he did? So, in the first experiment, just cut all the nerves. One good hand over here; this hand is paralyzed for life. Nothing happened. Puts a sling, and now this arm works after some

time. Then, he did something interesting. You know what he did? This guy cut all the nerves on both hands. Counterintuitive. In theory, these monkeys should be what, Brook? They should be paralyzed for life. It turned out that when you cut the nerves in both hands, after some time, they used both hands. It's counterintuitive science sometimes, isn't it? It's interesting.

4.17 Now, this brings us to a very interesting principle in neuroscience called 'learned nonuse.' So often, a lot of us fall into this trap called learned nonuse or disuse. When you have a stroke, for example, or you have damage—for example, these monkeys, when they did the surgeries to remove all the nerves to the spine, they had a spinal shock, and that spinal shock put them in a state where they had inflammation. It was difficult to use the hand for a while, and so they stopped using it. And guess what happened? That map in their brain corresponding to that region, fated. Use it or lose it. They weren't using it, so that map shrank. Sorry, shrink, shrink, shrink, shrink, became smaller and smaller. Is that interesting? It's because when they are forced to use it, they will start using it and the map doesn't shrink. Is that clear? The map doesn't shrink in this case.

4.18 Another example, actually. When you have a stroke, you have a similar situation. When you have a stroke, you have a cortical shock. So, just like you can have spinal shock in terms of the monkeys, you can have cortical shock. So, when you have a stroke, when the damage occurs, there's a lot of inflammation, there's a lot of chaos in your brain. Your brain is in a state of chaos at this point. And so, similarly, certain abilities might go away. You might not be able to have certain cognitive functions, motor functions, and other things, maybe speech. But at that point, if you stop using it, you fall into this trap. You have to use it, and that goes for any skill. You don't want to fall into the learned nonuse trap. All the time, just use it, use it, use it. Very good.

4.19 My colleague actually read this paper, and when he read this paper in the 1990s, of course, I was a very young child at that time; when he read this paper about these monkeys and what had happened with the 12-millimeter change, he was quite surprised. In fact, there was another finding in that study that was quite surprising. You know what that was? Guess what? The hand area in the brain is right next to the face area in the brain. So, what might happen over here? If the hand area in the somatosensory cortex, the touch is not being applied over here, so there's no touch, all this is dead, it's basically shrinking that area. Might the face area invade? If you touch the monkey's face, might it feel it in the hand? What do you think, Cassius? Might that happen? Absolutely. That happened exactly.

4.20 The face area invaded here. So, they mapped the brain. When they looked at the microelectrodes in the brain, they saw that. They could see the face area light up like a pinball machine when they touched the hand. Now, the monkey, of course, can't talk. So it can't say, "Doctor, I feel this in my face when you touch my hand." But my colleague in San Diego was studying people with phantom limbs.

- 4.21 Now, phantom limbs is a very interesting phenomenon. Mary, phantom limbs is this phenomenon where there's this guy, and he's in a motorcycle accident. He's going by, and then his hand gets injured. It gets damaged in the accident, so the doctor says, "Look, we have to cut the whole thing." So, they cut the arm, and then when they cut the arm, weeks later, he will say, "I actually feel the presence of my arm. It's there. I feel it. Sometimes, it waves." If they have a conversation, it might go, "Hello." It might scratch the head. Mind you, there's no hand. It's a phantom.
- 4.22 It feels like there's a hand; it does all these lively things. So why do you think, based on the knowledge you have, why do you think we have phantom limbs? Why do we experience a phantom limb? Ismail, what do you think? Could it be because we have a sensory region where the whole body is mapped? And so, perhaps when you cut the flesh, the map is still active and vivid. Alicia, could that be? Possibly, right? Absolutely. That's right. So, the physical hand is gone, but they still feel it. They still feel the vivid presence of an arm in many cases. This is called a phantom limb.
- 4.23 Now, a phantom limb had been a mystery for so long. The doctors didn't know what to do about it. Is it just nerve pain? And sometimes, you know what they would do? They would say, "Oh, it's just the nerve over here that's irritated. Let's cut more up." So, they cut a little bit more. "Oh, it still hurts, doctor. It still hurts, doctor." "Oh, let's just cut a bit more." It's like they're cutting a piece of cake. "Let's cut a bit more." And they would just keep cutting further back. That's crazy. So, they hadn't thought about this whole, that it could be a body image, perhaps.
- 4.24 So, my colleague was reading this paper about the monkeys and the 12-millimeter change, and how the face area had invaded the arm area, and he was like, "My God, this is striking. But what about human beings? Could this happen in humans? What about people with phantom limbs? They have no limbs, or they are missing a left arm, for example. What's going on there? Let me take a Q-tip," this is a simple guy, "Let me take a Q-tip and just touch his upper body.
- 4.25 So, he brought in Tom, the patient, and touched his body with the Q-tip. "Tom, what do you feel?" "I feel this on my stomach, doctor." "Good." "Tom, what do you feel?" "I feel this on my chest, doctor." "Tom, what do you feel now?" "Doctor, I feel this on my cheek and in my phantom hand." So, it turned out the whole hand was mapped on here. There's a pinky, index finger, thumb. The whole hand that was missing was up here. Interesting. They did a brain scan, they scanned the brain, and it turned out exactly: when they touched the face, the arm area would light up. It hadn't gone idle. It was being used.
- 4.26 So, this was the first, or one of the first studies showing human plasticity and how you have this change occur. Isn't that crazy? Using a Q-tip, you discover this major discovery. Absolutely. And then, well, he thought, "Is there a way for me now to treat these patients? Could I treat them? They have this phantom limb. Is there a way for me to get around that whole thing, that pain?"

- 4.27 Oh, did I talk about that? Maybe I didn't mention this, but a lot of the phantom patients actually have pain in their hands. This is a crucial point. I may have missed this. So, they not only have the phantom, which is, of course, spooky and weird, and it can do fun stuff, but they also have a lot of pain in the phantom. It feels like it's in a clenched position. It's often maybe paralyzed and things like that. It wasn't clear for a very long time why this would be the case. Why would you have—forget having a phantom, but why is it in pain? Why does it feel like it's paralyzed?
- 4.28 They looked at some of these patients and their records, and it turned out, actually, before the paralysis or before the amputation, sorry, it had been paralyzed, maybe in a cast for a very long time, and so 'use it or lose it.' What had happened over here: there was a plastic principle going on that the brain had carried that information up in the brain. You see what I'm saying? So, because it was initially paralyzed and then amputated, they carried that information with them in the body image. The body image had that wiring encoded, so even though you cut the arm, it doesn't learn that it's no longer in a paralyzed position or it's in a painful position. It was learned pain. It was a learned paralysis thing. Is that clear? Okay, very good.
- 4.29 Now, this brings us to pain itself. This is important because we have to understand what pain is. Now, we have certain receptors in the hand: thermal (temperature, vibration, touch, pain), nociceptors, different types of receptors. Activate them like this. Can you do like this? Not too hard, though. I don't want to get sued. Just a little bit. There you go. Good, good. Well, this is—a little harder, man. Come on. There you go. Then, your pain receptors will send a signal to your spinal cord, and it goes to the somatosensory area. But believe it or not, there are actually gates. This is called the gate theory of pain.
- 4.30 There are gates at various points in the spinal cord and in the brain where the brain decides whether that pain is worthy of feeling painful, like having that pain or experiencing that pain in a profound way. If it's like a light thing over here, you may not feel the pain. Your brain may say—the signal might go to the spinal cord, and at that point, at that gate, your brain may have an endorphin to relax the pain. So, the brain over here will decide whether there should be pain or not. Does that make sense?
- 4.31 Now, if the pain is a little bit more intrusive, if it's a bit more of a stronger signal there, your brain may say, "No, let me take it up one more gate." So, it goes a gate more, and so forth. Eventually, it may reach the somatosensory region and the pain receptors over there, and you send signals to the part of the brain known as the anterior cingulate and the insula. We talked about the insula yesterday. Insula is involved in disgust. Disgust, exactly. Disgust sensations. So, the insula then sends signals to the emotional part of the brain. Well, insula is emotional, too, but not in that sense. It's more the limbic and the amygdala. So, anterior cingulate.
- 4.32 This is a long journey, but I just want you to know this journey: receptors, the spinal cord, going to the somatosensory area, the body map, the homunculus, and from

there, at this point, it's very sensory. Of course, we said the endorphins from the neuropeptides can diminish this pain at any gate they want, but eventually, they may reach the sensory regions and the interior cingulate, and the emotional agony of pain. Not simply pain but the actual agony. "I feel pain. I am in pain, emotional pain."

- 4.33** Pain has a huge emotional component. If I was to take Cassius right here and dip his arm in hot water, very hot water, he would feel a lot of pain. He would say, "Doctor, don't do this. This is painful. Stop, doctor. I feel a lot of pain." Then, I hypnotize you over here. I can do these tricks. Baland does this trick. Hypnotizes Cassius over here or Ismail. I have you guys hypnotized right here, and I tell you, "Close your eyes. You no longer will feel pain. The pain is not real." You do this hypnosis, and then I dip Cassius's hand in this water again. Guess what? Less pain. Less pain. Pain goes down. What is that about? You tell me, what is that about?
- 4.34** It tells us that pain has a huge psychological component, and when these psychological parts of the brain, the anterior cingulate and the amygdala, cross-talks, chit-chats with the frontal regions of the brain, we can quell the pain. We can decrease the pain. We can make the pain less.
- 4.35** There are regions over here in the prefrontal lobes. In the fancy name I always use, it's called DLPFC, on the left side. It has connections to an area of the brain called the ventral striatum, and the nucleus accumbens, and the ventral pallidum. I want to make sure I get these Latin names to impress you guys. Over here, all the opioids and the endorphins dance with activity over here. So, when we use the frontal lobes, when we are happy, when stress goes down, guess what? We can activate some of these endorphins. The endorphins and these areas inhibit the activity of the amygdala, the emotional part of the brain. Is that clear? Or am I overcomplicating this? A little bit? Okay.
- 4.36** Prefrontal regions, particularly the left side, but also an area called the prefrontal gyrus. Remember, gyrus is the round thing? The gyrus. The left side of the prefrontal, the seat of intellect and thought, communicates to these areas that have a lot of dopamine, but also a lot of these endorphins that are very active; there are a lot of receptors for them. They become very active, and when they become active, they have a circuit, have fibers going to the emotional part of the brain saying, "Look, buddy, calm down. It's not that scary anymore." So, you go from a thought in the prefrontal to happiness and calmness and wellness, like feeling well, to inhibiting that amygdala. The amygdala then makes the pain go down in that way. Is that clear now? Very good.
- 4.37** Well, I didn't make the brain this complicated. What can I do? The brain has all these inhibitory, excitatory, on-and-off states that make it so many synapses. We talked about how many synapses there are. The brain actually—the number of brain states and possible permutations and combinations of brain states at any given time exceeds the number of elementary particles in the universe. And that's true. It is very complex.

No other object in the universe is as complex as the human brain. It's crazy. Goodness me. So this was pain. This is how pain works. Good. We learned pain.

- 4.38 Neurons that fire together wire together. So, you learn there's pain over here in my hand, so neurons in the somatosensory region fire, and pain neurons fire, and it becomes a learned pain. The arm is now gone. There's a phantom limb. The phantom is stuck with that pain forever, or maybe not.
- 4.39 So, you thought—and this was a clever trick—what if you have a board like this, but not really more of a barrier like this, it's like a board or a barrier, and you put a mirror over here. You put your left hand—let's say the amputation was here. The amputation is over here; you have a phantom over here. You feel there's a ghost hand here. Now, if you look in the mirror, what would happen? What do you see? You look from here. You look here. There's the mirror. Here's your hand. What would you see at this point? What's that? You would see two hands, right? I see two hands. That's right, Lydia. And the two hands would be—when you move this hand, what does it feel like and look like? What's that? It gives the optical illusion that you have optically resurrected the phantom. It's back. It feels like that. It's a spooky illusion. You should try it at home. It's a good party trick.
- 4.40 You do like this, you move it, and the patients would go, "My God, my phantom is—my arm is back. It's right there. I can move it again, and it feels good. It feels good." It's very strange. And then, after some time of moving it around and loosening it up a little bit here and there, weeks would go by, and they would say, "The phantom is gone, doctor." Initially, maybe they'd say, "Oh, it doesn't feel clenched anymore. It doesn't feel like it's in a spasm position. It doesn't feel like it's in pain." And then, eventually, many would say, it actually it's gone. So, he invented this mirror box. It's a very nice invention. What do you think? Isn't it pretty cool? It's very simple. So, he used a Q-tip to discover the thing, and then he used a mirror, a two-dollar mirror, to discover this, and this was a huge discovery.
- 4.41 Now, this other guy with the monkeys, remember the monkeys and all that, and the nerves that were cut? He also invented a very nice therapy. It's called constraint-induced therapy, CI therapy, which I think is a very elegant therapy because it uses that knowledge that he gained from the monkeys and how they overcame learned nonuse, and so on. He used that as a treatment for humans.
- 4.42 The principle is the following. Let's say you have paralysis in this hand. Paralysis is over here. You can move it slightly. If you can't move it at all, it's difficult; it's very difficult to treat in this way. But let's say you can slightly move it. So, you move it slightly. What he would do, he would put this right arm in a sling so you can't move—so you can't use this one. So, at this point, it's forced to use this hand here, and what they would do is that they would have the patient do things that they normally would just use the right hand for, like grab a cup or eat from a plate or eat a sandwich, grab something; they would have them use this hand. So, they were like, "Oh, this is difficult." Extremely

difficult. Extremely difficult for them, but they were forced to do it. So, this factor of constraining them was helping them. They would have great gains from this. Two weeks of this, they would do it for six hours every single day. Every single day, for six hours. Only two weeks of this, and they would be able to do twice as many things as a person who didn't do this therapy. That's a lot of functional improvement. Think about this. Just two weeks.

- 4.43** That gets us to another principle in brain plasticity, Moni. You know what that is? Massed practice. Did I mention massed practice yesterday? Well, I'm mentioning it now. It's called massed practice, and massed practice is the following. In brain plasticity, in terms of your brain, doing something for many, many hours in a short time is better than doing the same thing for one hour but for a very long time. Is that clear?
- 4.44** Acetylcholine, dopamine, you go there. The patient is trying to use that hand. "It's hard, it's hard," but then you reward them. "Here's an extra sandwich for you." Here's this for you; here's that for you. "I can't. It's difficult." But incrementally, a little bit more, a little bit more. "I can't, I can't." Normally, they wouldn't be able to do nothing at home; they would just sit there with their paralyzed arm. But now you're rewarding them, giving them dopamine. From the reward, they feel a bit happy. And also just the reward of seeing themselves progressing. "Oh, my God, I couldn't do this yesterday, but now I can." And it's so hard. They have to put all their attention on this thing. What is attention? Is that important for neuroplasticity? Acetylcholine system, nucleus basalis. So, they have all this focus, they have dopamine, and they're doing massed practice every day for two weeks, and they have all these massive, massive benefits.
- 4.45** Now, if you want to learn a new language, Cassius, do you want to learn Kurdish, my parents' language? It's a very difficult language, I tell you. If you want to learn Kurdish, and you look Kurdish to me a little bit. Now, if you want to learn Kurdish or any language, is it better if I give you a book, and you study Kurdish here in Miami for a year, and you look at this book? Or should I take you and just put you over there in Iraq, and you can learn Kurdish there in three months? What is better? Completely immersed in the culture. You can do nothing. Over there, they don't speak much English. So, you cannot—you're constrained, just like the sling. You're in a total sling over there. You are forced to speak their language, and you go simply from, "Ba ba ba ba ba," and then slowly, you add more and more and more. And then eventually, in three months, four months, you speak like my parents. Do you see the analogy over here?
- 4.46** When I was actually learning, in my teenage years, or maybe early 20s, around early 20s, I went to Egypt. I spoke no Arabic. No Arabic whatsoever. My parents are Kurdish from Iraq, originally, although I grew up in Denmark. But they always had Arabic channels on, like Al Jazeera. Al Jazeera was always on in our house. It was always going. Arabic all the time. I got nothing. Years and years and years, and I learned nothing. I could say, 'salam.' I could say, 'marhaba.' That was it. Nothing. It didn't work.

- 4.47 Then, I went to Egypt, and over in Egypt, I went to a university there. I engaged a lot with the people over there. Instead of hanging out with the university folks and all the fancy folks, I was with the people on the streets because I came from the streets, too, remember? I came also from a ghetto-like area. I was used to being with the common folks. That's what I love. So I was with the common folks, eating their Molokhia food, rice, and soups and just having fun in their little houses. It was so fun. I felt at home with these kids. I didn't speak English. I didn't speak Arabic, but slowly, I learned. [Speaks in Arabic], as we say in Arabic, I learned, and in three or four months, I was like [speaks in Arabic]. I was speaking so well Arabic. My mind exploded. I was literally—all these synapses. It was wild. I really learned Arabic in such a great way. Yeah, I'm so happy. I still speak it.
- 4.48 Now, it came at a cost because I'd learned Italian earlier, and I saw that language went a little bit down. You use the same circuitry often for various activities. It's one of those things. The brain is, as we talked about, the brain, when it has a circuitry system, and uses that for a certain function. Now, you don't use that language much; you learn a new language, and it uses the same circuit. A sloppy thing, but it does that.
- 4.49 It's the same thing when you have a wife; you divorce her, get a new wife, and you use the ex-wife's for the new wife. Bad thing, but same principle. You have the same circuitry for both people. It is true.
- 4.50 On that note, we didn't talk about neuropeptides, oxytocin. Have you heard about that one? It's relevant because similar to using the old circuits from before, oxytocin is this neuropeptide released from the hypothalamus, that region we said where aggression and sexual motivation are based, and some sleep systems are based there in terms of the biological clock. It's such a fascinating structure, the hypothalamus. Now, the hypothalamus—you also secrete these hormones or these neuropeptides, to be exact, called the oxytocin.
- 4.51 Now, oxytocin helps me bond. It helps me bond. If I get a son, I don't have a son, but if I get a son, I want to bond with him and I want to learn. My brain wants to get into a new state where I'm less selfish. I'm less self-absorbed. I want to be more; I want to live for that kid. And that's difficult. A lot of plasticity has to occur at this point. So, what happens is that your brain will rewire, and this neuropeptide will help your brain rewire, so you are more geared toward helping this child.
- 4.52 Let's say in animals, in certain animals, when they get an offspring, they have a lot of oxytocin. It helps them bond with the animal. What you can do is take away the animal, the offspring, and then introduce a new offspring from a different animal. Will it bond with this new offspring? Normally, no. It smells it. They have a very strong sense of smell; the sense of smell is profound in these animals, and they can smell, "This is not my animal." Then, you inject—the scientist comes along, injects oxytocin into this sheep's brain, and guess what happens? It will take the new offspring, and it will forget about the old one. So, it will take this.

- 4.53 This is what oxytocin can do. It's very profound. It makes you more empathetic as well. In many cases, if I take that, the amygdala activity, when you inject it into the nose, the area of the amygdala, which amygdala is what? Amari? Absolutely. It's fear. Fear goes down when it is injected. I feel a sense of calmness often that has this function. It's called the love hormone. That's what the reporters in the news call it: the love hormone. It's very interesting oxytocin here.
- 4.54 That takes us to the whole hormone system because the hypothalamus is actually very important for hormones and the release of hormones. We have our glands in the body, in various places. We have our gonads: testes in men and ovaries in women. They have certain hormones that are released directly into the circulatory system: testosterone, progesterone, androgens, all that stuff, estrogen being released. You have that.
- 4.55 You have other glands. You have your adrenals, glucocorticoids. You have your thyroid and so on. Now, how do these guys know how to release hormones? Well, there's a structure beneath the hypothalamus, called the pituitary, and it releases something called tropic hormones. These tropic hormones signal these glands to release hormones. So, this is the master gland in the brain, the pituitary. Okay, pituitary, but Baland, you mentioned the hypothalamus. What is that about?
- 4.56 Well, the hypothalamus is actually the master, master gland because it's telling the pituitary to release the hormones. But that wasn't known for a very long time because there are not that many fibers going from the hypothalamus to the pituitary. There are not many axons, so perhaps they don't really communicate well together. There are some fibers. But it turns out then that the hypothalamus releases something called the releasing hormones, and these releasing hormones tell the pituitary to release its tropic hormones, and then we release our actual hormones in the glands into the circulatory system. Is that clear? Does that make sense? The hypothalamus, pituitary glands, thyrotropin, for example, thyrotropin-releasing hormone. You see what I'm saying? These are the names. Releasing hormones, and so forth. Very good.
- 4.57 I do want to talk about mirror neurons. I want to ask you first, have you heard about mirror neurons? Okay, mirror neurons. There was this scientist, or actually a group of scientists, in the 1990s, and they observed the following. It was an accident, actually, it was a coincidence, a serendipitous event. These monkeys were reaching—or actually, this is how it works. The monkeys normally reach out for a peanut. Peanuts, fruit juice. That's what they love. Peanuts, fruit juice. The motor cortex becomes active. Reaching. But then, the scientist was doing the reaching. He reached out for a peanut. He reached out for fruit juice, and a subset of the monkeys' motor neurons would fire as if they were doing it themselves. Let me say this again. The monkeys were merely observing the experiment or caches reach for the peanut, and the monkey's brain, the motor neurons, a subset of them would become active as if the monkey was doing it. Okay? Mirror neurons, mirroring the other person's brain.

- 4.58 Then, they said, "What about pain?" What about pain? Now, a monkey pokes its hand, and the somatosensory area, the pain neurons in the somatosensory area, light up. Now, Dahlia, can you poke your hand? Let's say you have a needle; you're doing that with a needle, okay? It turned out the pain neurons, a subset of them, would become active in the monkey's brain by merely watching you do that action. Is that interesting?
- 4.59 Mirror neurons, not only for motor, movement, but also for pain, sensory stuff. This ushered in a whole line of research where in human beings, they would bring in human beings and they would say, "Look, buddy, look at this guy over here. This person over here is experiencing disgust. You simply watch him," and the disgust area of the brain in the person watching the other person would light up as if he was experiencing the disgust. Is that clear? So, it went from things like pain to more emotional sensory things like disgust. Is that clear?
- 4.60 Mirror neurons, though it's not officially known that these actually exist in humans, and there's a lot of debate in the literature about whether mirror neurons actually exist in humans because you can't or it's unethical to directly go into the scalp and measure individual neural activity in human beings. So, it's unclear whether they actually exist in humans. But functionally, meaning in terms of what you actually see, it looks like there is this perceptual insight, a subjective insight into somebody else's brain by merely watching his emotional states. I feel his disgust over there by watching him. Where is disgust processed in the brain? Insula. Cassius, Ismail, the insula, right? Absolutely.
- 4.61 Did you know that social disgust—when you tell me I should be disgusted by these people over here because they are invading my country, they are doing this, they are barbarians, they're evil, I have social disgust. I feel disgusted by them. I'm getting brainwashed or something. I'm experiencing social disgust. Did you know that the same area in the brain, the insula, also processes actual disgust? So, physical, actual sensory disgust, and social disgust are processed in the same part of the brain. That's why it's so visceral. That's why it really gets me when I feel disgusted, when I actually start to hate a group of people over there; well, there's an actual brain component that's very visceral. Is that interesting? So, perhaps—
- 4.62 Does that also means that you're going to feel it?
- 4.63 Absolutely, because the insula is connected with all the organs we talked about. It has a map of all the organs and even the receptors. It has that function. But it shows you how brain science works by looking at things like, "Why do we find a certain group as evil or bad in certain political scenarios?" And then you say, "My God, there's actually a brain substrate for this." So, by looking at the nitty gritty of the brain, we go all the way to how the wars have started in the world, potentially. That's what I'm trying to get at here by looking at how these centers are the same.
- 4.64 We mentioned yesterday how the hypothalamus might explain why people— why men rape women sometimes, and why some men are turned on by violence sexually. We

said the hypothalamus has neurons for aggression over there and then also for sexuality. They're in the same area of the brain, and when things are in the same area, there's a lot of what? Overlap, cross-activity, cross-wiring, right? Very good.

- 4.65 Now, of course, these mirror neurons inspired our work. I talked yesterday about our OCD research and how, by watching somebody else touching a contaminated object, you feel contaminated. And you watch him wash his hands, and you feel like your hands are being relieved. Well, that was directly inspired by that research, and that research was done together with this guy who invented that mirror and the Q-Tip guy. We did these studies together. But of course, in research, you don't only want to see how people feel relieved by watching somebody else. You want to think about whether there's a clinical application, especially in OCD.
- 4.66 OCD is such a crippling disorder. It's associated with so much agony and so much suffering. So we thought, "Well, what if you have somebody with OCD watch a video of himself touching contaminants." I may not have mentioned this yesterday, or maybe I did, but the main treatment for OCD today is touching contaminants. I did mention that. I did mention that. The way this treatment works, which is a general principle in neuroscience, is that when you do certain things that are aversive over and over, you eventually do what? You get used to it; you face your fear. You habituate. It's called exposure therapy. It's actually the way you treat a lot of anxiety disorders and things of this nature because if you have a lot of anxiety for something, and I expose you, often, gradually, you then overcome your fear. But of course, I have to make you do it, which is difficult.
- 4.67 Now, what I thought here was, what if I use a phone? Instead of having this guy touch the contaminant, what if I use a phone? And you watch it on the phone. It's like the rubber hand trick I talked about yesterday, but it's a phone version of it. He's watching this phone, he's touching this contaminant, disgusting feces, he's watching that for a whole week, four or five times a day. What would happen? Would he eventually habituate, and would his OCD symptoms go down? This was the question.
- 4.68 Another version of the same procedure. What if he watches a video of himself washing his hands? Perhaps, when he goes around, he walks around, and he feels, "Oh, my God, my hands are contaminated right now. I need to wash my hands. Oh, I have this video. It's on the phone. I look at it; I see myself washing my hands. Oh, I feel good." And then, perhaps over time, he may go, "Oh, I didn't die from this. Nothing happened to me, Brook, by me not actually going and washing my hands. I didn't die abstaining from the actual compulsion. So, perhaps there's a new learning taking place where I rid myself of the negative maladaptive behavior." That was the thinking.
- 4.69 And then, of course, we had a control app for experiments in science. I'm sure you all know this, but you use control conditions that are called the placebo. This would be a neutral thing, for example, so they watch a video of themselves doing random stuff. Nothing important. We had these videos for a whole week. Guess what happened? The

people with OCD symptoms had a 20% reduction in their OCD symptoms, in their symptoms after that week. Only one week.

4.70 And something called cognitive flexibility, being able to be flexible in your thinking, is a cognitive trait that is mediated or is processed—using fancy terms—here in the frontal lobes. In the frontal lobes, you have cognitive flexibility. In OCD, by the way, flexibility is out of the picture. No flexibility. You're having these thoughts bombarding you. We were talking about the orbital frontal cortex, the anterior cingulate, and the basal ganglia, this glue with no flexibility. It's just going around. "I'm contaminated, wash my hands. I'm washing them for five hours. Nothing's happening." They lack flexibility of thought, cognitive flexibility. We saw that go up by 40% after one week of using this.

4.71 This study was done at Cambridge with my colleague, Barbara Sahakian. A great study, of course, but it was done in subclinical people. You know what subclinical means? It means that they don't have the full OCD, but they have symptoms of OCD without meeting the full criteria. We usually start with these patients, these people. They're healthy. They're not patients, actually, because we don't want to hurt the patients. Initially, we don't want to do things that might have damaging effects.

Lecture 5: Memory and Genius

At a glance — Memory and genius.

- 5.1 Well, so this morning, I was like, what should I wear today? And I thought, you know what? These guys are heavy, so I may need to wear something heavy for them. So I'm wearing a suit. I can't just look like a slacker with your knowledge, the things you're grasping.
- 5.2 Which law am I tapping into right now with this suit? I just want to review a little bit. Contrast. Absolutely. There's a bit of contrast here, like the lions in the bushes, and I get an 'aha.' There's a lion. Or there's a gazelle. Contrast.
- 5.3 Any other interesting laws? Harmony, yeah, there's hopefully some harmony here. Yep, absolutely harmony. Definitely outlines. It's not all over the place; there's some outline, definitely.
- 5.4 Grouping. Did I hear grouping? Yes, I heard grouping. Grouping, okay. There's grouping, like the lions behind the bush. You see parts, and then you go, 'aha.' Or that's the mate over there, 'aha.' That's the meat—that's the meal, the next meal. 'Aha.' That 'aha' gives us a dopaminergic spike, a surge that the brain likes. The brain likes that.
- 5.5 In fact, do you know when—I don't want to be too—I have to be careful because this is a formal class, but if there was a woman over here and she was, let's say, naked. She was naked over there. Would I prefer her—would a general man prefer to watch a woman completely naked or partially naked? Or a woman watching a man. Partially? Why? Why would a man prefer to look at a partially naked woman versus a fully naked woman? It makes no sense. Your dopamine neurons should be more active if she's completely naked, correct? In theory. But why is this the case? A perceptual problem-solving goes back to the lion behind the bushes. The splotches, you put them together, and you go, "My God, this is a lion. My God, this is a gazelle. My God, this is a giraffe. My God, this is this or that." Your brain likes that 'aha.' It likes to problem-solve, so you can get that 'aha,' and with that dopaminergic spike you have, plasticity occur as well. "Oh, my God. I know where to look for mates in the future." And so when we translate that into just it's a pin-up or something like that, you actually—your brain prefers that partially naked woman.
- 5.6 Your auditory system is the same thing; you love solving. If you hear something that is partial, the language is kind of cryptic, and you have to sort things out, it's the same thing: you get sort of a dopaminergic 'aha' from that. The auditory system, hearing, is not that different from seeing. Well, it is very different, but it has many of the same wirings to the dopaminergic centers. Take music, for example. Music affects your whole brain; your dopamine system is just ramped up from music, and it turns out— and this was actually one of my colleagues, I haven't double-checked this, but one of my colleagues, I trust him, his name is Paul Smith at UC San Diego. This guy has written a

lot of books about brain chemistry, so I trust him. He told me that there are fewer synapses, or we know there are fewer synapses going from the auditory, from the cochlea, the microphone inside the ear.

- 5.7** So, basically, here's the setup of the ear. You have the microphone inside the ear, the cochlea, there are about 3000 different cells. They become active when there are different frequencies. It goes to the auditory nerve, and from there, it goes to something called the inferior colliculus, the medial geniculate nucleus, and then it goes to the auditory cortex. Is that clear? Medial geniculate nucleus, inferior colliculus, cochlea, and so forth.
- 5.8** There's this pathway, and from there, it goes to the auditory cortex, which is in the superior temporal gyrus over here. That whole pathway, nevertheless, is long. It has a few stops, like a few stops on the train, but it's still fewer stops if you want to go from the auditory cortex to the amygdala, the emotional part of the brain. There are still fewer stops than in the visual pathway, which has to travel all the back to the visual cortex and then split into these two pathways. Remember? It's still a bit shorter, and because of that proximity, music evokes more emotions than does looking at a painting.
- 5.9** Although we said da Vinci did evoke my emotions, so there is still some emotionality when you look at a beautiful Michelangelo painting. But music or auditory stimulation, which is converting sounds into electrical signals, going perhaps a few synapses earlier—synapses are the little gaps between neurons. Fewer synapses reach the amygdala—not the amygdala, the emotional part of the brain septum and dopaminergic structures. Is that clear? Am I being cryptic?
- 5.10** Are you getting what I'm saying? That is very important. That's why I'm here. I want to make sure I'm not just throwing out these Latin names, but you're actually getting what I'm saying. Right, Cassius? Good. Very good. So, we're getting the whole pathway.
- 5.11** Now, it is interesting that the brain can explain so many things. Things like this, my God, like music and why it's more emotional than a painting; that gets into some heavy stuff. And then, you can say, what about, have you ever walked somewhere, let's say, in Miami, or maybe you visited, I don't know, New York or something like that, and then you smell a perfume, a certain perfume, and you are reminded about somebody that you used to know in the past? Well, this could be maybe your ex-wife or it could be your mother, it could be anyone, but you have emotions coming, a lot of emotions from smell. We know this from the perfume industry; It's exploiting this and making trillions of dollars by all these smells that evoke emotions.
- 5.12** Yesterday, remember, we talked about how smell goes to the brain faster? It bypasses the thalamus. Remember that? Because it had an advantage in terms of survival. So, the only sense that has that gap missing and goes directly from the olfactory bulb to the peripheral cortex and the amygdala, the emotional part of the brain, is the smell.

That explains why perfume evokes these emotions. Is that true? Do you agree with me about the smells? Very good. Very, very good.

- 5.13** Now, the human brain also has— we generally take in the big picture, the Gestalt. It's called the Gestalt. The overall picture, that tendency we have goes back to the attentional system. In fact, there's this condition called change blindness that shows us that when we look at a feature of, let's say, an image over here, and there are a lot of things in it, but there are, let's say, there's two people conversing, two people chatting in that picture, but there's a big painting in the back. Now the picture is removed, and you ask, "Ballard, what has now changed in the picture?" You say, "Actually, I don't know because I was focusing on these two people, taking in the overall picture, and focusing my attention on these particular people, so I didn't notice the change." It's called change blindness; it's part of the circuit for attention. There's another feature: we tend to take in the overall picture. It's called the Gestalt.
- 5.14** Now, in terms of vision, there are some interesting studies. So, you have a mouse over here, and each time you show it a rectangular shape, you give it some cheese. "Here, take some cheese." Then, you show it a square, and you don't give it cheese. You don't give the cheese in this case; there's no cheese. Again, you show it the rectangular shape. Cheese. You keep doing that. Now, guess what? No cheese; which shape does it prefer of the two? The rectangular or the square? Rectangular, right? It gets happy; it gets excited. Guess what? Now you show it an even more rectangular shape, even more rectangular. Which one does it prefer? The more rectangular one versus the one that provided all the cheese, that led to all the cheese. Now, what is that about? Why might that be the case?
- 5.15** It learned a rule, the rule of rectangularity, that cheese equals rectangularity. That was the rule, the algorithm, the thing that the neurons learned. It didn't care about the actual shape, but it knew that the more elongated it was, the more likely it was to get the cheese. Just like the small birds, it learned that rule. So here's a rule: rectangularity. And so, the brain is using that rule to infer that then that more rectangular means better. It's more, it's better, it's more cheese. Now, let's go one step further over here.
- 5.16** Why do human beings feel that caricatures are so exciting to look at, so pleasing to the eyes, so fun? You know caricatures? When you have Donald Trump, and you paint him, and you make the belly bigger and the hair more orange—is he still the president, by the way? I don't want to get in trouble. So, you make the hair more orange, and then you make the face more round. You amplify Trump's features; you make him more Trump than Trump himself, and you find that extremely funny and very exciting. This is nice to look at; it's a caricature. You amplify the features, and your brain likes this. It's like the rectangular thing. You make it more Trump than Trump. And, of course, you don't know why you find it interesting, but you just kind of do.
- 5.17** Now, this takes us to one speculation about attraction, which is, might it be that early on, when we have our first amorous encounter— I'm not talking about amorous in the

freaky sense, but you might be a little kid, and you might have a crush on a girl in the kindergarten, you find her maybe a little pretty. You like her face, and you like her features for some reason, and so you like her; you have a crush on her. It happens. It has happened before; don't hide it. It has happened. Okay. Okay, very good. So, you had that little crush there. Now, your brain has created a template probably. It said, "Okay, I find these features attractive." It was probably paired with some nice personality features, and overall, you felt good about yourself; you felt good about the whole thing. You were happy when you were with her, playing in the sand. And so, her features became wired in your brain.

- 5.18** Now, later on, you've forgotten about this little Lisa you were playing with; you've forgotten about all that. You're older now; you're ready to get married. You're ready to get out and find a woman and get a wife and all that stuff, perhaps, and you find certain females attractive, only certain. And people might say, "Well, how come you find this girl attractive?" Or, "How come you find this one attractive?" It makes no objective sense, perhaps. She doesn't have to be a model or something like that; you just find her attractive. Do you agree that we have certain personal preferences in our attraction?
- 5.19** So, might it be, and this is speculation, but it's very plausible, based on what we know about the brain, that some of these people that we are attracted to are templates of earlier versions that we were attracted to as a kid, early on when we laid that template. And so, we had those features, but perhaps they're now amplified versions, like the caricature of Trump. They have more of that pointy nose or more of that bushy eyebrow, oh, sorry, that's me. They have more features that are perhaps more amplified than the original template, so you find it more attractive. See what I'm saying?
- 5.20** Now, this tells us that perhaps this is how attraction occurs. There's some early imprinting, meaning there's some early wiring in your brain. When it's very plastic, that early critical period, you have that template, and then that's how attraction evolves and also fits in. So, now we have a more specific, detailed version of attraction versus the attraction yesterday we talked about that human beings tend to have an innate attraction towards the human shape. Is neuroscience fun? What do you think, Alicia? All these things we can explain. Absolutely. That's why I got into this stuff. Very good. Okay.
- 5.21** I think that we did talk a lot about plasticity. Oh, let's jump over to memory because you want to know how I did that trick, don't you? How the hell did he do that? I'm no genius; I'm not smarter than the average. It was—well, one thing is that you can practice your memory, that's for sure. Actors have profound memories. They can look at scripts and just talk. I have all these notes. Look at me. I have all these notes. If I was a savant, I wouldn't need notes. But, basics of memory—there are two systems overall. There are two forms of memory overall. There's something called episodic memories—

sorry, sorry, sorry. I'm jumping ahead. There are episodic memories, but that's another form. I'm going one step. See, my memories are not being good to me?

- 5.22** We have something called procedural memories. They're very, you see them in kids. These are automatic, like learning how to crawl, learning how to ride a bike perhaps, or do certain things that are automatic, like going upstairs, up some stairs, and stuff like that. All the kind of automatic things. These are procedural memories, and they are processed in the—oh my God, you got it. There you go: cerebellum. That's right. So, these are the procedural memories, and they tend to be, as we said, very developed in kids, up to two and three; they become quite developed at this point. So, this is the procedural memory system.
- 5.23** But then, you have another memory system, which is called explicit declarative memory. Explicit or declarative memory. It has both names. These are memories of facts. I know I was here yesterday. I know that bananas are yellow. I know this. I know. It's facts about the world, about events and things like that. These are called explicit memories. These are very explicit and declarative.
- 5.24** Now, this is where I went with the episodic memory because then you can break it further down and say, well, there's another version which is called episodic memories, and these are a form of the declarative ones where you remember what you did on a particular day. I was eating ice cream when 9/11 took place. I was in this place. This is an even more different type of memory because now you have an emotional component, and emotions make you wire memories better if there's an emotional component. Why is this the case? The amygdala, what do you think its neighbor is called? The hippocampus: the memory part of the brain. It's the hippocampus.
- 5.25** We didn't even talk about the hippocampus, but the hippocampus is very interesting. The hippocampus—do you know what it's doing? It's taking memories from the short-term and placing them in the long-term, in the long-term state. So, you need that for that transformation to occur. In other words, I am meeting Farah for the first time. Let's say I'm meeting you; I have this experience. I have all this perceptual activation occur. I hear your voice, I may shake your hand, I may have all these various senses activated in my cortex. Maybe you say something nice like, "Oh, I like your tie," for example. I get a bit happy, right? So I have my emotional structures become activated that are beneath the cortex.
- 5.26** There's a whole orchestra of things: activation, activation. All of that is sent, then, to the hippocampus, which means seahorse. It's a little jelly roll- shaped thing, a seahorse-shaped thing, there in your limbic ring in the center over there. And it takes and says, "Well, this is important, let's turn this into a long-term memory." And, of course, the more emotional it is, the more salient it will be, the more detail you will register from the event because you have more attention, you have more acetylcholine, there's dopamine; there are all kinds of things going on. So, you have more long-term plasticity occur in this case. So, you have the hippocampus doing that, and then what it does, it

sends the information back to the original place and stores it over there. The hippocampus itself doesn't store the memory, but it simply says, "Okay, now you can store it in the original location."

5.27 So, the next time I'm closing my eyes and I'm thinking about my encounter with Farah, the same centers in my brain will become active as when I met her the first time. What I heard. The same centers in my auditory cortex, visual cortex, and so on, the amygdala, same. Is that clear the whole process? So, memory and actually experience, real experiences activate the same places of the brain. It's like having the same experiences, but this time inside your head, and the hippocampus is key.

5.28 Now, what would happen, Mani, if we remove the hippocampi? There are two of them. Hippocampi, on both sides of the brain, we remove them surgically; we don't want them anymore. What will happen if you remove the hippocampi on both sides? You cannot make new memories. You can stay in conscious awareness for about 30 seconds. Kathy, I can tell you the same joke over and over again, and my jokes are really bad. You would find it funny each time because you wouldn't have—you couldn't translate that into a long-term memory. Is that clear? Or if I'm married to somebody, I see her as if I'm seeing her for the first time each time I see her because there's no way for me to register that event. Is that clear? Or you can introduce yourself to this person the whole night, over and over, and it would be like the first time.

5.29 There's actually a case in psychology like this, in neuroscience. This might be the most famous case in neuroscience. Poor chap, he may have passed now, but his name was H.M. You have heard of him perhaps; it's a very famous case. He had his hippocampi removed surgically because, at that time, they didn't know that memories are stored there. You need that to store memories. So, they removed the hippocampi, and when they did that, he couldn't store new memories. His life was like every day was new, new experiences all the time. He was very open to the scientific community, so some of my colleagues would literally stand and introduce themselves the whole evening when they were talking to him. He'd forget. And he actually had this when his wife would come in; each time, he would go, "Oh, are you so beautiful," like he hadn't seen her for 30 years. Isn't that pretty interesting? Poor chap, but what can we do? Memory.

5.30 So, how did I do the little trick over here? Well, there's something called semantic memory as well. So, there's episodic and semantic. They're part of the declarative memory. So, the declarative memory. The semantic version is simply knowing that, as we said, a banana is yellow, for example. An animal, would it have semantic memory? Does it know a banana is yellow and a nut is brownish? Does it know this? It does, so it does have the semantic. Does a monkey have episodic memory? Does it know it was playing with some other monkey two days ago? Is this possible? Actually, they don't. They don't, and this was pointed out by Dan Schacter, a memory researcher, that they don't have episodic memory because you need a self.

- 5.31** You need an I, Amari. You need a self to anchor that memory. If there's no real I, monkeys are not conscious that way. They don't have names, they don't have your bank account and identity in that way. It's simply, "I exist in the present moment, and I'm gonna avoid pain and gain pleasure." Pretty much. So, they don't have episodic memories, but they do have semantic. They know facts about the world.
- 5.32** Now, in this case, when I did the memory demonstration, what I did was I took this information from short-term memory, from the working memory, and what I did was, in my mind, I placed the information in places I already knew about. So, I had places that I could imagine; it could be a childhood home, for example. You would imagine that, and then you would place the countries using certain associations. So, for Argentina, you can think about Messi, for example, and you think about him in a particular location. In that way, you can retrieve it very quickly.
- 5.33** Now, it's not a trick that you can just do. You need some practice and training because you can train your memory centers in the hippocampi and all that to do that. But the reality is that you can actually train your brain to do this very quickly, and I used to do this all the time. It helped me a lot in school and things.
- 5.34** Well, I can—so, here's the thing. You might ask yourself the following question, "What if you have this skill or this ability to memorize things because you have OCD?" Would it be a bad thing? Well, it turns out the OCD brain is kind of messed up, to be honest. There are hyperactive loops, and these hyperactive loops detract from your life. So, when you have OCD, and you remember things all the time, it comes at a cost.
- 5.35** Things in the brain are—if you become good at a certain skill, it can often mean another skill will be worse. If you become—so, event registration of this nature is the left hemisphere. Certain people with damage to their brain, so let's say that the right hemisphere is damaged; they become idiot savants. That's the name. These guys can remember everything just like the OCD folks, but it comes at a cost. But then you ask them, "Oh, what is the main theme of Shakespeare's Hamlet?" They go, "I have no clue. I don't know." Okay. "What was the main theme of our conversation?" "I don't know." "What is the main theme of this or that?" No. Why? Because the big picture stuff is registered where? In the right hemisphere, generally. That's where it's processed.
- 5.36** So, the brain has this balance between the hemispheres, and if you mess that balance up, what happens? You get overdrive in this hemisphere over here; it becomes overactive. You don't want that. You don't want an imbalance. OCD. Too much dopamine, usually. And then, OCD, the features of OCD, the overall architecture of an OCD brain, is hyperactivity in some of these loops, the motor loops, washing, and all that. But it rarely is one thing. Even though you might be able to memorize all these things, then in other spheres of your life, your OCD tendency will leak over. You become rigid about simple things. So, no, you don't want to be—having a human brain that can inhibit information when necessary is a very good thing. It's a very useful thing.

- 5.37** Think about dreams, for example. Every night, we dream. We sleep about eight hours, hopefully around eight hours, you sleep. In these eight hours, you pass through REM sleep, and we'll get to this tomorrow, and during REM, you have these dreams. These dreams are like blockbuster Hollywood movies. They're very, very, very vivid and crisp, and everything's all over the place. But it seems like you have an inhibiting mechanism, inhibiting these dreams from reaching memory. So, when you wake up, you won't be overwhelmed by all these dreams. The fact is that the little bit you remember is only a fraction of the dream in the morning, the few seconds before you wake up in the REM stage. Inhibition because you would be overtaken by all these emotions, all these dreams of Shakespeare on the moon, and all that stuff. You don't want that, you would become crazy perhaps, if there's too much of that, you see? So, inhibition is a good thing.
- 5.38** You have people with frontal dementia or damage to the left hemisphere. Bruce Miller at UCSF has studied these guys. They can no longer speak if it's in the temporal lobes, the Broca's area. Remember Broca? Broca aphasia: the ability to speak. Syntax: Wernicke. So, they can't speak now, but they start painting like Michelangelo. The details are impressive, striking, unbelievable. You have people becoming Da Vincis overnight from these—.
- 5.39** You might ask, "Why is this the case? Well, we said the left hemisphere has damage over there. The right hemisphere has that art module in the brain; it seems to be the inferior parietal lobule, particularly in the right hemisphere. The inferior parietal lobule. We talked about the superior parietal lobule yesterday. Well, the inferior parietal lobule, as the name might suggest, is right beneath. But here, that part of the brain is useful for conjuring up images of things and creating vivid imagery inside yourself. In fact, if you have a stroke over there, you won't be able to dream. Dreaming goes away. So, it's very, very critical for conjuring up images, the inferior parietal lobule. But it seems to be an art module in the brain.
- 5.40** Like, take this case. There was this case. Her name was Nadia. She had autism, a form of autism. When she was painting, she was drawing these hauntingly beautiful, striking images of horses and various animals. Now, these images, there's one of her images, and it's beautiful. It's this horse. It's leaping out of the canvas. You can just see the horse's muscles, its tail, its head. It's in movement, so it looks like it's moving as if she was recruiting movement parts of the brain, the MT, and it's wonderful. And then, you compare that with a painting of da Vinci. You compare it to a da Vinci painting. You know what? When you pull audiences, and you say, "Which painting is more beautiful?" They will point to Nadia's picture and say, "This is more beautiful." Even though she's idiot savant, an autistic child, eight years old, who can barely speak, who can barely tie her own shoelaces. She will draw more beautifully than Da Vinci himself, who spent his entire life painting and drawing, and doing all kinds of stuff. He did a lot

of things. He was the top man of the Renaissance, this is DaVinci guy. He was born on April 15. I don't know the year, but he was born on April 15.

5.41 You know, Da Vinci had this problem: he couldn't finish tasks. Perhaps he had OCD. He was very good. He was very good with details, but he kept procrastinating because everything had to be perfect, and perfectionism is actually an OCD trait. So I wonder, actually, whether da Vinci, you know, in his obsessive sort of focus on various details, had a form of OCD where he was—that perhaps made his talent, in a way. If he was just a regular person without this kind of OCD, he wouldn't be able to draw these beautiful paintings, for example. But this paradox about Nadia, isn't it interesting how she did that? Yeah.

5.42 Now, this really gets us to something I want to tell you. This is on my mind. It's very important. It's intriguing, an i ntriguing thing. Have you heard about Ramanujan? Sounds familiar? Ramanujan? Have you seen "Good Will Hunting?" Can we make commercials and advertise movies over here?

5.43 Now, there is this guy, mathematician Ramanujan. He lived in India, southern India, Madras, Chennai, these regions. He was very poor. Isolated as a child, he had no access to math books, libraries, nothing like that. No access. But he would sit in his room and just do math. How? Nobody knows. Math, math. S cribbles, scribbles, scribbles, scribbles. He was bad in school. Very poor—he would perform very poorly, wasn't good at English grammar or history or anything like that, only math. He would do this over and over.

5.44 Then, at one point during his—I believe, it must have been his teenage years—he sent letters of his math to various professors in the world. He did that, and people would say, "What is this crackpot? What the heck is this? Throw it out." But then, it landed in the hands of the leading mathematician of the time, G.H. Hardy. He was a professor at Cambridge at the time, and he was doing his math, and he looked at these scribbles and said, "What the heck is this? Strange. I've never seen anything like this. Bizarre stuff." He threw it away first and then went and played some tennis. Then suddenly, it occurred to him; he had the following thought: "If these scribbles were, in fact, not true, none could have had the imagination to make them up." This was his insight: if these scribbles were not true, none would have had the imagination to make them up.

5.45 He went back home quickly and double-checked the scribbles and the math, and it turned out to be right. Bizarre, strange math. Nobody had seen that before. So, he hurried up and invited this guy to Cambridge, Trinity College Cambridge, one of the Cambridge colleges. He brought him there, and together they made groundbreaking stuff. In fact, Ramanujan made groundbreaking stuff that is used today for things like understanding black holes and stuff. And he lived for a very short time. He actually died very, very early. He was there for a short time. Apparently, he was a vegetarian, and he couldn't get good food in England. So, he actually died very young, but he made profound ripples in the scientific community. Shocking, shocking.

- 5.46 They asked Professor Hardy. They said, "Hardy, on a scale from one to 100 of inborn talent, where do you stand?" He said, "I'm about a 20." "What about Ramanujan?" He said, "He's a 100." How is this even possible? You tell me, because I have no clue how this happens. The brain is complex. It's mysterious. Now, one clue might be, and this is one clue because we've got to have some clues here; otherwise, we can't give the lecture. So, one clue might be the following. We set inhibition of structure, so if the prefrontal lobes inhibit the part of the brain that's important for talent, for example, that goes away. The natural ability of the brain to conjure up images, like in dreams, comes to fruition. You can use that talent because it's no longer inhibited by the conceptual parts of the brain. Or the left hemisphere is no longer inhibiting the right, if that is the case in this scenario, if that is doing the inhibition, or vice versa. Normally, the right hemisphere, if it's holistic, now you have the damage there, and you become a left-brained savant that can calculate numbers. Is that clear?
- 5.47 But, the principle of isolation is also important. In fact, when you have a damaged region of the brain, let's say the left hemisphere, you also allocate all your attention, all your focus, to that region. So, you only use that all the time. So, somebody that has this savant ability will just draw all day, or the math person will just do math all day because there are no other modules to balance the brain. Overall, there's an isolation principle over here. And what happens when you keep using that part of the brain? It grows. It becomes larger, right? So, yeah, this might be an explanation for that. Very good. Very good.
- 5.48 So now, we shall proceed. Did I tell you about the spinning disk study? The spinning disk study. You should know about the spinning disk study because it's also a good example of this, of the brain coming larger. Now, if you—this is what we call a seminal experiment, a crucial experiment that was done. So, they had monkeys learn how to spin a disk, just like the mathematician doing it over and over and over again. It would do the spinning of the disk, and after about a month of doing this, the brain areas became larger for the spinning disk, the motor regions became bigger. Just like studies showing that people that are musicians, the finger areas of their brain that they play the instruments with become larger, their brain. And the longer they play, the bigger their area becomes. The bigger it becomes.
- 5.49 Do you know focal dystonia? This is a condition that some guitar players have. They move—they want to move this finger, but then this one moves, too. So, I go like, "I want to move this. Don't do this. Don't do this." I do like this, "Don't do this." Now, based on your knowledge of neuroscience, why might that be the case? Absolutely. Neurons that fire together wire together.
- 5.50 So, they use these so much for the guitar or something like that. Do you use two fingers for the guitar? Oh, the cello. Sorry. Okay, but let's say you play the guitar, all right? So, you probably use two fingers, but you have these fingers wiring together, and then every time you use this finger, you also use this one. What's the trick to getting out of

this? You use one, but you can't now because you're kind of stuck, you see. It's a good point; that's the logical thing. But how do I actually get out of it because—what's that? Tape it, you said? Well, actually, here's the thing. This is what people who do rehab here do.

- 5.51** So, what they will do is they will say, "Actually, maybe what we can do is have the person not play the guitar for a while." What happens then? The map for playing guitar, that map that is now merged between the two fingers, becomes a bit weaker over time. Use it or lose it; it becomes weaker. And then, slowly, incrementally, by reinforcing the person and giving them rewards to teach them to slowly move this alone. And then, you move this alone. And then finally, after some time, they start to uncouple the dissociation over there. Does that make sense?
- 5.52** London taxi drivers have larger hippocampi than the general population over in London. Did you know that? Ismael, did you know? Lydia knew. Okay, cool. It turns out that the longer you've been a taxi driver in London, that's where the study was done, the bigger your memory part of the brain, the hippocampus. In the hippocampus, by the way, there's another type of cell called place cells. Place cells are important for orientation in space, and preferentially, the right side of the brain does more of the computation to do with spatial navigation we talked about, but there are place cells in the hippocampus as well, telling me where things are in space.
- 5.53** Now, people with Alzheimer's, if you get Alzheimer's, God forbid, there's an erosion of the hippocampus. In certain cases, that's what you get. In Alzheimer's, you get erosion of the hippocampus. Now, have you noticed that people with Alzheimer's do not know where things are in space? They might get lost. But you ask them to remember some childhood event, cortical regions, they're stored up there, and they know what they did on a particular day 30 years ago. That's also important. Very good. Very good.
- 5.54** There are all these techniques to rewire the brain and really shape it into adulthood, really changing the brain. Meditation is yet another example. The insula part of the brain, regulating your visceral system, is larger in people that meditate, for example. So, just as an example of that. Even the CIT, the therapy I talked about before with the CI therapy, even that therapy, when you look at the brain—so, not only do they actually become better after using it for those weeks, the two weeks, but when you scan the brain, you also see neuro rewiring. So, you'll see typically something with the prefrontal cortex— the premotor area actually overtakes the motor area as the new motor area, in other words. So, before, it was only doing planning. No execution of actual behaviors, only planning them, but slowly, it would take over in those weeks. So, that's a neighbor takeover type of plasticity.
- 5.55** But then, you also have things like mirroring where—and this is very, I think, drastic because then you have the other side of the hemisphere taking over, too. So, the other motor cortex. So, if your damage is over here and this arm is barely moving, then the motor area of the different brain hemisphere will do the job for both hemispheres. Is

that interesting? So, there's mirroring. It says, "Look, buddy, I'm going to just do for the whole brain," and the original function of this arm is still doing its job, but it now knows how to do both hemispheres.

5.56 So, just like a child, we talked about earlier that children—babies use both hemispheres for a lot of things, and specialization only comes around year two, when the modules become more specialized. This also holds up for old people. When you get into the eighties or nineties, you'll find that some people, to a greater degree when you get older, you don't have the same—you don't use these various parts of the hemispheres in a lateral way, meaning that you don't use the left for this or right for that. It becomes more blended together because you depend on both hemispheres. Is that clear? So, that modularity of function diminishes with age.

5.57 So you may, when doing math, instead of relying on the math module, the angular gyrus, which we said was larger in Einstein, rely less on that region and more on the frontal lobes because you have to focus; your brain is atrophied. Of course, with age, your brain becomes smaller and less functional, but you can fight that with practice and things like that.

5.58 But, yeah, it does really beg the question, "Where does genius come from? Is there such a thing as a genius?" It seems to me that some brains are different, but do you have to have this damage or pathology in order to reach genius, or what we would say, a brain that is super functional in certain areas? Do you need to have that imbalance? Is that necessary? That is the question. People have actually considered using TMS. So, we talked about transcranial magnetic stimulation yesterday, which is a kind of stimulation. You zap the cortex, and you can sometimes—what you can do is you zap the brain in a way to inhibit the function of a particular hemisphere or a particular region of the brain. And so, if you do that to the prefrontal lobes, let's say, and those lobes are inhibiting the painting da Vinci module in the brain, well, can you then have some of these genius abilities occur in all of us, in you and I, for example? Is that possible? That is the question.

5.59 I believe there is a chap in Australia that has some data in favor of this, but I remain highly skeptical. But it could be possible, perhaps, if you inhibit the part of the brain, let's say the prefrontal, temporarily, you create the da Vincis, the paintings, and then you take it away, and perhaps, some of these abilities would stay on because you, particularly if you do it over time continuously, to train that region of the brain, it's possible, but now I'm going a bit on the fringes over here. This is not mainstream stuff, but we have to think aloud and speculate in science.

5.60 But what about the pruning process? What about the fact that when you have 50% more synapses in the baby brain, and these synapses are gradually pruned away, is it possible to harness the power of the brain at that point to make a genius? What if you were to take a kid and train him every single day: put him on the piano, give him math,

and make sure that his brain doesn't get rid of all these various pathways? Would that be possible, Mani? What do you think? Possibly, right? Yeah.

5.61 I believe there's a Russian couple, psychologists, who did this with their child. I think he committed suicide, eventually. They overtrained him. I remember. I think this was the case. This was a case I learned about many years ago, so I haven't actually fact-checked whether this—but this might be an outcome, for example. This might be an outcome if you train your child. He became very smart; he became a genius, but how the case unfolded, if he really— he wasn't tested objectively and all that. So, yes, but early stimulation, nevertheless, is very important.

5.62 If you look at mice and rats who are stimulated early on, so you stimulate their brains very early on and sometimes even later into their lives, you give them the running wheel, you give them various forms of stimulation—the running wheel is really the key trigger; the running wheel is very important for them. It stimulates their brains a lot. Like walking, what that does with the dopamine centers, it's profound, profound. It makes a lot of changes. In a month, they may have 15% (one five percent) more synapses, for example, after a month of just going around. So, there's early stimulation in that way.

Lecture 6: Brain, Body, and Belief

At a glance — Brain, body, and belief.

- 6.1** Well, you have so many questions. So heavy questions. It makes me a bit tired. All these angles on the brain you want to know about. It's heavy, even for me, to study this stuff. Do you study neuroscience? It must be. You have very good questions.
- 6.2** I actually got a very good question yesterday, which was, "How come some people freeze?" Let's say there's this lion attacking me, or I'm in a jail setting, or I'm in a park, and somebody runs at me. Now, this guy in the park runs at me with a knife, say. How come some people freeze? Actually, how come it is very common to freeze in such situations? It makes no sense. Why would you freeze when the person attacks you and wants to take your wallet over in the park? You're not supposed to freeze, right? You're supposed to run, but many of us actually freeze.
- 6.3** This is a question that is puzzling. But in many ways, it's not that puzzling because when that lion attacks us—early on, I'm meaning the Savannah guy, when the lion came, it made sense to actually play that you were dead over here, that you weren't alive; that you weren't an animal, but that you were a dead body so that it wouldn't eat you and chase you. I know it sounds counterintuitive, but animals actually get more excited when there's a moving object. Vision responds to movement; its attention is drawn towards it. So, playing dead over here is helpful. Does that make sense?
- 6.4** Basically, what happens in the brain is that you have the hypothalamus, the region of the brain that controls your hormones can shut down its outputs, so the amygdala, the emotional part of the brain, can communicate with the hypothalamus, shutting down its outputs. So you are, in a sense, playing dead. You go into a dead body situation. Is that clear? And that's why people nowadays would do that. They would go and freeze instead of running off, which would make more sense because a human being might not act like a lion or a tiger.
- 6.5** But more than that, actually, a lot of people that are raped, for example, women, would say, "When I was being raped, I was lying there and I was seeing myself from the outside. I was being depersonalized. I wasn't really there. I was just an object watching from the outside." A lot of people with trauma would say this. It's called deep personalization or a fragmentation of the self.
- 6.6** It may very well include some of the temporoparietal regions because when you zap them, when you disrupt the region, you will feel like you're floating out in space. Your self is detached from the body. And so, when you are lying there and being raped, it is easier to process, perhaps, by being objective. Now, if you are there and the self, Amari, the self is there and the self is being pretty much destroyed by this person, then it's easier for the whole sense of self, the ego, the I, the personhood, if I were to, in this temporary situation, look at the whole thing from the outside. And perhaps play dead,

too, so you have these circuits going on. So, shutting down the outputs of the hypothalamus, the hormone factory of the brain. Does that make sense? Is that a clear explanation for it? Very good.

- 6.7** Now, I see this a lot, actually, in sleep paralysis because it's similar to a rape scenario when you look at the brain. People will, actually, when they're lying there, paralyzed, unable to move or speak, and there's this demonic figure coming and attacking them, trying to kill them or experiment on them, they will often say, "Well, actually, I felt the whole thing from the outside. I was merely watching. I was merely observing as the tiger. I literally had this case of a person telling me, 'A tiger was eating my body, and I was just watching the whole thing.'" It's creepy, but it's an example of how the brain works. So, in trauma, you have a lot of this depersonalization, the self being fragmented. So, the self is unified. It can be unified by episodic memories. It can help you unify the sense of self. But you can also become fragmented in certain scenarios if it has an advantage, an adaptive advantage, for your survival, for your sense of self. Is that clear? Okay, very good. The mind is intriguing. Its limits are few. I mean, there are limits, but it's intriguing.
- 6.8** Think about this. They did a study where they looked at people. They invited this group of people and had them play the piano. They teach them the notes; they play. These guys have never played the piano before, I mean in an official manner. So, for two hours. Then, they split the group into two. One group goes over here and continues for five days, two hours a day, to play the piano and hear the notes and everything. They hear the notes, and they see everything. So, a total practice. Very good.
- 6.9** The other group simply goes home and imagines playing the piano and imagines listening to the notes. Now, five days later, they call these guys into the lab, and say, "Look, let's test you guys. But first, let's measure the activity in the motor cortex." They measure; they map the motor cortex: the same activity as if you've been playing the whole week. Then, they say, "Look, let's show us what you can do; show us your skills now." It turns out they're almost as good as each other, meaning the groups are identical in their ability to play the piano. All it took was two hours for the imagination group to catch up on the actual practice group. Does that make sense?
- 6.10** Let's say, Cassius, you're in a certain profession. You're a football soccer player. You're playing soccer. Now, might this have implications for you? Just imagine yourself playing and how that might improve your skills. And all it took was two hours. You didn't need a whole piano, no notes, no sound disturbing your sister. None of that. Just imagination. The motor cortex is the same, and only two hours, only two hours to become as good as the real group. I mean, that's the matrix, if you ask me. It's a profound experiment.
- 6.11** Now, an even more striking study. They had a group exercise their fingers for a whole month, five times a day for a whole month; exercise them. Do a certain contraction, lift a certain thing. A month later, the experiment is over. A thirty percent boost in muscle

strength. Thirty percent! Another group merely imagines the whole thing. Pure imagination, sitting at home, not going to the gym, not doing those reps. Sitting at home, and you got about a 22% improvement in muscle strength. Twenty-two from merely imagining sitting there and imagining you doing these finger movements and hearing the sound of a guy saying, "Come on, you can do it. Come on, you can do it," you have 22%. What is that about? Mani, is the brain mysterious? I'm saying, imagine. You're not doing anything. You're not going to that gym.

- 6.12** It makes no sense except when you are imagining a behavior, you're actually activating the motor cortex to a lesser degree, but you're still activating it. So, you're sending those signals to move, but they're being inhibited because the signal is very, very weak, but it's still sending those signals. So, when you are sending the signals, you are, in fact, practicing by merely imagining. Do you know that? When I close my eyes and let's say I dream right now, I have a dream, and I see myself running in my dream, I actually activate the motor cortex as if I was running, but this movement is being inhibited by glycine and GABA in my spinal cord, in the brainstem. Got it? Think about this. Imagination. This is mind-over-matter stuff.
- 6.13** But it's not only these muscle contractions and things that can improve in this way. I'm fascinated about the whole point of how your mind can influence mood in a way that is more—I let's say you're a person that is very stressed. You have a lot of stress, you're unhappy. You're an unhappy guy over here, Jose. You're not very happy. Now, it turns out people that are unhappy, unhappy people, if you measure a group, you take a group, and you take the most unhappy and those that are most happy, and you split them into these two groups, it turns out the people that are the least happy have 50% more cortisol in their circulation. Fifty percent. The cortisol, of course, has all kinds of influences on the immune system and inflammation in the body.
- 6.14** In fact, there's another study looking at people between the ages of 65 to 90. Here they looked at—for two years, we're going to monitor these people and see what happens to them, and then we'll know which ones are the happiest and who are the saddest, pretty much. It turned out the happiest lived twice as long as the saddest people. So we're going not only from merely having stress but death itself. And they control for all kinds of factors like smoking, alcohol, and things like that.
- 6.15** Or another study. They give somebody a cold virus, and it turns out the people—the whole group got the cold virus—and it turns out the people who are the happiest are three times less likely to catch the cold just from being happy. Again, cortisol and stress and all that stuff, the immune system. So, there's a strong link between your thoughts and your immune system, and how long you live. And I love this study.
- 6.16** Actually, this was a very interesting study. There was a study done where they looked at—this is a cute one. They looked at nuns who had lived a long time ago, and they had a record of the letters that they had written, and they analyzed, coding, the words they used. It turned out that these letters that were written when these nuns were quite

young—the more positive words they used in that letter, the longer they lived. It didn't matter how many negatives were there, like F and U and all. It didn't matter. Negative, just use whatever you want in principle. But the more positive, you lived longer. Dahlia, do you want to live longer? I certainly want to live longer. So, here you have a potential mechanism.

6.17 So, we talked about the DLPFC, the left prefrontal cortex, how it speaks to our opioids and opioid centers, dopamine inhibits the amygdala, the amygdala is the fear center of the brain, it communicates with the hypothalamus, the hypothalamus sends releasing hormones, CRH, and then we go to the pituitary and eventually, the adrenals, glucocorticoids, stress in the system. Stress is in the system, okay? Now, what happens when stress is in the system for a very long time? Your receptors change. There's a regulation of your receptors, and now you will have more stress in the system, and a vicious cycle starts over here. You become sensitized to cortisol. A lot of stress in the system. Does that make sense? What do you think, Brooke? Does it make good sense to you?

6.18 Now, on that note, yesterday we talked about the whole—the cortisol, the amygdala, and the hormone system are so intriguing. Now, there's this—if you look, if you have this frowning face—see, I'm adapting to the light over here a little bit. I'm taking these off slowly; you have to incrementally increase the exposure. So, if you have this frowning, frowning muscle, you're more sad and angry, versus if you don't have it, correct? We said that. This is a muscle interaction, muscle-mind-brain interaction. But even things like benzodiazepines and other medications for anxiety, how they work, they relax the muscles. You sit there; dripping down in your chair like you've been in a sauna for three hours. What's going on there? Well, you also feel you can't really be anxious at that point. Are you anxious in a sauna? No, because the muscles will communicate with your brain, and then you have less stress in the system. The amygdala goes down in activity. There's a great muscle-brain interaction.

6.19 Or this one is so fun. Who has a pen? Can I borrow a pen? This one is fun. If you read a comic book, and you have this in your mouth, like this, you will rate the comic book as more funny versus having the pen like this. Can you believe it? Muscles, smiling muscles. Is that nasty now? Do you have OCD? That's the exposure therapy. Exactly, exposure. Exposure therapy, right? You have it like this, you smile. Comic books, "Oh, my God, this is funny." You laugh. If it's like this, don't laugh. Might this work for therapy for depression? What if you watch a video of yourself smiling all day? Could it work? I don't know. I'm just thinking aloud here. You could do the experiment. I don't know. It may have some potential. So, there is this interaction that's so fascinating.

6.20 Also, something that I've encountered a lot in my work is the placebo effect. Now, we talked about the placebo when I referred to my OCD studies. You want to do an experiment. You say, "Mani, by doing a certain thing, jumping, you become good at the certain thing." Then, I give you the control experiment that looks like the original one,

but it's different to make sure that I'm not just imagining the whole thing but that it's actually a real thing. I manipulate one of the variables. Instead of seeing yourself touching a contaminant, you're just doing random hand movements to make sure that it's not merely you watching something on a screen, like hands doing the thing that's making the effect over there.

- 6.21** Now, a placebo is like—you all know the placebo, right? Am I right? You give somebody a happy—you say, "This pill is the newest, hottest thing in medicine. Depression is going to go away right away." You take it; it's actually a sugar pill. You feel better, for example. This is a placebo. Have you heard about the nocebo? It's the evil twin of the placebo. It's that I give you this water over here, and I say, "This is poison from Timbuktu." You drink it, you're dead, buddy. You're going to be over. And you react to it physiologically. There's a physiological negative reaction. You might get sick, for example. That's the nocebo effect. I encountered that in my work a lot.
- 6.22** We said that a placebo is very effective in pain, for example, because there's such a huge psychological component to pain. We talked about how hypnosis might change pain. But even having people in a hospital setting and giving them windows. These windows give you access to nature. Guess what? Less pain medication.
- 6.23** Another experiment. You have people where they can— actually, they have beepers so they can beep whenever they want their pain medication, and they will ask for less medication than if they don't have these beepers, if they're getting medications on a certain schedule where they lack certainty. Uncertainty is one of the factors that causes stress. Control and being able to manipulate your environment can reduce stress. So, by having that freedom, well, less pain medication, presumably less pain. In fact, if you see an IV infusion, doctor, if you see the IV infusion, versus not seeing it, less pain. You need less medication if you see it. So, a placebo is quite intriguing.
- 6.24** So, what about these magic potions? Can a placebo kill, potentially? This was a question that I actually asked myself a few years ago. So, I was doing research on sleep paralysis, which we talked about. I studied it in Egypt, Denmark, Italy, Poland, and various countries, and what I found was intriguing, really. What I saw was that in countries where people have a certain cultural idea about what sleep paralysis is—we said in Egypt, people say it's a genie of Aladin, though not that cute blue one, but a more dangerous, terrifying, vicious one that can potentially kill. Fifty percent of people say these will kill. Terrifying. They can cause trauma in that way.
- 6.25** In Italy, we did studies over there. A similar pattern over there where they will say that sleep paralysis is actually caused by the Pandafecche, a type of witch-like creature, a giant cat. It can transform into a giant cat, attack you, kill you, and do all kinds of things to you. Now, in Denmark, my home country, I did the research as well, and it turned out that in Denmark, people will say, "No, it's just a brain. It's just stress. It's just physiology. There's nothing exciting, no exotic demons coming at you. It's just your brain." So, I thought, "This is interesting." This discovery is quite interesting. But what

does it actually mean for the experience itself? Why is this important? You gotta know why this is important.

- 6.26** So, it turned out that when I compared Egypt to Denmark, sleep paralysis was three times more common. It was three times more common in people who had sleep paralysis versus Denmark. They had it much more, and the episodes were more terrifying. "I saw a ghost. It wanted to kill me. I was scared." It was potentially traumatic. The episodes lasted much longer, or they would say it felt like it lasted for ages. In Denmark, it was just a few minutes. Nothing too bad. It appeared as if their thoughts had changed the experience itself. The nocebo effect, potentially. Have their thoughts changed what it meant to have sleep paralysis? Are they experiencing trauma? In Italy, the same pattern. People would say it lasted longer; they were fearful, "I'm dying from this," compared to Denmark. That was quite staggering.
- 6.27** And then, it reminded me of a study that really was quite something. My colleague over in Boston, Magnali, had done a study years before me. We ended up working together, but years before me, he had people who thought they were being attacked by space aliens in their beds, sleeping. They're lying there, and as they're sleeping, this space alien is coming down. It's probing them. It's taking semen, potentially, and all kinds of eggs and things because it needs to recreate hybrid versions or something like that. It's quite a florid explanation. It will afflict pain to them sometimes, and all these kinds of things.
- 6.28** Now, what they did was—they did the following: they measured the physiological reactions of these people when they were explaining their encounters with the space aliens. They measured their sweating, heart rate, all that stuff, how much they sweat, their hearts, as they listened to their audio tape recordings of them explaining the whole thing. You know what? Their reaction was as profound as somebody with post-traumatic stress disorder, somebody who actually went to war. The difference was, presumably, that these people were just sleeping. Presumably, there was no space alien in their bedroom. Presumably. I don't know. I wasn't there. I can't be sure. If I was a space alien, I wouldn't tell you, of course, but they weren't there.
- 6.29** I'm open to space aliens. I mean, the universe is vast. There could be, but this seems like sleep paralysis. I'm paralyzed, and they were raped, potentially. There was some sexual stuff going on. Well, we know during sleep paralysis, the centers in the brain during REM, the sexual centers, are hyperactivated, and they are called up with the body image parts of the brain. So certainly, it seems logical because you have all these sexual dreams during REM that this was a nightmare of REM-ish nature, right? And we'll get to REM tomorrow, but my point is, they're having trauma when they should be safest at all in their bed, sleeping. You know what I'm saying? Recouping, resting up, all that stuff. Spooky stuff. But yeah, that's an extreme example of the nocebo effect. The nocebo effect, potentially.

- 6.30** Now, I created a theory called the Panic Hallucination Model, and it goes something like this to explain how the nocebo effect can work in these dramatic examples over here. So, you have little Lisa. Little Lisa lives in a faraway country, and over there, in that faraway country, she knows nothing about sleep paralysis, about anything particular at night. But then, one night, she has dinner with her grandmother. The grandmother says, "Look, my child, there is this demon. It comes at night. It attacks you, chokes you, strangles you, may even do sexual things to you, and maybe kill you." This girl has never heard about sleep paralysis, anything about being choked, or anything like that. She goes to bed, and what happens is she overreacts to this story. She overreacts to it, so her emotional centers are hyperactive, and she wakes up during REM sleep and has sleep paralysis. One time. She has it once. "Oh, my God, there was a demon." She projects the cultural narrative into it. "My granny said it looked like this, so I saw this."
- 6.31** It's because the top-down penetration, the conceptual parts of the brain, could influence what? The lower parts, hallucinations, and body maps. So, you hallucinate the creature the grandmother was saying—the granny was talking about. The next day, you have it again. "Oh, my God, it's coming at me." The fourth day, you have it. The fifth day, "Oh, my, this is personal now. I'm being possessed." She goes to tell other people around in her neighborhood about it, in that same vicinity, and they start having sleep paralysis as well, and it spreads like this. Does it make sense? This is sleep. Am I really making sense over here? What do you think?
- 6.32** So, your mind in that way can influence you to potentially develop trauma because now trauma and having some traumatic memories is not merely one traumatic event. But also, if you have repeated multiple traumas, they can constitute an overall big trauma, large trauma.
- 6.33** Now, it's interesting because in some of these cases—there was a group of refugees that came to America many years back. They seemed to die in their sleep from something called sudden unexpected non-eternal death syndrome. They would actually sleep and then die from something that looked like sleep paralysis or something like that. The question is, what's going on?
- 6.34** Or merely, what is going on when somebody gives you a magic potion, or does this curse on you? Which doctor in this country makes this curse? You actually see people dying from these curses. I'm not an anthropologist, but that's what's recorded in the literature: people die a short time after these curses. It's unclear. One explanation could be that the stress system is fired up from this, so much so that the vessels to the heart constrict, burst, collapse, low blood pressure, and heart attack. Something along those lines. What do you think, doctor? This is what they say could cause death in certain instances. From your mind, thinking that you're being possessed or there's a magic potion or death is already around the corner.
- 6.35** So, these are examples of the placebo and the nocebo, how your mind can influence your brain and your body. I just want to make it clear: when you talk about the placebo,

it's not merely through the stress system or opioids and pleasure systems. I mean, it's also through behavior. When you think about it: when I say this thing will make you—excuse me. If I say that a particular thing will make you healthy, what do you change other than your feelings and emotions, stress going down, and all that stuff? What happens? Your mindset, but you change your behaviors also.

6.36 So, if you have depression, let's say, and I give you this pill, "Here you go, Mani, this is a pill," although you're not depressed, but let's say, right? What happens? You think, "Oh, my God, now this pill is curing me, so I must do behaviors that healthy people do." You stop sitting and crying all day because, well, this drug is supposed to help me, so I'm supposed to be out for a run, maybe, or walking in the park, all the stuff that will trigger some of these nerve growth factors that will rewire your brain and trigger endorphins. Maybe I should be spending more time with my friends. Social connections that help the brain. You rewire your brain just from behavior.

6.37 I'm a huge fan of the placebo. I would suggest that occasionally we should give the placebo to people and say it's a real pill. In medicine, we have to be very practical. What do you think? We have to be very practical and say, what works? It doesn't matter if it's a placebo at times. Of course, except if the person has cancer, then you don't want to give a placebo. That might not be a very good idea.

6.38 The mind is very tricky. As I said yesterday, there was this idea for a very long time that all your emotions are tucked beneath your cortical regions, behind your ears. That's where all the emotions are. Cortex? No. It's intellect. It's thought. No emotions are being regulated from there. But then, you had these cases of people with damage to the prefrontal lobes; let's say you have damage over here, and changes would occur in their behavior. When the damage was here, you would have things like people would get sad, pathologically catastrophic reactions to the smallest thing. You'd talk to them, and they'd just cry. They thought, maybe that's an effect of the stroke, maybe some absurd delusional thing here. But then you see the stroke preferentially or selectively damaging the right, and you see the opposite scenario. The person has a stroke and is happy about everything. He's like, "I don't care." You can't move your hand. Let's say the stroke affected the whole right hemisphere. "Oh, I can't move; the hand doesn't work. I don't care." They're just laughing and happy. What's going on there?

6.39 Well, it turns out the left hemisphere, when we experience positive emotions, flares up with activity. Negative emotions like anxiety, the opposite scenario: the right becomes activated. A newborn baby—and this was a study actually that was done. They had babies kidnapped from the hospital when they were born, right at the point when they were born. They gave them sugar, and the left became active. They gave them citrus fruit. What happened? The right became active. Grownups— you show them a disgusting clip, something nasty, the right becomes active. You show them something humorous, something fun, Charlie Chaplin, for example, the left lights up. So, you have these activities, these tasks light different parts of the brain up.

- 6.40** It actually turns out this asymmetry of function, even in animals. You have the same thing in monkeys: make them happy, give them fruit juice, the left lights up. Over here. You do something bad to it, deprive it of something, maybe give it a small electrical shock, the right lights up. So, there is this innate asymmetry in the human brain.
- 6.41** We also talked about the prefrontal cortex and how it regulates these emotions. Now, the differences from person to person are staggering. You can have a person with 3,000% more activity. I'm saying 3,000% more activity in their left. These guys might be very, very happy versus another person. So, between people, there's a lot of activity to the state—to the degree of 3,000% change. That's crazy. That's amazing.
- 6.42** The left activity in the left hemisphere is associated with something called approach behavior. So, people who have approach behavior, approach the world in a way where they are not afraid of the world, they don't hide, they don't seclude or isolate themselves, but they have a go-getter mentality. They don't have what is called behavioral inhibition. That part of the brain—having the activity in the left is associated with this trait. So, it has a personality trait or a characteristic like this that goes with having a left activity. The right activity is the opposite thing, obviously. You have withdrawal, you have social isolation, you have behavioral inhibition, you tend to hide from the world, you are shy, perhaps.
- 6.43** The interesting thing is they've done studies where they look at kids, babies, kids. They measure the activity, and they say, "Okay, this kid, at just baseline resting, when he's doing nothing, I measure the activity, and the right is much more active for some reason." And then, ten years later, I check them again, and then the left is more active. So, it seems like a switch can occur, and the brain can change from being more right, activated in the right side versus the left. Does that make sense? It makes sense.
- 6.44** And by the way, this withdrawal of—and you might guess, which condition might have a very low activity in the left prefrontal cortex? Which mental condition, a neurological condition, might have less left activity? What about depression? Depression: very low activity in the left. What about something like anxiety, maybe PTSD? Could it have a lot of activity in the right? Exactly. Yeah, that is the case. And so, they will use, as I mentioned a few times, various ways to turn down these regions. Let's say if you have anxiety, the right. You want to reduce the activity in that region if you have anxiety by using stimulation methods. And if it's the left, you do increase your activity, you excite that region, you tickle the receptors, the neurons over there, so they become more excited.
- 6.45** It turns out you can change these circuits and the activity of the left and the right by doing things like meditation. So, meditation— am I a huge fan of it? I think it's good. It's shown a lot of things. It's one of those things like the mirror neurons that we talked about. For a while, they were like, meditation is good for everything. It can cure everything. "I have neck pain." "Meditate." "I have stomach pain." "Meditate." "I have depression." "Meditate." "I don't sleep." "Well, meditate."

- 6.46** Mirror neurons? The same thing for a very long time. Oh, we fall in love. It's because of mirror neurons. You mirror the other person's perspective. You see their life, you see that—you do a virtual reality simulation of their minds and become one with them. Everything was mirror neurons. Culture evolved because of mirror neurons, because of this. Everything was mirror neurons, although they may have a very profound function if indeed they exist in humans, if it's not merely a larger general system called the perceptual action understanding system. So, that's a debate. It's a complicated debate; I don't want to go there, but the point is that sometimes some of these neurons or some of these interventions, like meditations, get too much credibility or explain everything, and that's not the case.
- 6.47** But meditation is good. There were some of the earlier studies, very, very early studies that showed that if you do meditation and you do it, let's say, for eight weeks, and you do it maybe a couple of times, a couple of hours a day for eight weeks, well, you can change this asymmetry. So, if you were right, more towards the right, you can flip it over so you're more towards the left. Or, if the left is not very high, you can increase it by one-third; you can increase the activity of the left. That's still very interesting. So, they did these studies. They used EEG, as we talked about earlier in the first course, how they got to this. The point is you can change this activity. You're not born with—you are perhaps born with a certain tendency to have activity in one region versus another to a greater degree, but you can change the activity.
- 6.48** One way you do this, the one way you change the activity, is through something called epigenetics. You're a physiology expert, so you know. So, epigenetics is that we all have genes, of course, and these genes have a template function, which is this general function. Nothing, just replicate themselves. And then, you have what's called the transcription function. And over there, what happens is that in the DNA on the genes, a certain group, like methyl groups, a cluster of atoms, chemicals, can turn certain genes on and off depending on your behavior, what you do, what you actually are up to. Now, there are genes for anxiety and stress, so you turn them on or turn them off based on your behavior.
- 6.49** A great example would be 9/11. There were some women who were pregnant on 9/11; they were pregnant. They saw the horrible thing and potentially got some trauma there. Their cortisol levels changed. Later on, when the babies were born, and they were about 9 or 10 months old—they never witnessed anything; they were just inside the stomach, they were lying there, playing inside the stomach—they had the same cortisol levels as their mother. Is that interesting? Epigenetics. That's one example of that.
- 6.50** Or merely just when you take care of rats, and you give them—so, the mothers groom them, lick them. That's rat love. The mothers will lick them, so they get epigenetic changes, and their genes will turn on and off.

- 6.51** Or perhaps the most, what I think is one of the most striking examples of this is that you have a mouse. You have a mouse, or actually, the study was done in a rat, so let's say you have a rat. So, you have a rat, and this rat, each time it hears a particular tone, you give it an electrical shock. What do you think happens? It hates that sound. It has a conditioning going on. It pairs the sound with the electrical shock. Later on, you introduce the sound, no electrical shock, yet you still have the aversion; you have fear. Now, this rat has a baby. The rat has a baby later on. The rat hears the sound for the first time. It has never heard it before, never got any electrical shocks before, yet shows an aversion towards that sound. And that rat has a baby, and that baby, too, fears that sound. So, you see how fear can travel down generations? You see that?
- 6.52** My parents—I always say my parents—my dad was living in Iraq. He fled the Iraq-Iran war because they recruited him to the army. He was supposed to go into the Iraq army and fight with Iran, like a senseless war that he wasn't a part of. He didn't want to go to war. In fact, it was a good thing because those people that were called and recruited were bombarded by this plane, and all died. So, my dad would have been dead. But he grew up in this environment as an orphan, in fact. He was an orphan in Iraq, poor. He said he would go around the streets and he would do all kinds of work just to get a penny for food. That's my dad.
- 6.53** My mom, when she was 11, she saw her dad die from a stroke, but not any stroke. He called her in and said, "Look, daughter and wife," and the wife came in, my grandmother, and he said, "Look, I'm going to die now." And she was explaining that, and my mom was just watching the whole thing, watching him die. And they couldn't remove the body from the room. You know why? Because we had over there quarantine. There was a restriction, so you couldn't go out. During that time in Iraq, they would have these restrictions. So, she was just looking at that body for I don't know how long. Going crazy. Think about that. So, these are the genes that I got. You know what I'm saying? I don't know about you guys.
- 6.54** And then, I came to Denmark. I went to a refugee camp as a little baby. I was there playing soccer. I was presumably happy, but I think the way I escaped my genetic heritage and my epigenetics was probably because my parents were very kind and licking me like those mother rats. I got a lot of love, you see? I got stimulated, and even though I was in a foreign country—being in a foreign country, by the way, is one of the most difficult things. If you are an adult and you travel, not travel, but immigrate to another country, that's one of the most difficult things you can do. It's one of the most difficult things because really, you have this plastic wiring during your critical period and all that, and then you have to change all of that. New language, new rules, and then financial problems because, well, you come from a poor country, all that stuff. Then you have a child; how is the child?
- 6.55** We have certain customs in Denmark. When you go to kindergarten and during certain times of the year, like Halloween, you dress up. I would go without dressing up because

my parents didn't know they had to dress me up. So, I would just go with normal clothes and see the other kids. Well, this guy is Superman, and this is this, and Batman. And I had no clothes; I had just normal clothes. This was the environment I grew up in. And then, my mom would take off her scarf and put on a Zoro thing, "You can be Zoro, my son." And then, I was Zoro. But this is the whole thing. This is the environment.

- 6.56** Now, because of perhaps loving parents, Denmark is a safe country. The hoods over there, the ghettos, are not as bad as, say, Compton or Harlem or something. We don't have shootings, even though my best friend got shot 13 times in his house. It still was crazy. My neighbor got shot in the head, but it wasn't that bad, perhaps. I survived, but living on this—growing up in the streets over there, I learned a lot of skills. I, perhaps, learned to communicate a little bit because I had to survive. I was skinny, I was small, and I had to navigate that environment. I had to be a slick, smooth talker. Otherwise, somebody might beat me up, and I didn't have the muscles. So, I escaped my epigenetic destiny, so to speak, to some extent. But yeah, here I am. So, I did want to touch on that. Hopefully, I didn't get too—did I get too emotional on you guys? Or is it okay? I had to share my personal story. That's where I came from.
- 6.57** Compassion meditation. Alicia, do you have a favorite niece or a favorite uncle or something? You do. What you should do is, for two weeks, imagine somebody you love, like your uncle, your niece, your mom, in a suffering state, a state where they're suffering. And you tell yourself, you whisper to yourself, may they be better, may they heal, may they get over their pain, activating some of that oxytocin, some of those love hormones. The amygdala will actually reduce its activity by doing this over and over for two weeks, let's say 30 minutes a day. That was the study. For two weeks. The activity in the amygdala and in the fear-processing centers of the brain goes down. Now, if you do this over time, this becomes a structural change that will permanently change your brain, or at least for a bit of time, because it's a structural change in the circuitry.
- 6.58** By just merely imagining somebody in pain, somebody you love, and then imagining you're trying to heal them, you're giving them—and it's tricky because the brain—the thing is, when we are in pain, when you are really desperate, when you have an anxiety disorder, when you just want to, "I can't stand this anymore," when you have that, do you feel like you want to help other people or are you more like you just want to be concerned about yourself? "This is me. I can't think about my damn mother right now because I'm suffering." So, you trick your brain over here and say, "No, I may be suffering, but I can switch off my own suffering and take a more detached view here," and that is perhaps doing the thing because you're changing the perspective.
- 6.59** Now, I am giving something to you. I am being altruistic to you. So, the amygdala goes, "Oh, I may not be suffering that much right now since I have the energy and resources." What we know from nature is that energy and resources, brain resources—attention is very scarce. It's limited. So, you want to use it in an optimized way. You just don't want to throw it all over the place. You use it to help somebody, and then your brain says,

"Oh, that must mean I have tons of energy, resources, and attentional capacity; therefore, I must not be anxious right now." That's how it might work. It turns out that when you then ask them to see if they want to—you do certain financial games with them, and after the meditation, they become more charitable. They give more money. Same principle: if I can give money, that must mean I have excess money, even though you might be poor financially and materially. But giving money that way also is correlated with the behavior.

- 6.60** Well, I do want to tell you about something called the Capgras delusion because it helps us crystallize those circuits we use to understand how the brain works. It ties well into these emotions and emotional states we can have. Emotions are tricky. They depend not only on the amygdala but also on sensory information in a much more crude way.
- 6.61** An example is, if you have, let's say—actually, let me give you the case. There's this guy; he's in a motorcycle accident. After the motorcycle accident, a part of his brain is damaged. Now he sees his mother. His mother walks in. He looks at her and says, "My God, you are not my mother. You are an imposter. You are a copy mother, a duplicate, but not the real thing," mind you, this person is totally fine in other respects. You play chess with him, go on walks with him, nothing. He's not absurd; he's not clearly irrational, illogical, or delusional. Well, he's delusional in this sense only. But then, the mother calls the son, let's call him Joe, "My son, Joe, what are you up to?" Joe answers, "Oh, Mom, it's you. How are you doing? Are you well? I love you. It's been such a long time." So, on the phone, the delusion is not there. Only when I see the mother, the delusion is there. Do you see the dissociation here?
- 6.62** I want you to think for a moment, based on the information we've shared today, about what might be going on. Auditory circuits. What do you think? Could it be that the center that goes from the amygdala, the emotional part of the brain, to the visual part of the brain has been severed by the accident? So, there's less activity now, less fibers, and so when I look at my mother, I go, "My God, I'm supposed to have a visceral reaction to my mother. I'm supposed to feel something, yet I feel nothing. It feels as if she's not really my mother; therefore, she isn't my mother."
- 6.63** Did you know if I flash images on a screen of different objects, for example, I show you a chair, a table, or a TV, and then I show you your mother, you will actually show sweating and a physiological reaction to your mom? We all have this. So, this is a visceral emotional reaction. It's measured in the skin. There's sweating in the skin. Now, I don't have that visceral emotional, hormonal reaction to my mom; therefore, she must be an imposter. Does your brain hate incongruencies? Does your brain hate and dread these inconsistencies? Absolutely. Does it fill in the blank with delusions, even absurd ones at times? Absolutely. But what about the circuit that goes from the ear to the emotional part of the brain? Might that be intact still? Might that be intact, so you can actually, when you talk to her, have the right emotions because her voice is

triggering the right emotional and visceral response? This is called the Capgras delusion.

- 6.64** Some people even have it with themselves. They look at themselves and say, "Look, this can't be me because I'm not feeling anything." That's one version of how emotion can go awry, that it's not simply emotions in the more abstract sense, but sensory regions can affect emotion in this way.
- 6.65** Now, another version would be: you have your uncle. You love your uncle. You have pleasant thoughts about your uncle. You love him. Now, occasionally, if you have hyperactivity, excess cross-activity between the vision part of the brain, particularly the fusiform face area, where we identify faces, and the emotional parts of the brain, if that circuit is overactive, what might happen in certain cases? Well, it turns out there's a syndrome called the Fregoli syndrome, where you go about your life in Florida, you walk around here on the beach, and you look at each person, and they look like your Uncle Joe, each one. Woman, child, old, young, everybody looks like your Uncle Joe, each one.
- 6.66** And this is a real syndrome. It's called the Fregoli syndrome. The excess activity says to your brain, "Look, I'm feeling pleasant. I have good feelings when I watch these faces. There's something kind about them. Well, they look like Joe. That must be the reason. They all look like Joe, which is totally absurd, yet this must be the explanation. Your brain conjures up this idea. The Fregoli syndrome. So, that is the syndrome I want to end on. I think we covered a lot of ground. Tomorrow, we might do some review, and then we'll venture into dreams. Thank you very much.

Lecture 7: Sleep and Reality

At a glance — Sleep and reality.

- 7.1 Happy to be here for this lecture. As you can see, my notes have shrunk and become smaller because we're talking about sleep. I don't need damn notes for talking about sleep and dreams because I know that stuff very well.
- 7.2 So, I woke up this morning, and they brought me to the makeup room and put some makeup on me. Can you see it? I don't need all that makeup. I 'm already pretty, you know? But anyway.
- 7.3 Well, yeah, today will be fun. Now, I want to also use the time to revise a little bit and make sure that you learned because, especially in the first few lectures, one, two, and three, there was so much information. We went through the whole brain, the neurons, different circuits. We talked about the divisions of the brain and neurotransmitters. We talked about the cell, the axon, the dendrite. It's very important to actually have these concepts and crystallize them.
- 7.4 When I initially learned about neuroscience, you get this information; you learn about these different structures, but it doesn't really have any greater context. There's no greater meaning. You don't understand what are these things doing. So, you really need that. It's important. When you hear it over and over again, repetition, what happens is that eventually it crystallizes in your brain. It becomes more clear. Right, Mani? When you repeat over and over the names of football players, you know, Cristiano Ronaldo and these people, what happens then? You crystallize the information. You know the team players. You know that Cristiano is here, and Messi is over there; he's playing on the right wing and all that. Same with the brain. You need that information, and you need that repetition.
- 7.5 First of all, we talked about how the neuron looks. So, we have the cell, right, John? We have the cell, the cell body, the soma, right, Alicia? The soma, and then you have the axon. The axons can be very long. Some of these axons, go from the brain and then from the spinal cord all the way down to your feet, and they can travel very, very quickly; they're very fast. Electrical information is passed very quickly, between two and 200 mph sometimes. Think about it. When you move a toe, how quickly is that? So, think about moving a toe and then do it. How quickly is it, Michael? Do it. Fast, right? It's very fast.
- 7.6 So, we have that, and then we have the dendrite. What is the dendrite doing, Amari? It's receiving information. Receiving. So, on the other side, here, is a neuron, and we said the neurons can be multipolar, bipolar, unipolar, not bipolar disorder. That's different. But they can have different processes extending from the cell body. So, if it has multiple, it's called multipolar, and the vast majority of the neurons in the brain are multipolar.

- 7.7** The multipolar ones have some dendrites sticking out, and then you have the axon, and the axon sends the information. So, that's always how the flow goes. And then, there's the synapse, the little gap where all the magic happens. Here you have some liquid. Some vesicles exist at the axon terminal. That's where the neurotransmitter, the neurochemical, is released into the synapse, and it travels over on the other side. And then, boom, to the dendritic spine, and then it goes from the dendrite all the way to the cell body with the goal to elicit an action potential, which can be excitatory, inhibitory, on and off.
- 7.8** We talked about neurotransmitters of different kinds. Serotonin. Serotonin is what? It's involved in things like mood, regulates sleep—it's actually one of the neurotransmitters that has the most profound and global effect. The brain uses it for all kinds of stuff, and we'll see that with dreams because during dreams, when we have vivid, lively, and crisp dreams, the neurons that produce serotonin are turned off pretty much. So, the serotonin is not around to have its effect.
- 7.9** We talked about brain structures, and we have sensory systems, meaning areas of the brain where sensory information is processed. We have the frontal lobes, where you have the motor system, a motor strip of cortex. I need you all to say, precentral gyrus. That's the motor strip. Gyrus is a fat little shawarma-like thing. Right there. Yeah. You can laugh. I know. You can laugh, alright. So, you have that precentral gyrus, the motor strip, and in front of it, you have the—so, the motor strip is the main motor area, and then right in front of it, you have the premotor area, the area that plans action, planning the action, not executing it. The precentral gyrus is actually executing the action.
- 7.10** Then, you have the prefrontal lobes, where all the high ability of men comes about. Long-term planning, higher intelligence, decision-making, and a sense of self. I am Baland. I have an ego. Why do I have an ego, Mani? I'm not talking about a huge ego, hopefully, but I do have a sense of myself. I don't want you to say bad things about me now because I may feel that's hurting my feelings. An animal wouldn't have that sense of self, that ego. A baby doesn't have an ego at one; it doesn't know what ego means. It appears that ego itself, the sense of I, of selfhood, comes about in the prefrontal lobes, particularly the left prefrontal lobes. A child, as we said, is mainly right hemisphere geared. It uses mainly the right hemisphere.
- 7.11** Now, if you have a stroke in the right hemisphere, what might happen in some cases? We talked about somatoparaphrenia. The patient might deny—they might say that their arm belongs to somebody else if the stroke is in the right side. Where? In the body image side, the place that controls body image. But in terms of ego, sometimes these patients, when they have a stroke in the right side of the brain, what happens is that they will not only say, "My arm belongs to somebody else," but they will deny the paralysis because their ego cannot deal with the fact that the arm is paralyzed.

- 7.12** The ego is in the left hemisphere, and the right hemisphere is damaged by the stroke, and the right hemisphere's job is to inhibit the over activity of the left now. If the left hemisphere gets too much reign and activity over there, what happens is that it can even render the person delusional, manic, and all over the place. Having some ego and an optimistic attitude is good, but you don't want to go out and think you can buy a Rolex if you can't even afford a tiny thing over here. You want to have—you want to be realistic, and the right hemisphere seems to make you more realistic, big picture, more emotional, perhaps, seeing the overall picture. The left hemisphere: details. It can become manic, and delusional if it needs to, to preserve the sense of I.
- 7.13** But now and then, what happens is that if the right hemisphere is feeding the left hemisphere and telling it, "Well, the overall picture is that you are not this big shot lawyer," let's say. "You are not this big shot; you're not a multi-billionaire." It's telling the left to—I'm exaggerating, of course, but it's telling the left hemisphere to calm down a bit. Eventually, what happens is that the left hemisphere gets the big picture and rewrites the script about who you are. So, the right hemisphere's job is to look at the big picture and keep the left hemisphere in check.
- 7.14** But now there's a stroke there, and the arm is paralyzed. They might say things like, "Well, doctor, doctor, this arm is not paralyzed. It's completely fine" while it is paralyzed. And then, what they might do is that—one of my colleagues did this experiment with this patient whose arm was paralyzed. The doctor said, "Well, look, move your arm, Ms. D." And she said, "Well, I am moving my arm right here. It's moving." It wasn't moving, of course. She was denying the paralysis. Then, the doctor took an injection and injected the arm with salt water (saline). Then, the doctor said, "Now, Ms. D., move your left arm." You know what she said? "I can't, doctor, I can't move it because it's anesthetized. I can't move the arm."
- 7.15** You see? Do you notice the difference here, Maria? Now, she has an excuse as if somebody deep inside her brain knows that it's actually paralyzed, but now she can use the anesthetics as an excuse for the paralysis. So, they say things like that. Or you might ask them, "Which arm is stronger, the left or the right?" Knowing the left is paralyzed, they exaggerate sometimes and say the left is actually stronger even though they're paralyzed, even though the left is paralyzed. So, they have all these defense mechanisms, you might say, like reaction formation, saying, "Well, it's stronger," or a denial, and things like that.
- 7.16** The precentral gyrus was the motor strip. The postcentral gyrus is the place in the brain where you process your body image, not your body image in the more abstract sense, but the actual physical sense of self. I know I'm jumping a little bit, but we said the prefrontal cortex is the highest seat of learning, ego, and self. Then, we said the motor strip, the precentral gyrus, and then we went further back and said the postcentral gyrus. Are these clear? Okay, these centers are clear.

- 7.17** See, now, when you're done with the talk, and we're done with this whole course, when you go back to Poland, when you go there, you talk to your colleagues and friends, you say, "Look, I know. This is the prefrontal; it does this and that. And then, right there, you have the motor cortex right behind the prefrontal, and then you can do movements and plan them out. And then you go further back, you have the post central gyrus. Gyrus just means a strip. And then, there's a central sulcus, right in the middle, that separates it."
- 7.18** And then you say, "Oh, and by the way," you're impressing them now, you see? You say, "By the way, there's—all your body is mapped on to that strip of cortex called the postcentral gyrus, where if you were to touch the arm, that center would light up in the brain, and we can map that using unit recording, multi-unit recording, to map the activity of neurons directly." So we know that.
- 7.19** And then you might add, "Oh, and by the way, it's topographically organized, meaning it's organized like a man, but then it's not really a man. It's kind of a strange man, an absurd-looking man with a big tongue. Why does it have a big tongue? Well, that's because the tongue is more sensitive. It has more receptors." So, you might add that.
- 7.20** And then you say, "Oh, and by the way, it's also organized like a man because it's a 'use it or lose it' brain, and it's a brain that uses neurons that fire and wire together. So, the fingers are mapped next to each other like that. And we use the leg often with this part of the body, and so forth. So, that's why it's mapped like that." Is that clear? Am I talking too much, Jose? Is it understandable?
- 7.21** Did you know the auditory cortex is organized in the same way? It's called tonotopic because we tend to hear tones and frequencies in a manner where we hear low tones at the same time and higher frequencies at the same time, like when we hear talking or sounds. So, for that, the brain says, "Well, let me organize it with low frequencies here in the auditory cortex, and then higher and higher and higher," like a piano, for God's sake. So, even the auditory cortex has this system of this. Like it's a map. Good?
- 7.22** Where's the auditory cortex? Right here. It's right here. This area. Listening. Hearing. See, that's part of the sensory cortex. Everything behind the central sulcus, we said sulci are the furrows, the hearing center. We had the cochlea, the little microphones inside the ears with 3,000 cells vibrating, converting sounds into electrical signals going through the auditory nerves, the inferior colliculus, then going to the medial geniculate nucleus, and then going to the auditory cortex. We said that. See, I'm not just trying to impress you guys with these names, okay? There's actually—so, is it clear? Is it clear? Now, you hear it over and over again, and eventually, you know this is the part of the brain that does that. Okay, good.
- 7.23** Then, Amari, once we get further back in the brain, what area do we get to that's huge? That is so huge, in fact, that it has to be huge because it's so important for our survival, like spotting things—I'm not hinting— seeing, spotting things in nature and stuff.

Occipital lobes. That's right, my friend. Visual cortex. Vision. So, it's very important we have vision. Is vision processed only in the occipital lobes, or are there multiple pathways where vision is—multiple pathways, right? So, vision is initially in the visual cortex, and then spreads out.

- 7.24** So, the visual cortex we talked about, and we said that there are ancient highways, things like light, setting your circadian rhythm. Circadian rhythm is the rhythm that helps you know when it's light and dark, when it's time to sleep, and when it's time to be awake. And it seems like human beings when you put them in a cave-like situation where there's no light, there are no zeitgebers, triggers, and stimuli that can help you set your circadian rhythm, and even watches and sounds of birds and things like that will help you set your circadian rhythm. So, it's not only light. But even if you went to a cave in total darkness or something like that, you would still tend to fall into a pattern of sleep on a regular basis where it's, "I'm going to sleep around eight or seven hours, and each day I'll do the same thing." So, it's kind of wired in, this tendency, even if there's no stimuli. So, we got that. The occipital lobes.
- 7.25** We talked about the prefrontal; then we went deeper into the brain. We talked about the limbic structure. Now, do you all know what the limbic structure is doing now? Limbic: emotions. Amygdala: fear-processing. Now, emotions, limbic center: is it a ring? Absolutely. Limbic means a ring. It's like a ring. The amygdala is an almond-like structure. But it's so easy to just say that somebody with anxiety has a lot of activity in the amygdala. The amygdala is used a lot; it's an important structure, of course. But we often forget that, for example, when you have anxiety, let's say, and you fear outcomes in the future, "If this happens, if I go over there, this might happen," you focus on all the negative activities, there's a structure called the stria terminalis, and the bed nucleus with the strata terminalis is an extension of the limbic structure that's more concerned with future outcomes, with future things that might go bad.
- 7.26** What I mean here is that somebody with anxiety doesn't have a clear lack of fear. Even if you give them a lot of drugs for anxiety, if there is a lion, they will still have a fear reaction, but the anxiety may still be there. So, fear and anxiety are different. It's very important, and as you become more familiar with neuroscience, we make these smaller distinctions. We start to make these smaller differentiations between anxiety and what fear processing is about.
- 7.27** And then, we talked about the septum. The septum is inside the limbic center, and the septum is important for things like what? Septum. Caddy, you know, septum. Come on. Septum, septum, Mani? I know it's a tricky one. I only mention it, you know, a couple of times, but it's involved in things like excitement. It's involved in things like when you feel happy. Dopamine. It's a dopamine center. It's like the main dopamine and happiness center, the kind of happiness center we called appetitive pleasure. It's where you feel excited. Tension. You're about to have that meal: appetitive. And so here, the septum actually cross- inhibits the amygdala. There's all this inhibition, cross-inhibition

going on where parts of the brain are inhibiting other parts. So, the septum is doing that. On the amygdala, it's saying, "Look, buddy, don't overdo it. Don't keep doing it."

- 7.28** Now, if you have PTSD, a post-traumatic stress disorder, what happens? Often you see that—you know what you see, Abby? The amygdala might grow bigger. They actually become larger these amygdali. They become larger, and some of the structures and the septum can't compete. So, the competition, the amygdala wins automatically. Some of the wires going from the prefrontal lobes, that are supposed to constitute the brakes on the brain, become weaker. Some of the wires going from the prefrontal, the precentral gyrus over here—not the precentral gyrus, the medial part of the prefrontal and the DLPC, we talked about the prefrontal over here, they inhibit these parts of the brain. But they can't sometimes if the wires become too weak, if you have anxiety, if you have trauma, you simply—these wires have less fibers. You take pictures of the brain. There's less fiber there, what we call the white matter tract. The septum is very important. It's interesting.
- 7.29** So is, I guess, so is dopamine, because when you think about it, somebody with bipolar, somebody who falls in love, somebody on crack, sorry, somebody, but I have to—i t's true. Somebody on crack cocaine, you know what they have in common? High levels of dopamine. Happy. That's exactly true.
- 7.30** And what it seems like is that, or what's actually happening, is that these dopaminergic centers become sensitized. The threshold for when they start firing is lowered. So, it takes very little. When you fall in love, you're happy. Life feels so good. Not only Juliet, my beloved, makes me happy, but the whole damn world. I feel happy when I go on the streets and walk, and everything is poetic. I look at my ghetto neighborhood, and it looks like Champs Elysees, like France. Everything is beautiful. Everything is poetry. Globalization occurs where everything, there's this generalization, where everything just becomes exciting. Do you know what I'm talking about? You know what I'm talking about. You know it, right? You've had that feeling. Life feels poetic, and that happens. Why? Because the dopaminergic centers now become sensitized. This is a small little thing, and, whoa, yes, I love that. That's great. Yeah.
- 7.31** They take brain imaging scans of people that fall in love—who are just falling in love. So, it's a new relationship, and they show them the picture of the woman, and they look at the dopaminergic activity, and then they compare it to somebody on cocaine, and it's similar. Similar brains. Imagine that.
- 7.32** Now, we didn't really talk about addiction, did we? We should mention something. Overall, this is a course on the brain. Addiction—in fact, you can become addicted to things like love. When you fall in love, initially, it's very much like a drug. The person, you're with them, you feel happy, you feel excited, but then you are separated from the drug. What happens? Withdrawal. But then, you spend a lot of time with that person. What happens? You build up tolerance.

- 7.33 So, with drugs, usually what happens is that you need more and more of the drug in order to have that same buzz. Otherwise, it's not going to do it because your receptor is also desensitized to the dopaminergic centers. In fact, dopamine is interesting.
- 7.34 You might ask, how come somebody who just had a baby becomes depressed, or why is it that after that vacation where I was so happy and everything was so fun and I was playing and I was doing all these things, I come home from the vacation, I'm kind of depressed. I don't feel that happy. The reason is, Dhalia, that our dopaminergic centers have a limit. They have a limited supply. They can't be active all the time. I can't keep smiling all the time. At some point, these dopaminergic cells need to recharge.
- 7.35 What happens at that point is that you don't seem to go to baseline, but you go back to, you go beneath baseline. That's why depressive symptoms can even occur after having that baby because when you had that baby, everything was great, epic. My first child, I'm so happy. But then, you just can't feel happy afterward, and that's because you need the dopaminergic cells to replenish, and so they have to fall beneath the baseline. Actually, how much? If they went up this much in activity, they have to fall down this much, an equal amount in order to replace. It's not simply going to baseline. Is that clear? So, you can't always be happy.
- 7.36 What about neuropeptides, and when you have these painkillers that take away the feeling when a zebra is running on the savannah, and the guts are spilling out, and they feel nothing because the lion is there? The system is using these neuropeptides as anesthetics here. So, they're just running and feel nothing. And we can have that too through acupuncture, or sometimes it spontaneously occurs if you're shot, if you're Ronald Reagan. So, you have these painkillers, but even these have a limited supply. So, at some point, they run out. So, you can't always run around and use these neuropeptides. They run out.
- 7.37 Unfortunately, cortisol and stress don't run out. So, you can have that forever. And you're still going to be stressed and eventually die, probably from some disease because a lot of diseases come from stress.
- 7.38 Oh, and the addiction point. There's actually—it's interesting because these days people become so addicted to screens and Game Boy. Game Boy is a different generation. That's my generation when I was a kid, these Nintendo Game Boys. But people get addicted to these screens. They just need that buzz in more and more. Gotta watch that next video; keep going. But the thing is that with these things that just hijack the dopamine centers, when you feel like, "I have to do this." It's no longer just, "I like it," but, "I have to." That's how you know it's addiction.
- 7.39 A part of the addiction cycle is that I feel that other activities don't give me the same high. I don't feel happy anymore spending time with my wife when I can have this other wife on the screen, if you know what I'm saying. These nasty things people do; it's a big problem with these pornography sites. People get totally addicted, destroying societies. It

hijacks the dopaminergic system, and you cannot have pleasure otherwise from other places. That's a sign of addiction.

7.40 At that point, you become sensitized to that drug, and when sensitization occurs, something called DeltaFosB, a protein, accumulates in the cell, changing your epigenetics, changing the epigenetics in the genes. And desensitization—basically, what happens then is that when you just see a small reminder of that thing, what happens?

7.41 Let's say it's alcohol. Now, you go to the bar in the restaurant. What happens? You get triggered. DeltaFosB became sensitized over here; you became triggered. Actually, some of the changes by DeltaFosB can be permanent. It's a permanent destruction to your dopaminergic system so that ten years later, when you walk through that alley, and you smell that beer from that bar on the other side of the street, you have the craving. You want to just go there and just have that beer. Twelve years later. And that's why they say once an alcoholic is always an alcoholic. There you go. You want to treat your brains very carefully.

7.42 We went into the basal ganglia. We talked about the basal ganglia. What is the basal ganglia doing? You know this one. We talked about Muhammad Ali, and we talked about movement. Remember? It's the movement part of the brain. Do you remember smiling? Basal ganglia, smiling. You have two kinds of smiles. Basal ganglia, automatic movements: adjusting a shoulder before throwing a dart. Running down the stairs, walking, these kinds of things. Automatic muscle tone behavior. Where is the non-automatic conscious motor behavior? Where is that placed? Absolutely, in the cortex, the frontal cortex. Basal ganglia. So, we got that.

7.43 It's actually interesting with Parkinson's also. There's a motivational component, too, because dopamine, we said that in Parkinson's, the dopaminergic centers of the substantia nigra are destroyed. And so, you don't have enough dopamine in the basal ganglia, allowing you to move and have automatic movements. But dopamine is also involved in what? Motivation and pleasure. Is that true?

7.44 It turns out sometimes if these people with, let's say somebody with—it seems, this is a bit unclear, but if you have somebody with Parkinson's and there's a real motivation for movement, they will move. This is paradoxical, but they're walking like this, they have this shuffling gait, and then, they walk like this because their whole body falls down. But then, if they see somebody drowning there in the sea, they might jump in the water completely normally, just run normally, save them, and come back. So, very interesting. You have this revival.

7.45 And then, we said to help these dopaminergic centers and systems, you give them pills. Do you give them dopamine? Can dopamine cross the blood-brain barrier that protects the brain with dense blood vessels, and so you have this wall? No. L-Dopa, you take another drug because you have to have protection for the brain. Is this review good, John? What do you think? Is it helpful? Honestly? You look a bit skeptical. Are you sure?

- 7.46** I didn't mention this, but we have twelve cranial nerves. These cranial nerves allow the brain to communicate with other parts of the head. So, for the tongue, moving your tongue, moving your face, moving things like the vestibular system for balance, for a sense of swallowing, and things like that. So, we have these twelve cranial nerves. So, yes, and we said the vestibular system is doing what? Balance. Vestibular system. And it's where? It's in the ears. Well, you're becoming experts now.
- 7.47** Then, we went to the brainstem. We talked about the brainstem. What is the brainstem doing? It's involved in things like breathing, body temperature, heart rate, movement, automatic movements—a stroke there will probably kill you. That's it. A stroke there will kill you because there are so many processes there that are so vital for life. So, you have that.
- 7.48** Now, procedural memories, we talked about memories, but procedural memories tend to have a strong cerebellum component. Cerebellum movement. Now, we talked about two memory systems, the explicit and the implicit. Explicit is the things you know, facts about people, and then the implicit is the procedural. Walking down the stairs or walking up the stairs is what? Procedural. You got it.
- 7.49** Now, the thing is, when you are walking, let's say you're walking down the stairs, what would happen if you're using—or forget walking down the stairs. Let's say you're playing tennis. If you're playing tennis, you're using what? Procedural memory, probably. Now, if you want to win over your opponent, you know what you want to do over here? You tell him, you say, "Look, buddy, can you show me how you did that shot? Show me how you moved your hand like that when you hit the ball. Show me that." Now he's going to try to do that, to show you. But once he tries to show you, he goes from using procedural memory, unconscious memory, to using explicit, declarative factual memory. And that's a problem because now he's using the cortical regions, the cortex, and he can't. He's trying to show how he did that, and when he tries to focus, "What did I do?" You can't really do it right.
- 7.50** Or when you try to walk down the stairs and when you do it consciously. I'm going to use my left leg now. I'm going to use my right leg now. Does that work? No. So, these are two different brain systems.
- 7.51** Then, we have the spinal cord, we got to the spinal cord where the signals are sent to the body and come back. Efferent signals go out; afferent signals come back to the brain.
- 7.52** Of course, any neighborhood you want to know about, you need to have some clear names for where things are. The same for the brain and the neural neighborhoods in the brain. We said posterior. We said anterior, posterior. Dorsal, ventral. Lateral, medial. These are some of the signs.

- 7.53** We talked about the hippocampus and its role in memory. Memories are stored in their original location, but the hippocampus helps us take the short-term memories and place them in the long-term. So, we talked about that.
- 7.54** Well, we all sleep, and we spend one-third of our lives sleeping. That's a long time. But why? Why spend all this time sleeping when we can do other things? It doesn't make sense from a biological perspective when you think about it because, after all, the goal is to pass down our genetic material to the next generation. That is the biological agenda for life. So, why spend all this time sleeping? We don't know the answer to that, but it tells us, Michael, that something important must be going on that is important for life itself and sustaining our existence.
- 7.55** Every night, we pass through different stages. Initially, body temperature drops: stage one. Your brain is preparing for sleep; your heart rate goes down, and your body becomes calmer. Then, you hit something called stage three, and here you spend most of your time during sleep.
- 7.56** During stage three, something strange happens. Initially, you have the brain waves going down, just falling, becoming less active. Then suddenly, you have this activity with these spindles. Suddenly, the brain is becoming more active. Spindles, it's becoming more active now. It's preparing for sleep. It's preparing for a deeper sleep. And then comes deep sleep, what's called a slow-wave sleep. Here, you have waves going like this in the cortex if you measure. Delta waves. This part of sleep is used to consolidate your memories, put those memories in the right place in the brain, repair of tissue and skin cells, cardiovascular replenishment, and all these important functions.
- 7.57** We talked about the cerebral spinal fluid. Now, the cerebral spinal fluid, what it does during this deep stage, basically, bathes your brain. So, the cells in your brain, the neurons—the brain will shrink by 20% or so just to have that cerebral spinal fluid bathe your brain to get rid of these toxins because you have a lot of toxins accumulate over time. You might have things like tangles and plaques that lead to Alzheimer's. The deep stage of sleep. Muscular repair: the deep stage of sleep.
- 7.58** Then, you hit REM sleep, the rapid eye movement sleep. Something strange happens, Amari, at that point. Something very weird happens. Your eyes, first of all, go like this, but your body is paralyzed from head to toe. Then Amari asks, "That doesn't make sense. How come you're paralyzed, but your eyes can move? That makes no sense." But it turns out there's a different circuit for the eyes in the brain. So, you can still move your eyes, but your skeletal muscles are paralyzed. It's called atonia.
- 7.59** Now, Maria, the heart rate goes up. Your pulse. You eavesdrop on the activity of neurons during REM sleep, and you eavesdrop on the activity of neurons of somebody awake. The guy in REM looks more awake. His brain is consuming more glucose now by 20%. He looks more awake in REM. That's paradoxical. For many, many years, it was

believed that when you sleep and when you are going to these deeper stages; it simply means that your brain becomes less and less active. But here, we have a paradox because the brain is more active. There's more activity in the brain; the brain becomes more active. Your heart rate, you have this beta activity like somebody who's wakeful. Interesting, right? All this craziness.

- 7.60** To control for that, your brain paralyzes you. So, there are centers in the back of the brain called the brainstem. The pons and medulla are part of the whole brainstem. They release these inhibitory neurotransmitters called glycine and GABA. They signal the spinal cord, and you are paralyzed from head to toe, as we talked about. And that is to control the whole environment, and not only that because we have our most vivid and lifelike, crisp, crazy, strange dreams during REM sleep.
- 7.61** So, we do have dreams during some of the other stages as well. We have some fragments of dreams in deep sleep. For example, we have some of the nightmares. Some of these nightmares where you wake up, like, you jolt yourself awake, and you have a panic attack or something like that, not a real panic attack, but you have night terrors, that's in a deep stage of sleep. That's not REM sleep.
- 7.62** Or have you heard about people who come out of—they wake up at night, leave their bed, and start walking around their house? Have you heard of that? And then, I don't know why, but then they go to the balcony, of all places, they go to the balcony, or they might actually go to the car. It's crazy. So, they go out there and get this Tesla turned on, the Benz, and start driving on the freeway. What's happening there? Sleeping. What's going on in the brain? We're going back to deep sleep. We're not even going into REM now. We're taken back to deep. They're doing this in deep sleep because during REM, you're paralyzed. Right, Mani?
- 7.63** During these arousal disorders, you have a partial awakening of the cortex. The cortex: consciousness, awareness, being aware of your surroundings. Because you are in deep sleep, you can still move, you can still walk around, but because of that partial arousal of the cortex, what can happen now is you can have some knowledge of the environment. You kind of know what's happening, but not really because you're sleeping. The hippocampus is shut off, so when you wake up in the morning, you don't remember this.
- 7.64** I had my brother when we were—when he was a kid, he was in Iraq, and over there, they sleep on the roof. You don't do that here, do you? Maybe in Peru, you sleep on the roof if it's really warm? No, not really. It's an Iraqi thing, maybe. We sleep there. It's a thing; you sleep there. But if you have these conditions, it's not a good idea, as you can see. And so, he woke up and started playing soccer on the roof. It's very dangerous because there's no—you can just fall. So, he did that. That was so funny. Well, not funny, but interesting.

- 7.65** Sleepwalking is very common in children. The reason is that the cortex is not fully developed in kids. In kids, the cortex is still thin; it's not developed yet. In fact, that's why some of these kids will pee in their pants when they sleep because there's a circuit going down there, and if you don't have enough cortical inhibition, you might pee your pants or wake up and play with footballs on roofs. So, that's what's happening in these conditions.
- 7.66** There's another interesting one. It's called a sleep-related eating disorder. So, it's related. In this case, what they will do is they will go to the kitchen, and they will start, of course, while sleeping, they will chop vegetables, and then they will eat them. But of course, they're sleeping with a big knife. These are raw vegetables. Sometimes frozen food. Often frozen food. Carbohydrates. And they will just binge. Raw meat. They binge eat and slice their fingers, sometimes. Of course, if it's too hard, too much slicing, they will wake up, but they often have injuries. They often have issues with this. It's an interesting condition between the fringes of sleep and waking. That interface.
- 7.67** So, these are called parasomnias. These are arousal disorders, conditions that occur during deep sleep. But of course, REM is where all the fun happens. That's when it gets really crazy. The brain gets into a rabbit hole.
- 7.68** Now, during REM, we dream. Do you have any favorite dreams you remember from your life? Any favorite dreams? Good dreams? I had a dream once. I still remember this. I saw this girl. She was so beautiful. I was like, "Did she walk out of a Shakespeare poem, or what happened?" I'm serious. I was like, this can't be real. What's going on? Well, of course, it wasn't real, but I was like, "What is going on?" I'm seeing this happen. There's somebody that looks poetic, unreal, from another planet, in my dream, and had I met this girl before in real life, I would have known. Trust me, I would have remembered.
- 7.69** What happened? How could my brain create that construction? How is that possible? So, can your brain create new faces? First question. A completely new face? Supposedly not. She was hauntingly beautiful. I woke up the next day, and I really felt happy. I just felt so happy. Strange.
- 7.70** Another dream I had once—this is many, many years ago. That dream was in California. There was this random morning. I was sleeping in this new apartment on a mattress. There was nothing sexual about the dream. It wasn't like—we do have sexual dreams during REM; it's part of the whole REM thing, but it felt like I had a spiritual encounter. It was very meaningful. It was a very meaningful dream.
- 7.71** And once I had this dream. I was drinking this liquid in a forest, the forest behind our house back in Copenhagen, in the ghetto. There's a forest, and I was there, I was walking there, and there was this drink I had. It came from like a—it gushed from the earth, and I had it, and the taste was so good. I still remember that taste like it was yesterday. It was so good. It just kind of went down. I can remember how it was so

cold. It was like honey but cold, not like the smoothies you have from this world, but from a different world. It felt different, dramatically, qualitatively, excuse me, different from everything I've ever experienced. It was so good. So I'm talking about these kinds of dreams.

- 7.72** Have you had those? Very strange. But you can have these dreams, and they have these cosmic meanings that are very, very powerful, and of course, they tickle. You can have all your senses being tickled. You have activation in the hearing. You can have musical dreams. You can have sensory dreams. So, you touch something in a dream, and it feels like you're touching something in actual reality; it feels like that. Visual dreams, of course. You have all that.
- 7.73** Now, you have visual dreams because dreaming is very much a visual process. But it's more than a visual process. It's more like walking in art museums, like being inside an art museum and walking around and inspecting visual poetry. That's what it's about. It's more than semantics. Words cannot express what dreams are.
- 7.74** It's like you're walking, and then, in your dream, you have the sun and the moon there in front of you, and then over there, you have something other profound. It could be that you're then suddenly playing football with Messi. Everything is different. Everything is cosmic. It's significant. It's powerful, but you can't use linguistics and words to express it.
- 7.75** Now, when you have that dream, you just want to go tell people. I want to just go tell Dahlia that I had this dream. It meant this to me. It meant that to me. You want to tell other people. And people go, "Well, yeah, that sounds interesting." This guy's weird. What is he talking about? It's because it has that impact on us, on me, on the person. It's almost like it's triggering a subset of neurons and receptors involved in spiritual experiences. It's almost like half a spiritual experience at times, these dreams. Very cosmic.
- 7.76** Now, in fact, dreams, in this case, might be—you might talk about a different circuitry, and we talked a little bit about that. That there are certain circuits involved in these kinds of spiritual, cosmic experiences. And it might entail certain receptors of serotonin. Serotonin acts on certain receptors; one is called serotonin 2A, which—the mystical experiences come about when this receptor is tickled, you see?
- 7.77** So, it seems like that circuitry is activated in the REM stage in that dream because serotonin, as I mentioned earlier, doesn't work. The neurons producing serotonin do not work. So, there's a flipping of the neurochemistry of serotonin. Of course, when I say it doesn't work, it means that some of these activities of these neurons, of the receptors, can still become active, and at that point, when you have that cosmic significance, it has a lot of personal meaning to you.
- 7.78** This might explain a question—I got a question yesterday: how come we don't mix dreams with reality? We have all these dreams. We said we dream for about eight

hours. Let's say you sleep for eight hours, so you don't dream the whole portion, but you have a portion of that during REM for dreams. So, one-third of your sleep is spent in REM. So, at least you have that. Of the eight hours, one-third is at least spent in the vivid— where you have vivid life-like dreams. And we said you have some fragmentary dreams in the other stages, too. In the other stages, we have some dreams, but more boring, maybe some nightmares. But how come you don't mix that? You don't go and say, I had that dream, and then you mix it up with what you did in the office yesterday. It's a good question. It's worthy of answering.

7.79 Well, there's a condition called narcolepsy, in which this distinction, this wall between dreams and reality, can break; it can dissolve. They merge a little bit. So, narcolepsy — narcoleptic patients might say things like, they might go to the office and ask their boss, "Did I actually send out that report yesterday, or was that just my dream? Oh, did I make that phone call yesterday, or was that my dream?"

7.80 Narcolepsy is a disorder that occurs as a—it's certain neurons in the hypothalamus, for example, that are not working as they should. So, the sleep-wake cycle becomes totally mixed up, and when that occurs, they fall asleep during wakefulness, during wakeful hours are the words. So, I usually say, "If you fall asleep during one of my colleague's lectures, that is understandable. Not mine, but one of my colleague's, that's understandable. But if I, the lecturer, fall asleep during my talk, I may have narcolepsy." Do you see the difference? Context. So, they might walk in a park and fall asleep in narcolepsy. So, they have problems there with the sleep-wake cycle, and they have these symptoms. They have sleep paralysis as one of the symptoms of narcolepsy. So, they lack this distinction between these realities.

7.81 We talked a little bit about how, actually, in dreams—dreams are so strange and bizarre. We're using different circuitry, and so probably because of using that different cosmic circuitry, that's also why we don't mix the memories. We don't mix them as well. We don't mix them. So, that's one part.

7.82 But I have argued, I claim, that sometimes the reason for having that feeling when you are—so, you have this dream. I'm doing something strange. I'm in a different world doing something bizarre. Then, weeks later, I have a different dream, and I remember the first dream. Do you know that dream? This condition, this experience? It's very interesting. You have a dream, and then you forget that dream. Weeks later, you have another dream, and you remember that other dream. It's like déjà vu, but dream déjà vu, you see? It's different, and it has to do with the circuitry now because you are in the deep dream stage, in the dreaming stage, Cassius. You are in the dreaming world, and so you're using dream circuitry, allowing you to tap into that circuitry and that network more easily to retrieve the old dreams.

7.83 So, sometimes what happened to me, for example, I had one dream: I was doing something, it was fun, and then I forgot. Then, I had another dream where I suddenly

remembered, "Oh, that dream happened. Was it this dream? Was it now?" You get confused sometimes. It's interesting. So, you have that.

7.84 So, what should we call this phenomenon? I don't think it has a name. Let's coin a name for it, so we can go into the science books. What should we call it? Double dreaming. I like that, I like that. Double dreaming.

Lecture 8: Dreams and Beyond

At a glance — Dreams and beyond.

- 8.1** You know, after having my first sleep paralysis experience as a teenager, that experience where I had the ghost come to my room and choke me and strangle me, I was terrified. Remember that? It was a terrifying experience, and it changed the course of my life. I went into science. I thought I had to study the brain. But even before this, in high school, and I hinted at this earlier, through my entire school years, in fact. But high school was just a nightmare. I hated science, chemistry, physics, math, biology. Hated it. So boring.
- 8.2** But right about the time I had that sleep paralysis experience, I remember I was just in a vacuum. I was just like, "Nothing makes sense. Why do I exist?" And this is going to philosophy almost. But, "Why do I exist? What is the purpose of me being here? Why am I trapped in this body? Why do I see myself like this? This vantage point of me looking down on my feet and my arms. Why am I trapped here? Who is it that is trapped in there?" I was asking, "Why here? Why now? Why me?" Does that make sense? Or is it too much?
- 8.3** But I was having that feeling a lot. It was quite something. I had that feeling all the time, and that got me even more away from school. I had 25% absence. They wanted to throw me out all the time. "We don't want this kid. We don't want this kid. We don't want this kid." College? Forget it. My parents had a pizza shop, a pizza restaurant. You have this in Miami, right? That's what I was doing. I was just going there sometimes. So, no college education in my family, nothing. You know, where I came from. You know, my parents' story.
- 8.4** I was like, "What do I do? There's no purpose to life. What is the meaning of all this?" And then, around that time, I had that sleep paralysis episode, which got me really triggered, and I had to study what was going on in the mind, the brain. What was all this? And suddenly, I found myself more in the library than in the ghetto, going around and doing things, doing my things. I was more in the library. I was reading but for my own purposes and out of my own interest.
- 8.5** Nobody was forcing me to do it. I didn't—nothing like that. I was just reading through this section, that section. "Oh, what about psychology? Oh, what about memory? How come we remember things?" I wanted to learn. "Why do I exist? I have to learn about the brain because that is the place where existence comes about. That's what my teachers are telling me."
- 8.6** And this gets to a point like the soul. Do we have a soul? What is the soul about? John is a soul, right? It's a big question. I don't have the answer to that, but some of these things in the brain look like and mimic some of the things that our soul is doing. Now, I'm a spiritual person. I'm a Muslim. I believe in more than we can see with the eye. As

a scientist, you can have a personal belief, and you can believe in the brain. In fact, what I do for a living is detangling and looking at the brain and saying, "Well, look, this is not supernatural. It's the brain. That 's why you see a ghost. This is not your soul. This is your brain doing this and that."

- 8.7** But you can still have your own beliefs. That's fine. But at that time, I was questioning everything. Everything was being questioned under the loop. Scrutinized. I had that sleep paralysis—I actually wonder sometimes whether that sleep paralysis was because of my own thinking. I wonder because it came around that time; I had that experience.
- 8.8** Then, I saw my vocabulary grow. My memory became better. I sought out mentors, scientists—first, not scientists, just friends who were more learned. I learned. They said, "Well, read that book, watch that movie. Do that. That has knowledgeable components." Things that I could learn and stimulate my brain. Just like mice running, stimulating them in a good environment. I was being stimulated now.
- 8.9** In the hood, I was also stimulated because we had to be on guard all the time. So, I was stimulated, too, when we had a lot of philosophical conversations with my folks. Trust me, when you walk around the streets over there, it's not simply things like, "Let's go do this now. We've got to do this." No, it was like, sometimes, "Is God real? Do we have a soul?" We would ask these kind of questions. "Your parents are from Lebanon, and my parents are from Iraq, Kurdish. We landed here in Denmark. What is this about? Are we really Danish?" Questioning our identity and existence and all that kind of thing.
- 8.10** So, I think this took me on a journey. All these questions, all this vacuum, all this turmoil got my brain stimulated, and then, eventually, I ended up, I guess, here at this very moment. At this very moment, I ended up here. I know I'm being very abstract now, but I just want to tell you this is how I feel. I want you to know who I am. That's how I ended up here.
- 8.11** For some reason, in my life, things would happen. So, I went to California. Well, before going to California, when I was in Egypt at an American university, a professor over there, when I was just a kid, happened to be the grad student of Mihaly Csikszentmihalyi, the guy who invented flow. The flow theory is that when you do something that's very engaging, you kind of forget everything. And so, this is very good for people that are stressed and worried because when you do something like play soccer or something, you forget about all your worries and you're in the flow. Intrinsic motivation. The guy who invented that, one of the founders of positive psychology—I ended up with him, went to California, met my mentor, things would happen like that, serendipitous encounters like that. And I ended up here.
- 8.12** But all this questioning—on the way, my brain got shaped in various ways. And by the end of this talk, I'll give you some of the, I guess I want—because some of you came far. Some of you came a bit far, and I want you to have some actual practical things that benefited me that I know helped my brain become sharper. My memory grew. I used to

have the worst memory, but my memory became better. I became sharper. I became better at this and at that.

- 8.13** Some talents I did not have, and I just said, "Well, look, I don't want to have them because it's not something that is worth me spending time on, for example." And then, I used the isolation principle of the autistic child and just focused on whatever I was good at and became better at that. I'll give you some clues at the end. Is that a good idea? This might be beneficial.
- 8.14** Now, dreams are strange and bizarre. They emerge out of REM sleep, we said. During the night, as you cycle through these different stages, you have maybe four or five cycles of 90 minutes of stages one, two, three, four, and REM. Some people say it's only four, meaning you say it's REM and non-REM. So, there's REM and then stages one, two, three, maybe, and REM. So, it depends. Some people exclude some of the earlier stages and just lump them. So, you have these different stages, and you have about four or five cycles of these during each night, give or take, let's say if you sleep seven to nine hours.
- 8.15** Now, of course, bats sleep about 18 hours. Smaller creatures sleep a lot. Bigger creatures, like an elephant, might sleep about four or five hours, for example. Or a giraffe. They don't sleep that much. So, we cycle through these stages.
- 8.16** You have more deep sleep: the restorative muscular, restorative cell, and restorative memory stage during the early part of the night, during the first part of the night. That means if I have a football match the next day, and I'm playing, and I want to be good, and I want my muscles to work well so I can dribble like Messi, then what? I want to make sure I get a lot of sleep in the first part of the night because, during the first part of the night, you have proportionally more deep stage sleep versus REM, the dreaming stage.
- 8.17** You might sleep in the first cycle or in the second cycle, and there's maybe only—of the 90 minutes, only about ten minutes of REM, maybe even a few minutes, ten minutes, and then, you have a lot of deep sleep, whereas by the morning, you have maybe one hour of REM sleep, and REM is more concerned with emotions, emotionality, and emotional therapy. That's REM. The early part of the night: muscular restoration and things like that. So, if you're playing that football game, even if you don't get a lot of sleep in the morning, it might be okay.
- 8.18** The memory part. That's very interesting, the memory part, if you look at it. If you have somebody who is, let 's say—because of the memory component of sleep and deep sleep—if you have somebody teach themselves to play something like the piano at night, by sleeping on that in the morning, do they become better or worse or the same? Better, worse, or the same? Better by sleeping? Come on. Better? What do you think? The same? Better? Exactly, they become about 20% better just by sleeping. Twenty percent better just by sleeping. Sleep is important if you have that exam and you just

want to stay up all night and just cram for that exam, not a good idea. Get that sleep in at that point.

- 8.19** Of course, during REM sleep, you have dreams, and then you might ask, Jose, dreams are strange and bizarre. Did we answer why this is so? As I mentioned earlier, that's because a part of the brain that constructs our logic, our sense of constructing a reality for myself, I'm here giving this talk, I'm wearing this jacket here in Miami, I construct a reality for myself, but that part of the brain is deactivated during sleep. It's called the dorsolateral prefrontal cortex, DLPFC. The shorthand is right here, in the prefrontal.
- 8.20** That part of the brain gives us a reality that's actually logically coherent, that's bound together in a meaningful way. Now, that part of the brain turns off during the REM stage of sleep; and therefore, dreams are illogical. That's why you are on the moon having a snack from the fridge, chatting with Queen Elizabeth, and playing soccer at the same time. Time, places, people, everything is warped, and that is the reason: because of the illogical structure being deactivated.
- 8.21** Now, do dreams feel real or not? Dreams feel real when you're in them. Do you know when you're dreaming when you're dreaming? Sometimes. What about most of the time? Do you know when you're dreaming when you're dreaming? No. So, when you're in a dream, everything is strange and bizarre. There's the flying elephant. There are all these crazy things. Do you know it's a dream even though everything is illogical? So, you lose your sense of agency. Your agency is lost. Why is that the case, Ismail? Tell me. This region of the brain: DLPFC again. Agency. The sense of—having a sense of self and being aware of yourself, like being actively aware. Brooke, do you agree? It makes sense, right?
- 8.22** So, now tell me, Brooke, how do we know this is not a dream right now? And I'm being perfectly serious, no joke at all. No playfulness. How do you know this is not a dream right now? How do we know? Pinch yourself, right? I don't have the answer. It makes logical sense that it's not a dream. I mean, I don't have an answer to that. But you may ask yourself—some people have argued that life is like a dream, and it's only when we die that the real life—this is the Chinese philosopher, we don't want to go there.
- 8.23** Are dreams emotional? Are they instinctual? Are they ecstatic at times? Why might this be the case? The dopamine center. So, one of the reasons is that the mesolimbic dopamine circuit, the frontal circuit going through the limbic system, is active. And if there's a stroke in this part of the brain, guess what? Mark Solms has shown that dreams go away. So, dopamine is important for dreams.
- 8.24** Not only that, it turns out that acetylcholine—have you heard about acetylcholine in this course? You've heard about it. What is acetylcholine doing? Attention. So, it gives you attention, focused attention, and plasticity. It turns out acetylcholine is hyperactivated during dreams. So, it hyperactivates the lower parts of the brain: the instinctual parts and the emotional parts. Acetylcholine is very excitatory, but

serotonin, the neurochemical, is very inhibitory and fuels, as a gasoline, the cortical regions. That part, that neurotransmitter, is off, we said. It's not working. The part of the brain called the raphe nucleus, the dorsal raphe nucleus, the hub where all these serotonergic neurons are produced, is turned off. It's called off.

- 8.25** So, serotonin is not being produced, so the gasoline is not there for the prefrontal. The CEO of the brain, the prefrontal cortex, now cannot compete with all the hyperactivity of acetylcholine flaring up from the lowest part of the brain. So, that's why dreams are so instinctual. Plus dopamine. You have this combination, and now, dreams are ecstatic and instinctual. They are all over the place.
- 8.26** When I see a pretty girl, for example. Sorry, I'm using girls all the time; I'm not being immodest, but I'm just being real. Now, if you see that pretty girl in your dream, what do you do? You may pour your heart out all at once. "I love you. You are so pretty." There's no filter, there's no inhibition. Why? Because of the brakes. You're like Phineas Gage now with your prefrontal being out of the way, out of the system. No prefrontal cortex here being able to dampen down the activity of your limbic emotional brain. Acetylcholine, dopamine. Does that make sense? Agree?
- 8.27** What about your volition? Have you ever had that dream where you just want to get away, but you're stuck, you're not moving, and there are monsters chasing you? Have you had that dream before? You've had it? What is that about? You're heavy. You all have this. Surely there must be some brain component, some brain explanation for this. It turns out your motor cortex, a part of the cortex, of course, needs serotonin to do its job of making you feel like you're moving. But at the same time, as we said, emotions are flaring up. So, in a way, the monster represents your emotional brain. The monster, the lion, whatever it is. You not being able to get away is simply your motor cortex not working, not having the right gasoline, not having serotonin, and so forth.
- 8.28** But then you might ask, "Why are dreams so spacey? Things are unfocused. They are all over the place." And surely, we talked about things being illogical, but this feels so bizarre—not bizarre, but it feels spacey. In fact, if you wake somebody up from REM, if you have somebody sleeping in REM sleep, he's sleeping in REM, you wake this guy up, and you ask him, "Sun and moon," or "Thick and thin." You ask him about these two words and see how fast he is to pair them versus saying, "Thick and chair." Two unrelated words. He is faster with the unrelated words, whereas you and I, when we are wakeful, are much faster comparing thick and thin, sun and moon. Does that make sense?
- 8.29** So, the brain, when you wake him up in REM, is primed towards linking seemingly unrelated objects, as if it has some functional logic in the system, as if it means something, or as if it's good to do that. This is where you can talk about the brain during the sleep stage as Indiana Jones here. It wants to explore things that normally you can't explore in an easy way. You want to take items and test them together. I

want to see myself on the moon. I want to see myself in these strange places with people in an emotional context.

- 8.30** And usually, there are a lot of people around. In dreams, you see a lot of people. They're populated by people, and usually, a lot of conversation is going on. But before I get to the people part, let me talk about the related words and stuff.
- 8.31** So, it turns out the chemical called noradrenaline is the chemical body of adrenaline. Adrenaline makes you focused; focused attention and things like that, you have that. Well, that part of the—that is a way. It makes you—it helps you with logical structuring. So, you don't have that. So, in other words, things are spacey. You have attentional focus and plasticity, but in an unfocused way; it's a paradox.
- 8.32** But everything about sleep is paradoxical. We talked about serotonin not being there, yet you have certain receptors of serotonin being active in other parts of the brain. Things are paradoxical in the dream stage. And so, you can relate things in new ways. You can think in out-of-the-box ways because you have that opportunity, John. It's a unique chance because normally, during life, you can't manipulate objects like this. So, it's a unique chance to do that.
- 8.33** I sometimes wonder. I talked about Ramanujan yesterday, the mathematician and genius. Do you know he would actually say that his math would come to him through his dreams all the time? How do you know Ramanujan? You don't have any access to books. You don't have libraries. How do you know these things? "In my dreams—I see these things in my dreams." How? I don't have the answer. It's very strange, but it seems like you have this ability to problem-solve, and in a way, you think about things in a totally novel, new way, and that experimentation is very important and a part of the whole dream thing.
- 8.34** Edison himself, at times, what he would do, he would sit, have a spoon in his hand like this, and then a plate would be on the floor, and then he would dose off to sleep, dose off to sleep, dose off to sleep. The plate would break. He would wake up and write up his ideas. Write them up. "I have this." There's almost—in that stage of access to dreams, we have the ability to think in novel ways. We have new ideas. We are not as inhibited by the cortical regions.
- 8.35** We talked about this at length yesterday. The cortex, the conceptual part of the brain, what Freud called the superego—I'm not a Freudian, but what he called the superego—is important to have, so we don't do crazy things. It helps society to have a structure because people have roles, and they're not all over the place like my talk. They're more systematic, and that is part of the? That is part of what? That part of the brain is not working as it should, so we have more freedom to explore in dreams. Edison would do that as a way.
- 8.36** Now, dreams are populated by a lot of people, we said, a lot of encounters. And it appears that we have a center in the brain called—actually, we do have a center in our

brain—called the theory of mind module. The theory of mind is a big concept. It covers a lot. When I'm chatting with Caddy, I 'm chatting with you; surely, I must have some theory of mind, meaning a theory of other minds. "What is Caddy thinking right now? Might she think this is boring? Might she think this is good, interesting, fun? What are her intentions right now? Agenda?" All that stuff.

- 8.37** You create this theory of mind all the time for survival reasons. "What is the lion thinking? What is the other guy in the savannah thinking?" Theory of mind. "What is the potential mate thinking?" Theory of mind. You need a circuitry in the brain in order to tap into other brains. Of course, not literally here, but you want to tap into other brains, Ismail, and understand, "What might that person think right now?"
- 8.38** That structure in the brain, it's unknown exactly where it is. Some have evoked the mirror neuron system. It tends to be in the—people talk about the frontal-parietal regions, going back to the parietal regions where the self is constructed, because in order to have an idea of you, I must first have an idea of me. What does it feel like to have a plan, an agenda? I know how it feels, so therefore, I can also sense how you might feel. It is about how you might feel. So, I need a sense of self first, so I create a theory of mind.
- 8.39** Now, for the theory of mind, in order to not have a disinhibited theory of mind, what is very, very important is to have inhibitory structures, structures that inhibit that. So, when I go out, I empathize with Amari, I empathize with him. There's empathy. There's—I 'm trying to understand what Amari is thinking right now. If he feels some pain, I may feel it, too. Not literally, but I may understand what it's like to have that.
- 8.40** Now, if my prefrontal cortex was out of the picture, perhaps your pain and your suffering might be more literal to me. I may feel like I'm actually kind of merging into you, the selves. This is not a joke. Think about it. Self-other distinction. We talked about the self-other distinction in the temporal parietal junction up here, back in the brain. In some people, like schizophrenics, it might not work. So, you feel like, literally, you are merging with Michael. The boundary between me and you is being blurred. It's no longer as crisp as before. Theory of other minds.
- 8.41** Now, in this dreaming state, is the cortex disinhibited? Is it weaker? Yes. It doesn't have serotonin, the gasoline it needs in order to inhibit the rest of the brain. Much of what the prefrontal does, as we've said, tends to inhibit the more lustful and excited brain of the limbic lower structures. So, that's part of it.
- 8.42** And also your skin, actually. Your skin, muscles, joints, and all of that stuff, all these receptors, give feedback to your brain, telling it that it's now in this body, right? But during REM, because of the lack of movement and the lack of sensory feelings in the REM stage, in the dreaming stage, you are not getting feedback from your body telling your brain that your brain is anchored right here. So, it's more loosened up. We also mentioned that.

- 8.43** So, because of these two components, because of the lack of inhibition from the prefrontal and overactive emotional brain and overactive empathy parts of the brain, you can create other theories of mind. You create theories of mind and project intentions and agency, agenda into these guys, and also emotionality. This person might have this emotion, and so on and so forth. Is that clear? Now, does that make sense? Does it really make sense? I want you to know this point. It's very important. Is it clear?
- 8.44** This explains why in your dream, you can have a very, very intimate conversation with another person. Have you had that before? A deep conversation? I had that before when I was—I remember having some of the most profound conversations with other people, presumably myself. Presumably, this girl I was conversing with, the beautiful one, was myself. It's a projection.
- 8.45** I had a dream once. This guy told me a joke, and I found it so funny. I was laughing. I was laughing, ha, ha ha, to the point I woke myself up from laughter. I went, "Haha," and then woke up. It was so funny. It was hilarious. But then I thought about it. If this is myself telling myself a joke, how come I find it funny? Jokes are only funny when you don't know the jokes. The whole story plot has to be hidden. That's the point of a joke. There's a story plot, and then there's a punchline. But if you know the joke, you're not supposed to find it funny. So, how come this happened?
- 8.46** You tell me, because I have no clue other than perhaps your brain is suppressing parts of your own data, and your brain works in layers, like the cocktail party effect, where you monitor the entire environment, but yet not really. You are in the overall sense : if somebody says your name, you're there. So, your brain can inhibit information at various points. So, it seems to be doing that here in dreams. Is that interesting, Michael, to think about? I find it usually.
- 8.47** Absolutely, it may, absolutely. The hippocampus may be involved, and the hippocampus is also involved in the sense that we forget our dreams. Ninety-five percent of dreams are forgotten. We forget these dreams. Now, you may ask, "Why?" Well, the hippocampus shuts down. And we also talked about why it's important to inhibit these drastic, and cosmic, and violent—it's like the matrix on steroids all night. You don't want to have that every day. You go to work and have all these dreams for eight hours, all these dreams pouring out of your REM sleep. You don't want that, so you inhibit it.
- 8.48** Now, the serotonin component. Serotonin is a factor that is important in your hippocampal region as well. And that is why when—and this is a good point. If you want to remember your dreams, what you do is that when you wake up— the wakefulness process itself involves the brain being flooded with serotonin. That's part of it: noradrenaline and serotonin, these chemicals. Noradrenaline comes from the locus coeruleus in the brain stem, in the lower part of the brain. These parts become

activated once you wake up. They give you arousal. So, come and pour on your brain, and that's why you wake up. Does that make sense?

- 8.49** So, you have a little memory when you wake up, at the moment when you wake up. At that point, Jose wants to write down his dreams right away, at that point, because you only have a little bit of time. If you wait too long until breakfast, what happens? You forget it. So, you want to take that opportunity.
- 8.50** Is there anything else I can do to remember these dreams? You tell me. Well, what if I drink this big glass of water before going to sleep? If I have a big glass of water before going to sleep, will I tend to wake up a bit more? Yeah. Will I tend to go in and out of REM sleep a little bit more? Yes. So, this might give us the opportunity to actually remember our dreams more. So, people who have a very solid architecture of REM in the sense that they just forget the dream, dream very easily. One trick is to drink a big glass of water, and that might help you have a bit more fragmentation in the sleep cycles, in the stages. So, you tend to wake up a bit more during the REM stage. That's one thing you can do.
- 8.51** But, yes, you can say that dreams are very much about updating the self. You update the sense of self. The ego is reboosted. It's like these updates you do on your computer, your laptop. You update it. It's like that each night. The REM stage allows you, through these dreams, to update the sense of self and engage in emotional interactions with other people, usually negative ones, because it has more of a survival benefit.
- 8.52** If I was dreaming about princesses all the time, it wouldn't help me survive much. I need some negative encounters. I need some battles there. I need some lions chasing me, too, and by me having these dreams of a lion chasing me and seeing myself running on the savannah and being chased and jumping over that thing, that little piece of tree, that stone, "I'm jumping, I'm running," it is reinforcing new circuitry. It is laying down circuitry, and because of the DLPFC being deactivated, I have no clue that it's only a dream. So, it feels even more realistic and emotional. So, what my brain does in this situation is it puts down new circuits. It lays down new circuits in a very immersive way, much more immersive than virtual reality because you have no clue that it's not real. And then, in that case, I don't need to practice on the real lion because I had the dream. Right, Mani?
- 8.53** You have a dress rehearsal for real life in your dream. And this is a good example of it right there. You can experience things without actually experiencing them. The same thing with encounters with people. You learn how to interact with people, what these feelings might be, you see how a person is talking, and oh, that facial expression. You might learn a little bit here and there. So, it has these components.
- 8.54** Now and then, you may have a fleeting thought that perhaps you are awake in your dream. You feel like, "Well, this can't be a dream; I kind of know I'm dreaming right now." What's going on there? In these dreams, called lucid dreams, you can meet all

the people you want to meet and engage in all the fantasies you want to engage in—because some of you are minors, I don't want to say what the most common theme is during these dreams. But during lucid dreams, you can create and manipulate your environment. You can fly, and people do that. That's one of the main things that people do. And many of us, about 20%, have had one of these lucid dreams. During these lucid dreams, you know you're dreaming. You know that this is a dream, and then you can start doing things. You can start experimenting in a much more conscious way. I'm just wondering: any lucid dreamers here? Oh, wow. This is a lucid dreaming club. Wonderful.

- 8.55** Well, the question is, why lucid dreams? Why do we have them? Are they a quirk, a phenomenon? Are they just something that—a spindle, something that's there because it just happens to be there, or does it have an advantaged purpose? Why do we have lucid dreams?
- 8.56** Think about the ego thing. The ego thing, the ego updating, has a lot of purpose. Having an ego is very, very important. Having a sense of self and being interested in that self allows you to survive, go and seek food, and all that. Having that is important. What about the lucid component, lucid dreams? What do you think?
- 8.57** Well, I argue that perhaps lucid dreams allow us to be more creative. We do research in labs, but here, you have a lab where there are no limitations. The laws of physics, Newton, are non-existent. You can do whatever you want. Pretty much. Of course, sometimes things can get out of hand, but you have more creativity. It's like the normal dream, but allowing you a bit more agenda, a little bit more manipulation there. Right, Amari? Is that right? Farah, what do you think? You can manipulate a bit more; you can do a bit more things.
- 8.58** So, now the question is, what's going on in the brain when we have these lucid dreams? How is the architecture and the structure and function of—well, not the structure, but how's the function altered during the dream state, during the lucid dream state? What do you think? Absolutely. The frontal cortex, the prefrontal cortex, the dorsolateral prefrontals become activated even though you are in REM. They become partially activated to the extent that you know you're dreaming when you're dreaming.
- 8.59** And I can tell you from lucid dreams that it's not always—because I have them sometimes. I have some of these phenomena. So, I have lucid dreams once in a while. But not all lucid dreams are equally lucid. Some are just halfway there, so you feel like you're lucid, but then, "Oh, this wall, I want to control it, but I can't. It's taking over. " So, there's this battle going on, in a sense, between your cortex and the lower parts of your brain and the agency's area that goes like this: this wrestling match between wakefulness and sleep. Does that make sense? So, that's what a lucid dream's about.

- 8.60** So, that's what I'm saying. A lot of these bizarre and strange phenomena that people find very uncanny, we explain in the brain, and not in an arrogant way, but simply pointing to the neurocircuitry. For example, out-of-body experiences. When you have these, we point to the TPJ regions, the self-regulation regions. These parts. And when these parts don't work, as in dreams because the TPJ, the superior parietal lobule—you know these fancy names by now, right? Body image and body construction don't work during dreams; that part of the brain turns down as part of the cortex. It goes down, and when that goes down, the self is loosened up. I may see myself chatting with an elephant, or I may become the elephant.
- 8.61** I swear to you, this is true. In a recent dream, I had one of my very good friends come in the body of a blackbird, chatting with me. I had no clue that it was a bird, but I knew it was my friend. I didn't find it strange that my friend had become a bird. That's what I'm trying to say. Is that clear? Everything was just normal. We were having this chat, conversing, "Omar, what are you doing here? This is great. This is a great conversation," but he was just a bird for a minute.
- 8.62** And that's something strange about dreams, this whole thing. We are in unfamiliar environments. There's this great theme in dreams: unfamiliar environments, like palaces and moons and other planets, perhaps. But they feel very familiar. You may have a dream of your bedroom, yet it looks like you're in the palace of some king five hundred years ago. It looks like that, but it just feels familiar; that's my room. So, it feels very familiar, even though it's epic and cosmic.
- 8.63** But what about waking up during REM sleep and having sleep paralysis? What's the purpose of that? Seeing a demon? Does it have a survival purpose? Does it teach you anything? Sleep paralysis is such a fascinating phenomenon because it tends to revise so much of the stuff. You have so many components of the brain right there. In one experience, you have motor paralysis: glycine and GABA paralyzing you during REM, but then you have the wakefulness process occur.
- 8.64** You start waking up in the frontal region enough for you to actually feel fully awake, sometimes semi-awake, but mostly you are conscious. And by the knowledge you now know, by the knowledge you've now accumulated, you know that the cortical regions, the cortex, the part of the brain that makes you know that you are awake and aware, are in a continuum. And if you have enough arousal there while still being stuck in the REM stage of sleep, you find yourself in sleep paralysis, this dual existence where two worlds collide : wakefulness and sleep. There's a collision of these two separate worlds.
- 8.65** So, what your brain does, because it comes out of REM sleep, you have something called REM mentation, REM imagery, the imagination of REM. All your dreams can easily be projected into your living room, which makes it quite scary. There's a portal there. There's a portal that's open that 's not supposed to be open, you can say in a way. Now, it helps us understand the motor system because you're firing all these

signals, but there's no feedback coming back, as we said. That's why you have the paralysis.

- 8.66** By the way, this is a great point. Do you know that during sleep paralysis, people will feel like there's chest pressure? Do you know that? Have you had that before? Anybody? It feels like there's chest pressure. It feels painful. During the REM stage, our physiology is like this. We breathe like that. That's a part of the REM physiology. When you wake up during sleep paralysis, many of the components of REM sleep carry over. They spill over into the REM experience because REM physiology is active; it's still there.
- 8.67** When you try to control that, it will feel like something is pressing on your chest. As we mentioned in the first lecture, when you try to move, you can't, so your brain creates hallucinations to fill in the gap. Incongruencies here; it fills it in. And when you try to control your movement and breath, what happens is that it will feel like something is holding you down because of the pain. You keep sending signals, "Move, buddy. Move, buddy. Move, buddy," but there's no actual movement occurring at this point. You need that proprioception, the feeling—proprioception means the feeling that comes back to the brain telling you that your body is moving—when that is lacking, it will feel like—it gives you a spasm-type of feeling. It feels like something is pressing on your hand.
- 8.68** My breathing is automatic during REM. Now, I'm consciously aware, and I'm trying to control it. But because it's automatic, when I control it, it feels like something is pressing on my chest. So, something is pressing on my chest. Something is holding me down. My sense of self is being loosened up. The TPJ regions are floating. They're less active. So, I feel like I'm floating. Perhaps my whole body is projected out there.
- 8.69** Now, what do you think? Does your brain like to create narratives and stories and link things together? Think about this whole course, the rubber hand illusion, all these things. Does your brain take random events and say, "Well, look, this couldn't happen by random chance. There must be some story behind this. It must mean something in a gestalt, overall way." Does that make sense?
- 8.70** Your brain is a great storyteller during REM sleep and during dreams, in fact, a lot of the things that are going on over there are dreams, and these dreams have stories. They have story plots—complex story plots. And so, your brain loves building narratives. It's part of the whole process of survival. You embed whole memories about episodic memories about building narratives. So that happens during the dreams as well, and so it has this narrative component.
- 8.71** And then, when you have all these features, seemingly random: something is holding you down, something is pressing on your chest, I can't breathe, I'm being strangled, I'm seeing my legs flying up and down, my whole body is spinning in the air—does it give grounds to build a narrative around that? About a ghost coming and chasing you? And

may you project higher-level cortical conceptual regions—we said vision and mind are when you go up into higher regions; down is more conceptual—might that spill over and give you a more complex narrative? What do you think? It might. So, you have that during sleep paralysis.

8.72 As I mentioned yesterday, people are terrified. They have all these ghostly explanations about what's going on in the brain. This is one of the factors that I've suggested: all these factors put the ground for your brain to create a narrative around the whole thing and explain why we see this.

8.73 During that stage of sleep, during the sleep paralysis stage, occasionally, what can occur is that you can have a sleep paralysis episode that's only partially negative. So, about 90% of these episodes are terrifying, trauma-like, dangerous. Why? Because your brain is in a vulnerable state. It's lying there. You're scared. You can't move. So, your emotional part of the brain is hyperactive, but you seem to have an activation of those receptors that make you have fear and things like that. So, you have that. But sometimes, though, and I've had this, it's not terrifying, and this is where we get into things like the soul.

8.74 I'm not saying this is the soul, but what I'm saying is, I had a dream once. I was in California, sleeping on a mattress again. I had these nice mattresses or not-so-nice mattresses. And as I was sleeping one day, I found myself paralyzed. I couldn't move. I was trying to move. Nothing. Scream, nothing. No words. Nothing came out. Nothing would come out of my mouth. I said, "Oh, goodness, another sleep paralysis episode."

8.75 But then, suddenly, I kind of jolted out of my body, and I became a ghost. It sounds strange for a scientist to say this, but I became a ghost, and I was walking around in my apartment in California at the time. I was walking around as a ghost. I'm not kidding; this is actually true. I was walking around, seemingly in my apartment, and I was walking around in my apartment. It occurred to me that I had to do an experiment. I had to test this.

8.76 So, I found this scrap paper on the floor, and I put it in my pajamas. I put it in my pj's, and I said to myself, I'm going to go back to sleep now, Mani. I'm going to go back to sleep, and if this is still there in my pj's when I wake up, then I would have shown something, that something is happening beyond the dreaming world. I went back and jumped back to my body. I woke up.

8.77 I want to be completely honest with you. Was it there or not when I woke up? Unfortunately, it wasn't there; otherwise, I'd be giving this talk at the Nobel conference, alright? It wasn't there, but at least scientists like me can say that we're working while sleeping. We have that thing; we have that privilege. No, unfortunately, it wasn't there.

8.78 And that brings me to near-death experiences. What about near-death experiences? Alicia, we were talking about this this morning. What's happening in near-death

experiences? Is the soul leaving the body? What's going on? People will say that they're lying there and they're about to die and then—apparently about to die—and then they feel like they're seeing their whole room in front of them, and they sometimes go into a light and they can see a light in a tunnel and they have some spiritual experience. What's going on?

- 8.79** So, we don't actually know for sure, but we do know that depriving some of these regions involved in the self-construction of oxygen can also create some of these bizarre self-jolting experiences where the self is jolted out, and you feel like you're scanning the room. Of course, you know how the room looks because you were there; you were there the day before. So, you may have some component of that in your memory. So, you're reconstructing that. That's what scientists would argue.
- 8.80** That's what the *prima facie* explanation would be. When you look at it firsthand, that's what it looks like. And then, you have endorphins potentially flooding your brain because you're in surgery, giving you maybe a sense of calmness, a sense of well-being that's beyond this world. And then, you might have that combination. So that's one explanation.
- 8.81** Now, you could test this in an experiment, but how would you test this? All these people having out-of-body or near-death experiences—how would you test this? Let's say, what could you do in the hospital to test it? So, my friend and I—Ramachandran, had this idea. We said, "Look, what if in the hospital, the various hospitals, you put some text, like newspaper or something, on the ceiling as a way the person can read it, and then when they have the experience, they have to simply say what it was up there and give the information. Is this going too far? We did have this chat, conversation about doing this experiment.
- 8.82** Look, I'm not saying there's no soul. The soul is metaphysical. In fact, what I always do in science is say, "Look, metaphysics, beliefs, spirituality, which scientists like myself can have, do not conflict with science and studying the physical world." In fact, you should always be 100% honest and go to the science and study what we can see. That's what science is all about. So, that's very important, but it doesn't negate other types of reality. We don't make fun of these things. You know what I'm saying? It's not like, "Oh, this is just the brain. It means you're crazy or something." These are two explanations, two ways, of describing the world. But as scientists, by understanding the brain and what it does, we can begin to understand it in a more natural way. Is that clear? I just want to make that distinction. It's very important.
- 8.83** There's another experiment that we thought about, me and my friend in California, the phantom mirror guy. You may have seen "Inception," the movie "Inception." And we wondered, "Is time, the passage of time in dreams, the same as a passage up here in this life?" I mean in wakefulness, of course. "Or is it slowed down there? Is an hour in dreamland corresponding to a thousand—is it thousands of years down here?" So, an

hour up here in our wakefulness state, is it like thousands of years? Like in the movies? How is the passage of time? Is it slowed down? Is it the same? Is it faster?

- 8.84** Well, we had an idea to test this. So, if you have this patient, we call the patient Joe. Joe is lying here, and as Joe is sleeping, we measure the activity of his brain. We ascertain through the muscle activity and all that, that he's in REM. So, he's in REM sleep. Sleeping. And presumably, now he's, of course, dreaming. Not presumably. He is dreaming, but he's having a good dream right now. Imagine this: in this dream, while he's dreaming, I, the wakeful experimenter Baland, can be mischievous, get close to him, and pour some water on him. I pour some water on the sleeping subject. Now, the question is, would he be able to incorporate that stimulation into his dream?
- 8.85** First of all, this is important. We kind of know that you can infiltrate dreams by external stimulation. So, if you are in a cold room, Amari, you may feel cold in your dream. If you are in a smoke, if there is smoke outside your apartment, then you may feel like there's a fire in your dream. So, we know that. So, infiltration itself is no problem. It should happen.
- 8.86** But if he's walking in his dream here in Miami. He's walking around, and then one of those days, it's raining a lot, so he takes out his umbrella, opens it, and walks. And as he's walking, the question here—that spraying of water represents the rain. It represents the rain in the dream because we made an infiltration. So, here's the question. In that time, between him seeing the rain, from before it comes until the rain comes, is it a quick gap? In other words, will he say, "Oh, the clouds came out, and then I texted my mom, and I called a friend saying, 'Oh, remember, we have to cancel this appointment because it's raining.'" And then, slowly, the rain came." That's one scenario. Or scenario B: you pour the water on him, and he will say when you wake him up, "No, I was in a dream, and a tsunami hit me. I was hit by a tsunami. Water just came all over the place." Which two explanations do you think might be the case? Mind you, it's very important, this point that the guy is sleeping, he's dreaming, you pour the water on him, and you wake him up in milliseconds, maybe a second after the water, you wake him up. So, it's very quick.
- 8.87** Now, the question is, which one is it? Is it like, "I was dreaming and the tsunami." Or does he have time? Maybe a minute or two? What feels like a minute or two or three or hours in there? You can potentially ask this very important inception-like question by doing this experiment. So, these are the things we're thinking about in our labs. Not only boring stuff but things like this, too, to study this.
- 8.88** Now, there's a problem here because, in my time in Boston, there was a dream researcher over there. His name was Hobson, Allan Hobson. Unfortunately, he passed away. He was one of the greatest sleep or dream researchers of the last century, pretty much after Freud. I was lucky to get to know him when I was there. It was actually the early days, my graduate school days that I spent some time there.

- 8.89** One day, I just had this feeling, "I know of Hobson. Let me just contact him and see if he wants to chat with me." I was interested in dreams and had done this work in California but hadn't done much research. A little bit of sleep paralysis, but I mustered the courage and contacted him. Kind of like *carpe diem*.
- 8.90** So, I called him, and it was very nice. We talked on the phone, and he was like, "Tell me what you're up to and what you're doing." And I said, "Well, look, I'm interested in these experiments on dreams." And before I could talk about that, he was more like, "No, tell me about you. Who are you? What are you doing? What is your life about?" And I was very—it felt very warm. "Are you Kurdish? Oh, tell me about their history; tell me about that." And then, after all that time, he said, "Okay, now you tell me about the experiment." I was very moved. Having mentors like that is quite something, isn't it?
- 8.91** Well, he wasn't really my mentor, but somebody that I got to know briefly, unfortunately, not very long. I mentioned this experiment to him, and he said to me, "There is a problem in this experiment because in the REM stage of sleep, when you're dreaming, there's a sensory blocking barrier, meaning that if you pour water on him like that, either it will be too low potency to actually have any effect on his dream. In real life, yes, you may have certain instances of this occurring, but in a dream, it may be very difficult in an experimental setting. Or, if you pour too much on him, he will wake up," he said. So, I said, "Oh, that's too bad. And we can't do this experiment. It's not good."
- 8.92** Then, hours later, I got a phone call from him. He said, "Baland, I have figured out the problem that has puzzled scientists for centuries." These are the kind of calls you get out there. This is the—I have gotten around this problem, the sensory barrier problem, that has puzzled scientists for centuries. I have a solution for you; what you can do." Now, the solution, I'm going to mention it after the talk. It's a secret because otherwise, somebody might scoop it and do the experiment. But it entails using—I'm just going to say this now—TMS, so you can stimulate the cortical regions, the sensory cortex, to make it just the right sensitivity if you want to open up some of these gates to dreams.
- 8.93** But truly, my encounters with these people, with people like Hobson and Ramachandran, changed my career. Especially Rama, my countless walks with him on the beaches of San Diego, talking about science, the history of science, talking about the brain, talking about people having mystical experiences, and how the brain works. He's a very open-minded guy. It really moved me, and it really shaped me and shaped my understanding of the brain.
- 8.94** So, some of the principles that I've come up with, my rules for life, or my rules for the brain, are the following. What helped me was the following. First of all, I got out of my little room and the hood I was in because the environment does a lot, and when you change the environment, what happens? A lot of neuroplasticity. Acetylcholine, dopamine, everything is new. That was a huge trigger for me.

- 8.95** Then, you bond with people. Social bonding is so important. This is not only back then but now, at any given time, because of how social bonding and oxytocin eradicate old circuits and allow you to lay down new circuits and diminish the activity of the amygdala in this way. It's a great way to, first of all, become less self-centered but also just live in a more happy state, be more happy, be more fulfilled. You don't have to be in that dopaminergic state all the time of excitement of the next goal, the next million dollars, as they say in America. Just chasing that dollar. No, you got to live. You also have to live because your brain, dopamine, dopamine neurons run out of what? They need to be replenished. So, if you are always in that state, you may burn out. So, that's another part of it, allowing yourself, your brain, to have that plasticity in that way.
- 8.96** Practical things: if the running wheel works for the mouse, it also works for human beings. Exercise is one of the ways to trigger BDNF in a very direct way. Brain-derived neurotrophic factor. Remember the lady, Rita Montalcini? The Jewish lady in fascist Italy who discovered these nerve growth factors? Well, it turns out exercising, cardiovascular exercise, things that would make your heart pump, will trigger this. So, this is a very good thing to do all the time to get more plasticity.
- 8.97** So, those are my main, I guess, the key points. I think, other than that, is having an open mind and realizing—this is, again, getting to some deep stuff, but realizing that this very moment is a gift also. Appreciation. Get some of the right neurons active.
- 8.98** For example, when you think about it, had your great-great-grandmother coughed when she was intimate with your great-great-grandfather, you wouldn't be here today. Just a simple cough. In fact, anyone in that chain of great-grandfathers and mothers, if anybody had a cough, you wouldn't have been here today. The fact that you are here is statistically, highly, almost totally impossible. You're not supposed to be here today, Jose. No way. You're not supposed to be here. It's statistically not supposed to happen. Yet you are here. You competed with so many other cells to get here, and so many events had to occur in a way that was very improbable. Forget the lottery. This is the next level. Yet you are here.
- 8.99** This allows appreciation, greater context, and understanding that we shouldn't take life for granted. Being happy. Happiness makes you what? Live longer. Stressed people have what? Forty-eight percent more stress level. Stress affects what? The whole system, the amygdala, everything. It's said that health and immune systems are destroyed by stress.
- 8.100** So, these are the points that I wanted to cover with you today. I do want to thank you so much for coming and listening.